MEASUREMENT OF ALPHA CAPTURE REACTIONS ON ^{10,11}B

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Abstract

by

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Nucleosynthesis in the first generation of massive stars offers a unique setting to explore the creation of the first heavier nuclei in an environment that lacks any heavier nuclei impurities from earlier stellar generations. In later generations of massive stars, hydrogen burning occurs predominantly through the CNO cycles, but without the carbon, nitrogen, and oxygen that allow this catalytic reaction sequence, these stars would have to rely on the inefficient *pp*-chains for their energy production. However, their may be other reaction chain sequences that utilize only light elements act as alternative pathways around masses 5 and 8 that activate under these special conditions. One such reaction chain could be ^{2}H$(\alpha,\gamma)^{6}$Li$(\alpha,\gamma)^{10}$B$(\alpha$,n$)^{13}$N. A competing reaction chain could also be ^{10}B(p,$\alpha)^{7}$Be$(\beta\nu)^{7}$Li$(\alpha,\gamma)^{11}$C$(\beta\nu)^{11}$B$(\alpha$,n$)^{14}$N that would provide a neutron source to this early stellar environment. In this work, improved measurements are reported for both the ^{10}B$(\alpha$,n$)^{13}$N and ^{11}B$(\alpha$,n$)^{14}$N reactions. The measurements of the ^{11}B$(\alpha$,n$)^{14}$N reaction utilizes a standard ^{3}He counter, while those of the ^{10}B$(\alpha$,n$)^{13}$N are made with a state-of-the-art deuterated liquid scintillator. *R*-matrix analyses are performed for both reactions in order to facilitate a comparison of the underlying nuclear structure with reaction measurements and for reaction rate calculations. The resulting rates are used to estimate the effects of the reactions on first generation stellar nucleosynthesis.

CONTENTS

FIGURES

TABLES

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CHAPTER 1

INTRODUCTION

1.1 Nuclear Astrophysics

1.1.1 Stellar Evolution at Solar Metallicity

The mass of a star, as well as its initial chemical composition, determines its evolution track. Numerous work has been done to investigate the evolution of stars at solar metallicity in the past decades. Metallicity is defined as the abundance of elements that are heavier than hydrogen and helium. In this section, we will discuss the nucleosynthesis and burning phases related to different evolution stage of stars with different mass at solar metallicity.

All stars begin their journey on the main sequence by starting central hydrogen burning. Depending on the initial mass, the hydrogen burning phase can undergo different paths, pp-chain or CNO cycle. Both processes fuse four protons into a single helium nuclei, and release an energy of Q = 26 MeV [61].

Stars with initial mass $M \leq 1.5M_\odot$ ($M_\odot$ is solar mass) have their energy generation dominated by pp-chain. This process is initiated by fusing two protons into a deuterium, followed by the reaction $d + p \rightarrow^3 He + \gamma$. Depending on the temperature, density and composition, this process can be followed by three different reaction chains (shown in Table. 1.1). The energy production of each chain is slightly different, due to the energy carried away by neutrinos. The effective burning temperature of the pp-chain is T = 4-16 MK. After hydrogen is exhausted, the core starts to contract, leading to core temperature increases and eventually shell hydrogen burning is

TABLE 1.1

PP-CHAINS

pp-I chain	pp-II chain	pp-III chain
	$p + p \to d + e^+ + v$	
	$d + p \to^3 He + \gamma$	
$^3He +^3 He \to \alpha + p$	$^3He + \alpha \to^7 Be + \gamma$	
	$^7Be + e^- \to^7 Li + v$	$^7Be + p \to^8 B + \gamma$
	$^7Li + p \to \alpha + \alpha$	$8B + \beta^+ \to^8 Be + v$
		$^8B \to \alpha + \alpha$
$Q_{eff} = 26.19 MeV$	$Q_{eff} = 25.65 MeV$	$Q_{eff} = 19.75 MeV$

Notes: Three different types of pp-chain reactions. pp-chains schema adapted from Ref. [57].

ignited. For stars more massive than $1.5M_\odot$, helium burning is initiated through the triple-α process and the $^{12}C(\alpha,\gamma)^{16}O$ reaction [32], when the temperature reaches 10^8 K . This is the last burning stage for stars with mass less than $8M_\odot$. These stars will end their life as carbon-oxygen core white dwarfs.

High-mass stars undergo the hydrogen burning stage dominated by the CNO cycle, in which carbon, nitrogen and oxygen serve as catalysts to convert hydrogen into helium. A higher temperature of $T \geq\sim 20$ MK, and the existence of C, N, O in the initial environment, are the prerequisites for CNO cycles. Once the core hydrogen is exhausted, stars enter the core helium burning stage and continue burning hydrogen in the shell. These burning stages continue with core carbon burning, shell helium burning, core neon burning, shell carbon burning, core oxygen burning, shell neon burning, core silicon burning, and shell oxygen burning. The burning processes result in a onion-like structure (as shown in Fig. 1.1), and stop when the core is mainly composed of iron group elements. At this point, further nuclear reactions can not release energy to maintain the inner pressure to resist core collapse. Eventually, a type II supernova event happens leaving behind a neutron star or a black hole, depending on the initial mass of the star. During this explosive event, nucleosynthesis products are ejected into the universe and will be used to form new generations of stars. This matter cycling process between stars and interstellar medium leads to the fact that younger stars are more heavy element enriched.

There are three distinct stellar generations, which are known as Population I, II, and III. They are classified by age and metallicity. Population III stars are the first generation stars, which are believed to be metal-free. The zero-metallicity feature makes Pop III stars a unique site to explore the creation of the first heavier nuclei in the primordial environment.

1.1.2 Massive Star Evolution at Zero-Metallicity

The nucleosynthesis in first generation stars is fueled by the primordial abundances of hydrogen, helium, and lithium isotopes produced in the Big Bang. The energy generation in massive primordial stars is therefore characterized by a completely different nucleosynthesis reaction pattern. The lack of carbon and oxygen in primordial material means that hydrogen burning is constrained to the *pp*-chains without any CNO cycle contributions [110]. Since the energy production through the *pp*-chains is limited by the slow, weak interaction based fusion of two protons

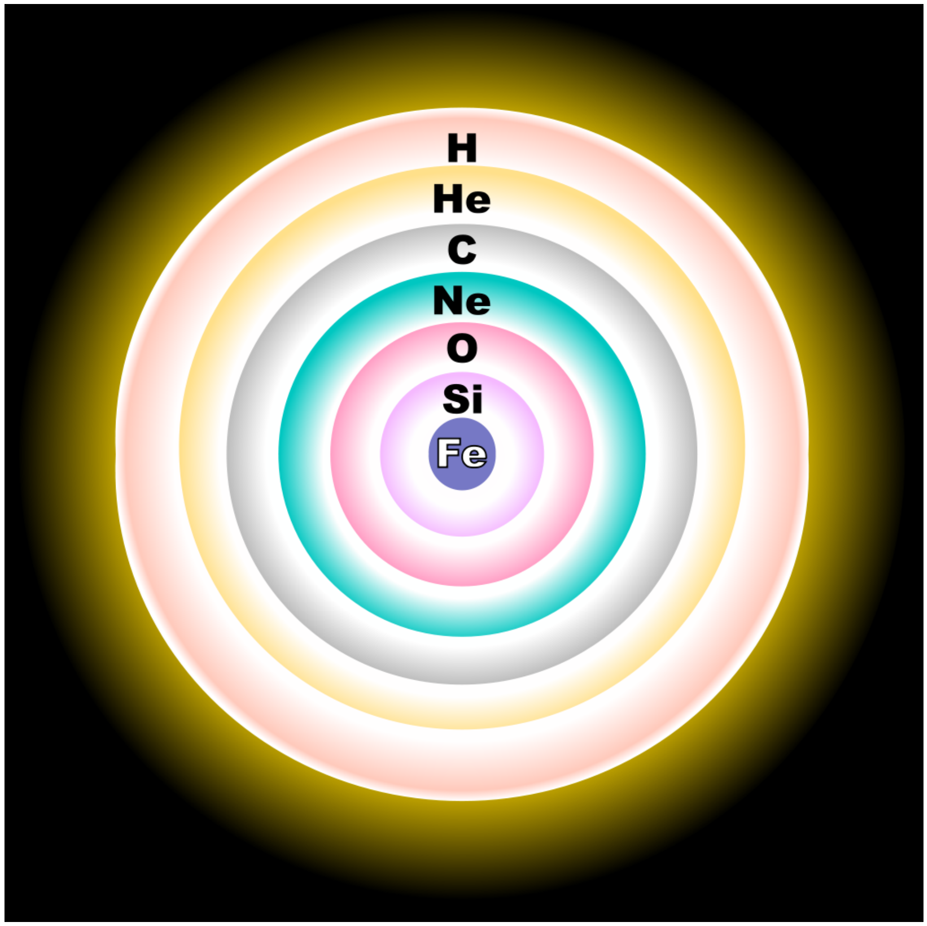


Figure 1.1. Shell structure of a massive star before its core collapse (layer size not to scale). Figure credit to Wikipedia .

to deuterium [11], the internal energy release cannot compensate the gravitational contraction of the massive star causing a gradual increase in temperature and density until helium burning ignites through the 3α-process [51], producing the first generation of ^{12}C and subsequently ^{16}O through the $^{12}\mathrm{C}(\alpha,\gamma)^{16}\mathrm{O}$ radiative capture reaction [32]. Both ^{12}C and ^{16}O are observable in pronounced abundances in the oldest - second or third - generation stars in our universe [43, 63].

The 3α-process presumably serves as the main reaction link between primordial ^{4}He and ^{12}C, ^{16}O abundances. A previous analysis identified the conditions in primordial stellar environments where alternative reaction patterns feeding CNO elements may occur [109]. However these reaction links were handicapped by insufficient amounts of primordial deuterium to emerge as a suitable alternative. Here we consider two further reaction sequences linking ^{4}He with the CNO mass range, these are reaction branches based on primordial lithium abundances ^{2}H$(\alpha,\gamma)^{6}$Li$(\alpha,\gamma)^{10}$B$(\alpha$,n$)^{13}$N or alternatively ^{10}B$(\alpha$,d$)^{12}$C. One should be aware that this cycle has strong leakage due to competing hydrogen induced reactions such as ^{6}Li(p,α)^{3}He and ^{10}B(p,α)^{7}Be, but to determine the efficiency of this reaction path both more information about α induced reactions on ^{6}Li and ^{10}B are necessary as outlined below.

A second possible reaction branch is based on the primordial abundance of ^{7}Li with the ^{7}Li$(\alpha,\gamma)^{11}$C branch with the latter rapidly decaying to ^{11}B. Again, it is expected to be a weak branch in view of the strong competing ^{7}Li(p,α)^{4}He branch closing the second pp-chain. However, depending on the reaction rate and the primordial abundance distribution, the α induced reaction branch may emerge as a further link to the CNO range due to the subsequent ^{11}B(α,n)^{14}N reaction, which would also provide an appreciable neutron flux to the first star environment.

This work is focused on studying the strength of α induced reactions on ^{10,11}B at very low energies, in order to explore the impact of these two reaction chains.

1.1.3 s-Process

During the burning stages described in Sec.1.1.1, many nuclear reactions take place and synthesize elements heavier than A $\geq$ 12 [57]. Elements lighter than iron are mainly synthesized via charged particle induced reactions. This type of reaction becomes increasingly unlikely with increasing atomic number, due to the rapidly increasing Coulomb barrier. Neutron capture process is the mechanism that is responsible for the synthesis of elements heavier than iron. Depending on the neutron capture rate and the beta decay timescale, neutron capture process is categorized into s-process (slow neutron capture process) and r-process (rapid neutron capture process). The latter one will not be addressed here.

The s-process undergoes a series of neutron capture reactions hence neutron rich isotopes are synthesized. At some point, an unstable nuclei is formed, which has a faster β-decay rate than its neutron capture rate. As a result, this nuclei is converted to its isobar with proton numbers increased by 1. Therefore, the s-process proceeds along the valley of stability.

The prerequisite for the s-process is an initial stellar environment with neutron densities $N_n \simeq 10^8 cm^{-3}$ [43]. These free neutrons are mainly coming from 2 reactions, $^{13}C(\alpha, n)^{16}O$ and $^{22}Ne(\alpha, n)^{25}Mg$. The reaction rates of these two reactions play an important role in the determination of the s-process yields. Therefore, extensive investigations have been dedicated to measure the cross sections of both reactions.

For the $^{13}C(\alpha, n)^{16}O$ reaction, the effective burning temperature is around $T \sim 0.01\ GK$ [61]. Recent measurements on the S-factor from Drotleff et al. [36], Davids [30], Bair and Haas [13], Brune et al. [22], Harissopulos et al. [49], and Heil et al. [52] are shown in Fig. 1.2. So far, this reaction has been measured down to the center of mass energy of $E_{c.m.} \sim 280$ keV, which is higher than the Gamow window (see Sec. 1.3.1) ranging from 120 to 250 keV. Thereby, the S-factor in this astrophysically important energy region relies on extrapolation of the existing data. The discrepancies between different measurements give rise to an order of magnitude difference [26] in the extrapolations towards the region of interest. Similarly, the cross section of the $^{22}Ne(\alpha, n)^{25}Mg$ reaction has not been measured at relevant energies yet.

One of the challenges of very low energy measurements has been neutron-producing background reactions of light-Z nuclei. As the energies of incident alpha-particles decreases, the possibility of tunneling through the Coulomb barrier decreases exponentially, hence the cross section, as shown in Fig. 1.3. An important candidate nuclei is boron, because boron is a very common target impurity [34] with small atomic numbers, and the $^{10,11}B(\alpha, n)^{13,14}N$ reactions have positive Q-values.

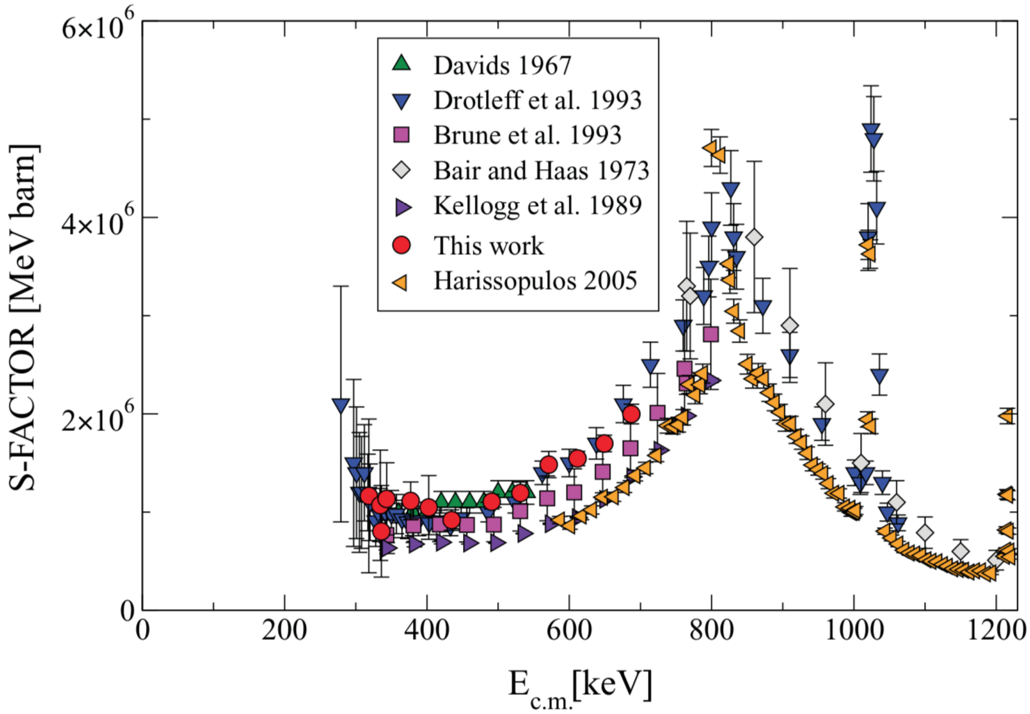


Figure 1.2. Previous results for the S-factor of $^{13}C(\alpha, n)^{16}O$. Figure from Ref. [52].

1.2 Applied Nuclear Physics

The National Ignition Facility (NIF) [79] provides the unparalleled opportunity to study matter under high temperature and pressure conditions that are not currently achievable through any other means. These conditions are achieved by focusing 192 high powered lasers, delivering a total power of 500 TW to a target system, and ultimately onto a single point a few millimeters in diameter. The conditions are comparable to those found in the heart of a star or at the epicenter of a nuclear detonation. In this environment, material is in the form of a plasma, whose physical properties remain poorly understood. NIF offers an experimental laboratory in which to study this environment first hand. Yet, the timescale of a NIF implosion (shot) is

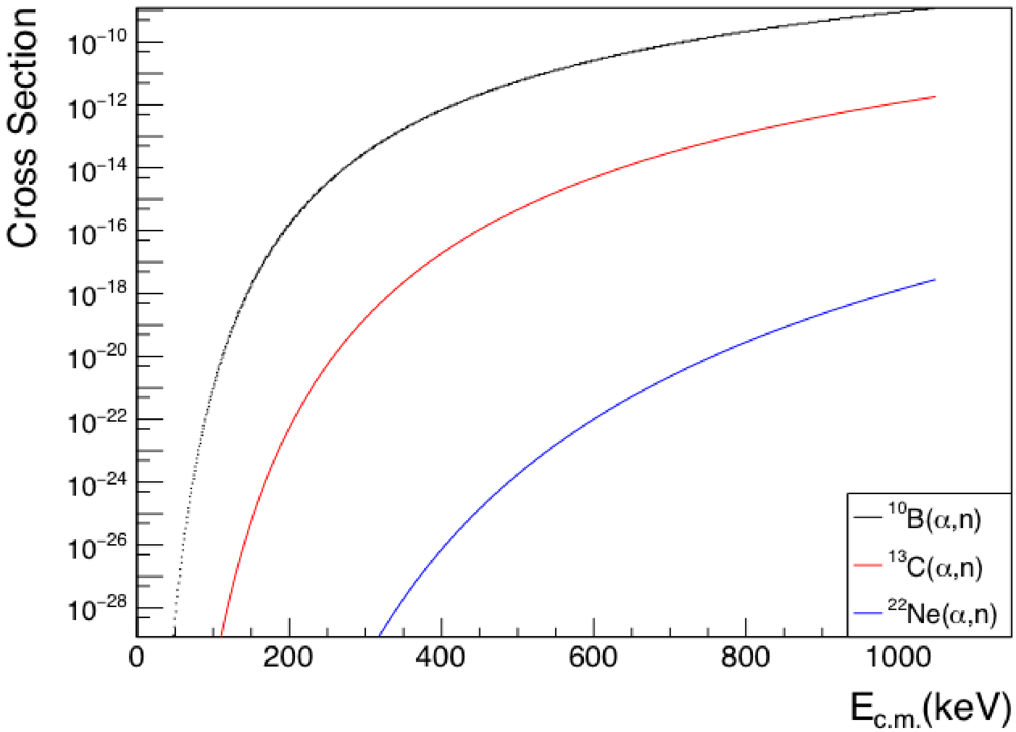


Figure 1.3. Coulomb barrier of different reactions.

only a few nanoseconds and the repetition rate is of the order of a few shots per day, making conventional measurements very challenging. Therefore, indirect and novel methods must be developed and utilized. For recent reviews of the high-energy-density physics at NIF see Hurricane and Herrmann [56] and Cerjan et al. [23].

One such indirect method for obtaining diagnostic information is that of activation, a longstanding technique in the nuclear physics community. NIF implosions often use equimolar DT as a fuel mixture, producing energy through the very exothermic (Q-value of +17.6 MeV) $d(t,n)\alpha$ reaction. The reaction produces neutrons at $E_n \approx 14$ MeV and α-particles at $E_\alpha \approx 3.6$ MeV. These α-particles are rapidly stopped in the high density environment of the compressed capsule material [56], dumping additional energy into the system. This so-called alpha-boot-strapping effect is required to increase the energy budget towards ignition, but has only been indirectly tested. Within the short range of the α-particles, dictated by the stopping conditions, α-induced nuclear reactions can occur on fuel, ablater, and dopant materials in the capsule environment. These reactions can be utilized for testing the alpha-bootstrapping concept and to verify theoretical predictions of the phenomenon. Possible reactions in unprocessed fuel isotopes such as ^{2}H$(\alpha,\gamma)^{6}$Li and ^{3}H$(\alpha,\gamma)^{7}$Li, may occur, but the low number of reaction products will remain unidentifiable compared to the natural lithium abundance in NIF materials. A good candidate for a dopant reaction is ^{10}B$(\alpha,n)^{13}$N because it produces a radioactive isotope that can be easily analyzed as a NIF product. The reaction product ^{13}N has a half-life of $t_{1/2} = 9.965(4)$ minutes [5]. This half-life is long enough that the ^{13}N material can be collected after the shot using the Livermore Radiochemical Analysis of Gaseous Samples (RAGS) system [96]. It is important to note that only reactions populating the ground state of ^{13}N will contribute to the final yield. This is because excited states of ^{13}N are proton unbound, and will instead immediately decay to stable ^{12}C. Therefore we need to study this reaction in the energy range between 1 MeV and 3.5 MeV.

1.3 Reaction Theory

1.3.1 Reaction Rate

The knowledge of the reaction rate of a nuclear reaction is important in determining the energy production in stellar environment. The rate is dependent on the probability that a nuclear reaction will take place, which is the definition of cross section (σ). The cross section σ can be expressed as Eq. 1.1,

$$\sigma = \frac{N_R}{[N_b/A] \times N_t} \tag{1.1}$$

where N_R is the total number of interactions occurred per unit time, N_b is the number of particles that incident on area A per unit time, and N_t is the number of target nuclei within the beam. Generally, cross section is energy dependent, hence velocity dependent. As cross section can be measured experimentally or estimated theoretically, one can calculate the reaction rate with Eq. 1.2.

$$r_{12} = \frac{N_1 N_2}{1+\delta_{12}} \int_0^\infty \sigma(v) v \phi(v) dv = \frac{N_1 N_2}{1+\delta_{12}} \langle \sigma v \rangle \tag{1.2}$$

where $N_{1,2}$ represents the number of interacting particles of type 1 or 2 per unit volume, v is the relative velocity of particle pairs, $\phi(v)$ is the relative velocity distribution, and δ_{12} is 1 if the interacting particles are identical, 0 otherwise.

Since stellar plasma is considered to be in thermodynamic equilibrium, the relative velocity distribution $\phi(v)$ of particles follows Maxwell-Boltzmann distribution,

$$\phi(v)dv = (\frac{\mu}{2\pi kT})^{3/2} 4\pi v^2 exp(-\frac{\mu v^2}{2kT})dv \tag{1.3}$$

where k is the Boltzmann's constant, T is the temperature, and μ is the reduced mass given by $m_0 m_1/(m_0 + m_1)$. As a result, the stellar reaction rate can be rewritten as Eq.1.4,

$$\langle \sigma v \rangle = (\frac{8}{\pi\mu})^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma(E) E exp(-\frac{E}{kT})dE \tag{1.4}$$

For charged-particle induced nuclear reactions, such as (α, n) reaction, two positive charged nuclei must overcome the repulsive Coulomb force to make the reaction occur. Gamow [45] proved that it is possible for particles with energy lower than the effective Coulomb potential $V_C = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 (R_1+R_2)^2}$ to penetrate through the barrier. In general, thermal energies are far below the Coulomb barrier. Under this condition, the nuclei radii is much smaller than the classical turning point R_C, as shown in Fig. 1.4, the tunneling probability, which is also known as Gamow factor, can be approximated by Eq. 1.5,

$$P(E) = exp(-2\pi\eta) = exp(-2\pi\frac{Z_1Z_2e^2}{\hbar v}) \tag{1.5}$$

where η is called the Sommerfeld parameter, Z_1 and Z_2 are charge number of the charged particles, v is the relative velocity of the particles, $\hbar$ is the reduced Planck constant. Due to the exponential component, it is possible that the cross section of nuclear reactions involving small-Z nuclei would have much larger cross section than those involving high-Z nuclei.

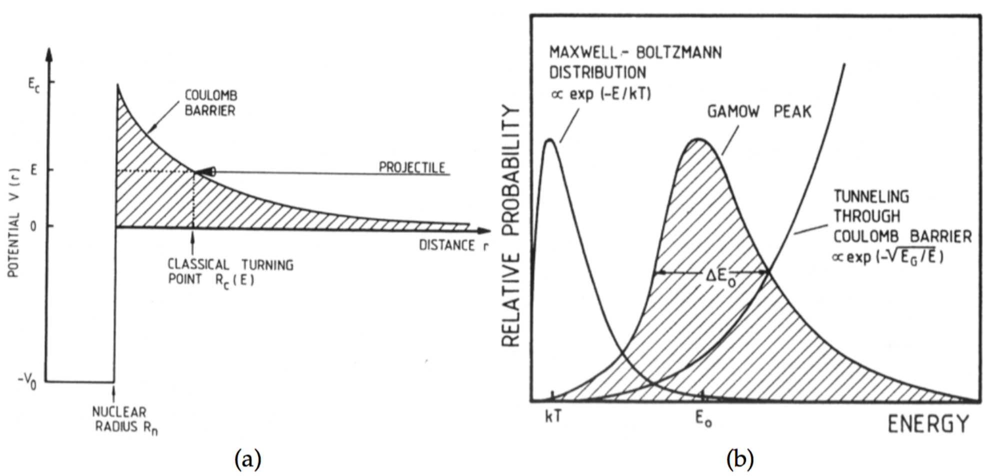


Figure 1.4. The combination of the high energy tail of the Maxwell-Boltzmann distribution and the penetrability function through the Coulomb potential gives rise to the Gamow peak with its maximum probability at the effective burning energy E_0, which is much higher than the thermal energy kT. Figure is take from [89]

In addition to the tunneling effect, the cross section is also dependent on nuclear effects. The cross section can be factorized as Eq. 1.6,

$$\sigma(E) = \frac{exp(-2\pi\eta)}{E} S(E) \tag{1.6}$$

where S(E) is referred to as the astrophysical S-factor. Its dependence on energy is much smoother for non-resonant reactions than cross section, especially in the low energy region.

Using both Eq. 1.1 and Eq. 1.4, the reaction rate can be rewritten as Eq.1.7,

$$< \sigma v >= (\frac{8}{\pi\mu})^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty S(E) exp(-\frac{E}{kT} - \frac{b}{E^{1/2}}) dE \tag{1.7}$$

where b represents the barrier penetrability $b = (2\mu)^{1/2}\pi e^2 Z_1 Z_2/\hbar$. For non-resonant reactions, the S-factor is often treated as a constant over the narrow energy region. Therefore, the reaction rate is dominated by the exponential term in this integral, which is the product of the Maxwell-Boltzmann distribution and the tunneling probability. Mathematically, one can show that this integral leads to a peak with its maximum located at energy E_0. This peak is known as the Gamow peak (Fig.1.4), and E_0 as the effective burning energy, which can be calculated by Eq. 1.8,

$$E_0 = (\frac{bkT}{2})^{2/3} = 1.22(Z_1^2 Z_2^2 \mu T_6^2)^{1/3} keV \tag{1.8}$$

where T_6 is temperature in 10^6 Kelvin. Generally, it is difficult to measure the cross section at such low effective burning energy E_0 (see Table. 1.2). In such case, phenomenological approaches like R-matrix is utilized to extrapolate the higher energy experimental data to the energy E_0. The $1/e$ width of Gamow peak is referred to as

TABLE 1.2

EFFECTIVE BURNING ENERGIES FOR $^{10,11}B(\alpha,n)^{13,14}N$ REACTIONS.

Reaction	T_9 (K)	E_0 (keV)
$^{10}B(\alpha,n)^{13}N$	0.5	506.19
	1	803.53
	1.5	1052.93
$^{11}B(\alpha,n)^{14}N$	0.5	510.65
	1	810.61
	1.5	1062.20

Notes: List of effective burning energies for $^{10}B(\alpha,n)^{13}N$ and $^{11}B(\alpha,n)^{14}N$ reactions. T_9 is temperature in 10^9 K. schema adapted from Ref. [57].

Gamow window Δ, where stellar burning is most likely to happen.

$$\Delta = \frac{4}{3^{1/2}}(E_0kT)^{1/2} = 0.749(Z_1^2Z_2^2\mu T_6^5)^{1/6}keV \tag{1.9}$$

1.3.2 Resonant Reaction

The nuclear reactions of interest in this work are resonant reactions. By definition, resonant reactions can only happen when the sum of the energy of the projectile and the Q-value equals to the excited states energy in the compound nuclei. When beam energy is “on resonance”, the cross section is at local maximum.

For a single-level narrow resonance, the cross section can be described by the

Breit-Wigner formula [89],

$$\sigma(E) = \pi\lambda\frac{2J+1}{(2J_1+1)(2J_2+1)}(1+\delta_{12})\frac{\Gamma_a\Gamma_b}{(E-E_R)^2+(\Gamma/2)^2} \tag{1.10}$$

where λ is the de Broglie wavelength, J is the angular momentum of the excited state in the compound nucleus, $J_1(J_2)$ is the spin of the projectile(target) nucleus. Γ_i is the partial width of the compound state which has the total width of $\Gamma = \sum \Gamma_i$. The total width is related to the mean life time t_0 of a compound state through

$$\Gamma t_0 = \hbar \tag{1.11}$$

Substituting cross section term in Eq. 1.4 with the Breit-Wigner formula gives

$$<\sigma v>= (\frac{8}{\pi\mu})^{1/2}\frac{1}{(kT)^{3/2}}\int_0^\infty \sigma_{BW}(E)Eexp(-\frac{E_R}{kT})dE \tag{1.12}$$

For a narrow resonance ($\Gamma \ll E_R$), the term $Eexp(-E_R/kT)$ is nearly constant in the resonance region. Thus, the calculation of reaction rate can be simplified to:

$$\begin{aligned} <\sigma v> &= (\frac{8}{\pi\mu})^{1/2}\frac{1}{(kT)^{3/2}}E_R exp(-\frac{E_R}{kT})\int_0^\infty \sigma_{BW}(E)dE, && (1.13)\\ &= [(\frac{8}{\pi\mu})^{1/2}\frac{1}{(kT)^{3/2}}E_R exp(-\frac{E_R}{kT})][2\pi^2\lambda_R^2\omega\Gamma_a\Gamma_b\int_0^\infty \frac{1}{(E-E_R)^2+(\Gamma/2)^2}dE], \\ &= [(\frac{8}{\pi\mu})^{1/2}\frac{1}{(kT)^{3/2}}E_R exp(-\frac{E_R}{kT})][2\pi^2\lambda_R^2\omega\frac{\Gamma_a\Gamma_b}{\Gamma}], \\ &= (\frac{2\pi}{\mu kT})^{3/2}\hbar^2(\omega\gamma)_R exp(-\frac{E_R}{kT}), && (1.14) \end{aligned}$$

where w is the statistical factor,

$$\omega = \frac{2J+1}{(2J_1+1)(2J_2+1)}(1+\delta_{12}) \tag{1.15}$$

and γ is the width ratio,

$$\gamma = \frac{\Gamma_a \Gamma_b}{\Gamma} \tag{1.16}$$

1.3.3 R-matrix Theory

As mention in Sec. 1.3.1, the cross sections in the typical energy range of stellar burning are usually not measurable in most laboratory conditions. Thus, the determination of reaction rates at low energies depends on reliable extrapolation of the higher energy cross section data. Experimental cross section data often shows complicated structures consisting of several resonances and interference effect between resonances with same spin parity. This complex structure can not be reproduced by the Breit-Wigner formula. R-matrix theory has been developed to address such problems [69].

The fundamental assumption of R-matrix theory is that the configuration space can be divided into internal region with nuclear interactions and external region with Coulomb interactions. The two regions is separated by a boundary surface(S), at which the wave functions of two regions and their derivatives must match. R-matrix is a powerful tool to describe the relationship between complicated energy dependency of cross sections of possible reaction channels and wave functions through collision matrix, as well as the relationship between collision matrix and resonance parameters through the R-matrix. The theory will be briefly discussed below, for more extensive reviews one can refer to Azuma et al. [9], Lane and Thomas [70] and Vogt [102].

As only Coulomb interaction is considered in the external region, an analytical solution determined from Coulomb wave functions (F_c and G_c) exists. The solution can be written in terms of incoming and outgoing waves (I_c and O_c),

$$\phi_c = (\frac{1}{v_c})^{1/2}(y_c Ic - \sum_{c'} U_{cc'} y_{c'} O_c) \tag{1.17}$$

where $c = \{\alpha l s\}$ is an unique reaction channel identified by particle pair α, relative angular momentum l and channel spin s. v_c is the relative channel velocity. $\mathbf{U}$ represents the collision matrix, and the term $\sum_{c'} U_{cc'} y_{c'}$ is the amplitude of the outgoing wave.

In the internal region, the total wave function is the summation of the product of the radial wave function for each channel ϕ_c and the wave function appropriate for the corresponding quantum numbers ψ_c. It can also be expanded into a complete set of eigenstates with eigenvalues of resonance energy (E_λ).

$$\Psi(r) = \sum_\lambda C_\lambda X_\lambda(r) \tag{1.18}$$

Therefore, the total wave function satisfy the Schördinger equation and boundary condition,

$$-(\frac{\hbar^2}{2m_c})\frac{d^2 X_\lambda}{dr_c^2} + V X_\lambda = E_\lambda X_\lambda \tag{1.19}$$

$$r_c \frac{dX_\lambda}{dr_c} = B_c X_\lambda | S_c \tag{1.20}$$

where m_c is the reduced mass, and B_c is the boundary condition number appropriate to channel c. Solving for C_λ gives,

$$C_\lambda = (E - E_\lambda)^{-1} \sum_c \gamma_{\lambda c}(\phi_c' - B_c \phi_c)(\frac{\hbar^2}{2m_c r_c})^{1/2} \tag{1.21}$$

where $\gamma_{\lambda c}$ is the reduced width defined by,

$$\gamma_{\lambda c} = (\frac{\hbar^2}{2m_c r_c})^{1/2} \int \psi_c^* X_\lambda dS \tag{1.22}$$

and R-matrix is defined as,

$$R_{cc'} = \sum_{\lambda} \frac{\gamma_{\lambda c}\gamma_{\lambda c'}}{E - E_{\lambda}}. \tag{1.23}$$

Then wave function takes the following form,

$$\phi_c = (\frac{m_c a_c}{\hbar^2})^{1/2} \sum_{c'} R_{cc'} (\frac{\hbar^2}{m_{c'} a_{c'}})^{1/2} [\rho_{c'} \phi'_{c'} - B_{c'} \phi'_{c'}] \tag{1.24}$$

where $\rho = k_{\alpha} a_c$, a_c is the channel radius, k_{α} is the wave number, and ϕ' is the derivative of ϕ with respect to kr.

Matching the value and first derivative of the internal and external solutions at the boundary surface gives the collision matrix,

$$\mathbf{U} = \rho^{1/2} \mathbf{O}^{-1} (1 - \mathbf{R}\mathbf{L}_0)^{-1} (1 - \mathbf{R}\mathbf{L}_0^*) \mathbf{I} \rho^{1/2} \tag{1.25}$$

where $\mathbf{L}_0 = \rho \mathbf{O}' \mathbf{O}^{-1} - \mathbf{B}$. The transition matrix $\mathbf{T}$ can be calculated with

$$T_{cc'} = e^{iw_c} \delta_{cc'} - U_{cc'}, \tag{1.26}$$

where w_c stands for Coulomb phase shift. This yields the final expression for angle integrated cross section for a $\alpha \to \alpha'$ reaction,

$$\sigma_{\alpha\alpha'} = \frac{\pi}{k_{\alpha}} \sum_{Jll'ss'} g_J |T^J_{cc'}|^2 \tag{1.27}$$

where g_J is the statistical factor shown in Eq. 1.15.

This work utilizes the code `AZURE2` [10, 100], a multi-channel, multi-level R-matrix code, to perform R-matrix calculations. One can achive good agreement between experimental cross section data and $\boldsymbol{R}$-matrix fit by iteratively adjusting reduced width and level energy. Including data from different reaction channels helps to better constrain the fitting parameters.

1.4 Previous Measurements of the $^{10,11}B(\alpha,n)^{13,14}N$ Reactions

1.4.1 The $^{10}B(\alpha,n)^{13}N$ Reaction

The ^{10}B$(\alpha,n)^{13}$N reaction was one of the first nuclear reactions studied that itself produces a radioactive product [29] and α-induced reactions on boron were used to make early measurements of the mass of the neutron [24, 28]. Early measurements of the cross section were hindered as only radioactive α-emitters were available to induce the reaction. The first accelerator based measurement was by Bonner et al. [20], but they were unable to clearly disentangle contributions from the ^{10}B$(\alpha,n)^{13}$N and ^{11}B$(\alpha,n)^{14}$N reactions as the detection system was only sensitive to the number of emitted neutrons. This is a significant issue because the ^{11}B$(\alpha,n)^{14}$N reaction is approximately an order of magnitude larger than that of the ^{10}B$(\alpha,n)^{13}$N cross section over the energy range of interest [103]. Further, the moderated proportional counter used for that study could not distinguish between the different neutron channels, making the conversion of their reaction yields into cross sections impossible. A later study by Gibbons and Macklin [46] was able to more clearly obtain yields for the ^{10}B$(\alpha,n)^{13}$N reaction, but still used a moderating counter-type detector (carbon sphere) that could not separate the different neutron channels. Van der Zwan and Geiger [101] then used stilbene scintillators to investigate the reaction. These detectors could separate the different neutron groups using spectrum unfolding, but measurements were for the most part limited to three angles ($\theta_{\text{lab}} = 0$, 90, and 160°). Because of the complicated angular distributions of the reaction products, total cross sections could only be calculated at a few energies where additional measurements were made. More recently, Prior et al. [84] have again made measurements using a proportional counter setup. With the aid of detector modeling using `MCNP`, and the approximate branching ratio information of Van der Zwan and Geiger [101], a better determination of the total cross section could be obtained between $E_\alpha = 2.2$ and 4.9 MeV.

In this work, an experimental technique similar to that of Van der Zwan and Geiger [101] is utilized, except that detailed angular distributions are extended to the entire energy range of interest allowing for a determination of the angle integrated cross section of the ^{10}B$(\alpha, n_0)^{13}$N reaction from $1.5 < E_\alpha < 5.0$ MeV. The present measurements utilize deuterated liquid scintillators. Like the stilbene detectors of Van der Zwan and Geiger [101], they are sensitive to the neutron energy by analyzing the recoil energy given by the scintillation light spectrum. Deuterium has been found to be preferable over hydrogen based liquid scintillators because the light response spectrum presents a peaking at the highest recoil energy improving the condition of the detector response used in spectrum unfolding [39, 71]. Using this method, the ^{10}B$(\alpha, n_0)^{13}$N cross section can be measured.

1.4.2 The $^{11}B(\alpha, n)^{14}N$ Reaction

The $^{11}B(\alpha, n)^{14}N$ reaction was populated over a low energy range that corresponds to an excitation energy range of $10.7 < E_x < 12.9$ MeV in the ^{15}N compound system. This energy range has been of interest for study because of the similar proton ($S_p = 10.207$ MeV), neutron ($S_n = 10.833$ MeV), and α-particle ($S_\alpha = 10.991$ MeV) separation energies in ^{15}N [106]. The energy scheme of ^{15}N used in this analysis is shown in Fig. 1.5. For the ^{11}B$(\alpha, n)^{14}$N reaction, there have been only a few studies [80, 113, 107], where that of Wang et al. [107] was the most comprehensive. Studies of the ^{11}B$(\alpha, p)^{14}$C reaction have been made by Refs. [31, 99, 107] and the ^{11}B$(\alpha, \alpha)^{11}$B reaction by Refs. [74, 72]. Studies of ^{14}C+p reactions include measurements of ^{14}C$(p, p)^{14}$C [50, 53], ^{14}C$(p, n)^{14}$N [80, 111, 90, 93, 46, 107], and ^{14}C$(p, \gamma)^{15}$N [14, 111, 47, 53]. Finally, studies of neutron induced reactions on ^{14}N

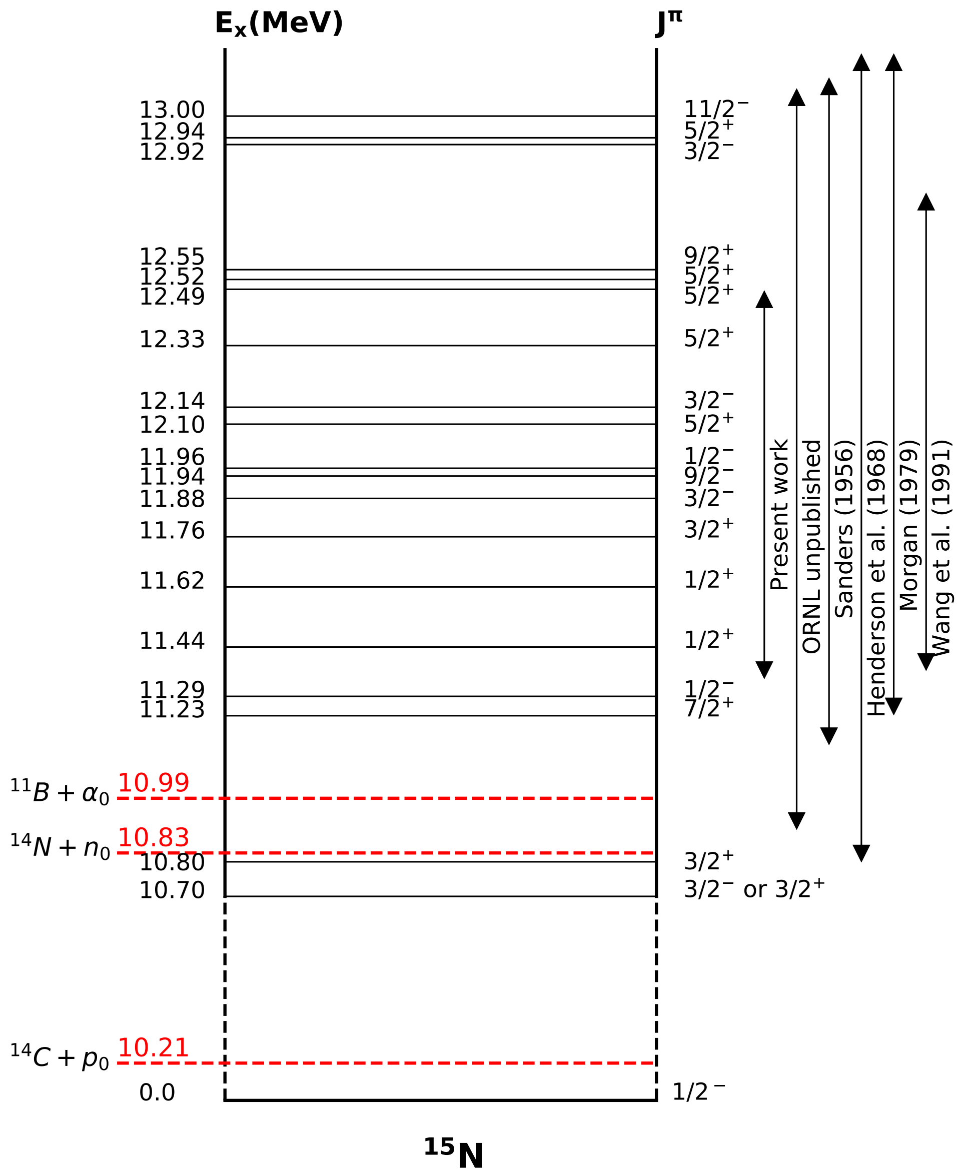


Figure 1.5. Energy structure of ^{15}N. The double-ended arrows indicate the energy range covered by previous works. The energy levels and spin-parities are taken from [5].

include ^{14}N$(n,n)^{14}$N [42], ^{14}N$(n,p)^{14}$C [44, 46, 59, 78] (fast neutrons), ^{14}N$(n,\alpha)^{11}$B [44, 59, 78], and ^{14}N(n,total) [54, 60, 62]. There have also been several studies of the thermal ^{14}N$(n,p)^{14}$C cross section, but for the analysis presented in this work only the comprehensive measurement of Koehler and O'Brien [66] is considered. In addition, a highly accurate and comprehensive measurement of the ^{14}N(n,total) reaction has also been made at Oak Ridge National Laboratory, but remains unpublished. Some of the data from this measurement are considered in the analyses of Jurney et al. [62] and Hale et al. [48].

CHAPTER 2

EXPERIMENTAL SETUP

The reactions ^{10,11}B(α,n)^{13,14}N were studied using three different experimental setups. The measurement of the ^{11}B(α,n)^{14}N reaction was done using a ^{3}He counter neutron detector. The detection system for the low energy measurement of the ^{10}B(α,n$_0$)^{13}N reaction consisted of two deuterated liquid scintillators, one of type EJ-301D(size: 7.6 cm diameter $\times$ 7.6 cm thickness), one of type EJ-315(size: 7.6 cm diameter $\times$ 5.8 cm thickness), and a high-purity germanium detector(HPGe) with a relative efficiency of 54%. The high energy measurement of the ^{10}B(α,n$_0$)^{13}N reaction was done using a similar setup to the low energy measurement, but with three newly developed deuterated liquid scintillators of type EJ-315-M(size: 7.6 cm diameter $\times$ 5.8 cm thickness). Different detection systems were used in the measurements of ^{10}B$(\alpha,n)^{13}$N and ^{11}B$(\alpha,n)^{14}$N reactions due to the very different cross sections and the enrichment constrains of the targets. All experiments were performed between January 2016 and August 2018 at the Nuclear Science Laboratory (NSL) of the University of Notre Dame.

2.1 Accelerator

The α-beam used in our experiments was produced by St.ANA (Stable ion Accelerator for Nuclear Astrophysics) at the NSL. The accelerator is commonly known as the $5U$ for its five individual acceleration sections. The schematic layout of $5U$ is shown in Fig. 2.1.

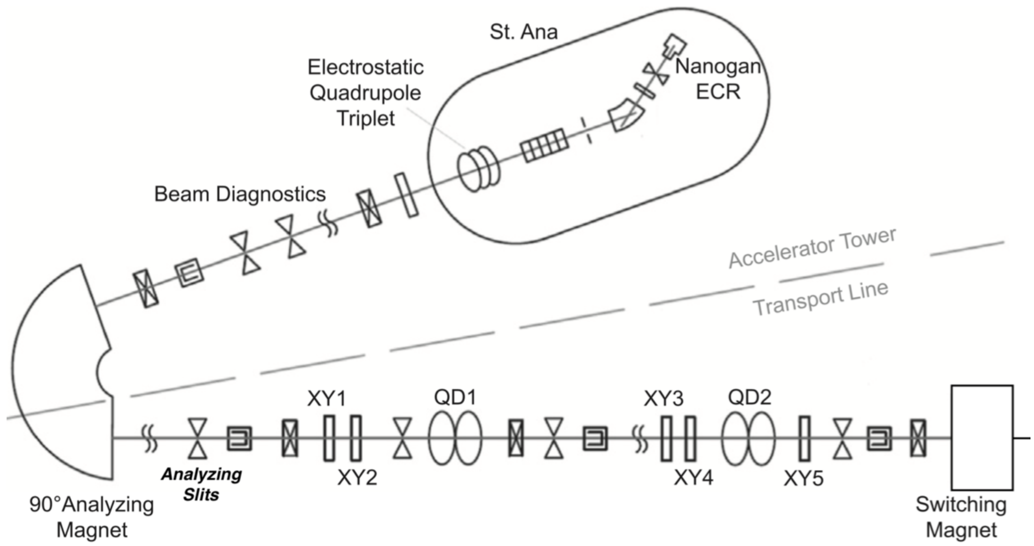


Figure 2.1. Basic layout of the $5U$ accelerator and its transport line, where XY are steering elements, QD are magnetic quadrupoles. Figure taken from [75].

High intensity ions are created by the ECR(Electron cyclotron resonance) ion source. The ECR ion source operates under a constant magnetic field, in which electrons orbit at a fixed frequency. When this frequency matches with the frequency of the injected Microwave, the Electron cyclotron resonance phenomenon happens. Therefore, it leads to the kinetic energy increase of all electrons. These energized electrons will eventually collide with the gas molecules, and ionize them if the kinetic energy of the electrons is larger than the ionization energy of the gas molecules. This process produces positively charged ions. These ions are then pulled out into the acceleration tube by a negative field generated by the extractor. With a large voltage difference between the terminal shell and the bottom of the accelerator, the acceleration tube in between makes use of a series of resistors and plates to create an electric field gradient to transport the ions along the tube.

The ions accelerated to experimental required energies will then go through the analyzing magnet with a 90° change of direction. According to basic physics, the force experienced by a charged particle in a magnetic field is given by Eq. 2.1

$$F = qvB = \frac{mv^2}{r}, \tag{2.1}$$

where q and v are the charge and velocity of the charged particle, and B is the magnetic field. With simple transformation, one can get

$$B = \frac{\sqrt{2mE}}{qr} = k\sqrt{E}, \tag{2.2}$$

where E is the kinetic energy of the charged particle, and $k = \sqrt{2m}/qr$ is unique for every charged particle. During the high energy measurement of the ^{10}B(α,n$_0$)^{13}N reaction, the constant k was calibrated with the well-known resonances at $E = 1054$ keV, 1336 keV in ^{13}C(α,n)^{16}O (see Sec 4.5.1).

2.2 Germanium Detector

γ-rays can interact with detector material in the following three way: photoelectric absorption, Compton scattering, and pair production. In the photoelectric absorption process, photons transfer their most of the energy to the bound shell electrons of the absorber atom. These electrons leave the atom and lose their energy to the absorbing material. The amount of energy transferred from photons to electrons, E_{e^-}, is given by Eq. 2.3

$$E_{e^-} = h\nu - E_b \tag{2.3}$$

where E_b is the binding energy of the electron in its bound shell, and ν is the frequency of the incident photon. The probability of this process has an empirical expression,

$$\tau \approx constant \times \frac{Z^n}{E_\gamma^{3.5}} \tag{2.4}$$

where n varies between 4 and 5 [65]. After the photoelectron leaves the atom, electron rearrangement happens inside the atom and liberate the binding energy. As a result, the full energy of the incident γ-ray is deposited, and it appears as a full-energy peak on the pulse height spectrum.

Compton scattering is the elastic and inelastic scattering between photons and electrons. The energy deposited to the electrons is dependent on the scattering angle, and can have an energy ranging from zero to $\frac{E_\gamma}{1+2\frac{E_\gamma}{m_0c^2}}$. The γ-rays can either scatter off the detector and escape, which result in a continuum spectrum (Compton continuum), or experience secondary scattering, which result in a full energy photopeak.

In pair production, a photon can decay into an electron-positron pair, if its energy is larger than two times the rest mass of the electron (1.022MeV). The total kinetic energy of the electron and positron is described by,

$$E_{e^-} + E_{e^+} = hv - 2m_0c^2 \tag{2.5}$$

where E_{e^-} and E_{e^+} are the kinetic energy of the electron and positron, hv is the energy of the incident γ-ray, and m_0c^2 is the rest mass of an electron. Both electrons and positrons can quickly deposit their energies in the detector. However, the positron can also annihilate with another electron and produces two secondary γ-rays with $E_\gamma = 511$ keV, when its energy is very low. When the detector size is large, both secondary γ-rays will be absorbed by the detector, which results in a full energy peak. When the detector size is intermediate, only one of the secondary γ-rays can escape and the other is absorbed by the detector. It results in a single escape peak at hv-511 keV. Finally, if the detector is small enough, both secondary γ-rays escape and 1022 keV will be lost from the detector. This gives rise to a double escape peak at hv-1022 keV on the pulse height spectrum. The cross section of pair production is proportional to Z^2 [2].

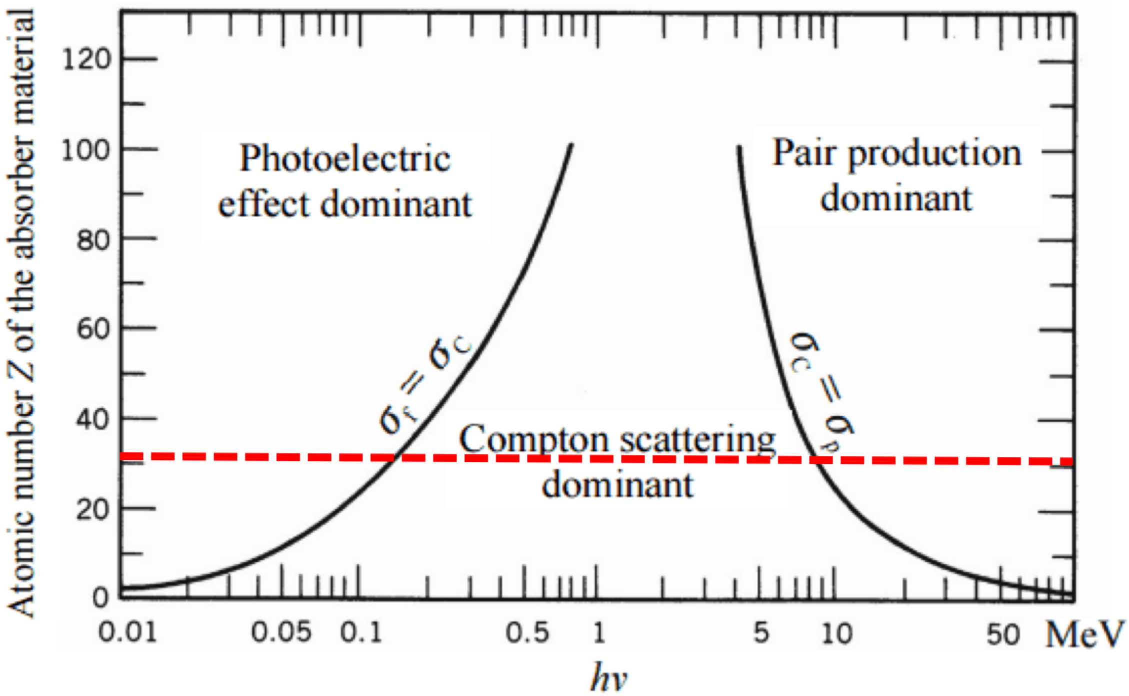


Figure 2.2. The dominant region of three types of γ-ray interactions in terms of atomic number Z and the γ-ray energy $h\nu$. The red dotted line, where Z=32, shows the different energy regions in germanium. Photoelectric absorption is the dominant γ-ray interaction for E_γ below ~0.1 MeV. Compton scattering is the dominant interaction mechanism in the E_γ energy range from 0.1 MeV to 10 MeV. Pair production dominants for E_γ above 10 MeV. This plot is adapted from *The Atomic Nucleus* by R.D.Evans. Copyright 1955 by the McGraw-Hill Book Company.

In this work, the detector used to detect γ-rays was a high-purity germanium (HPGe) detector. Fig. 2.2 demonstrates the relative importance of the three γ-ray interactions for different combinations of absorbing material and γ-ray energy.

HPGe is a type of semiconductor detector. It works by converting photons to electron-hole pairs. Under an electric field, the electrons and hole can be collected at corresponding electrodes, hence a pulse is generated. The number of electron-hole pairs is proportional to the energy deposited by the incident radiation. HPGe is maintained at liquid nitrogen temperature to reduce electronic noise from thermal excitation of electrons.

2.2.1 Detection Efficiency

The efficiency of the HPGe detector was determined with standard sources ^{60}Co, ^{137}Cs as well as the ^{27}Al$(p,\gamma)^{28}$Si reaction at the well known narrow resonance at E_p = 992 keV [6]. The detector was at a distance where geometric effects and summing could be neglected compared to the precision of the measurements.

The full energy peak efficiency of a γ-ray detector is defined as Eq. 2.6,

$$\epsilon_p(E) = \frac{N_p(E)}{A \cdot I_\gamma(E)} \tag{2.6}$$

where $N_p(E)$ is net area of the full energy peak, A is the radioactive activity, and $I_\gamma(E)$ represents the branching ratio. By placing a radioactive source with known activity at the target position, the HPGe detector can measure the number of emitted γ-rays. Hence the full energy peak efficiency can be calculated. However, the energies of γ-rays emitted from most commonly used sources are limited to low energies (below $E_\gamma \approx 3$ MeV). With the aid of the ^{27}Al$(p,\gamma)^{28}$Si reaction, it is convenient to map out the efficiency curve in the energy region from 0.5 MeV to 10 MeV, in which the branching ratio of γ-rays are well determined [8, 95, 76, 6]. Another important benefit of utilizing this reaction is that the yield of 1.78 MeV γ-ray can be easily converted to an absolute efficiency [6] using Eq. 2.7,

$$\epsilon_p(E = 1.78\ MeV) = \frac{N_p(E = 1.78\ MeV)}{C \cdot N_{incident}} \tag{2.7}$$

where N$_{incident}$ is the number of incident proton particles, and C = $1.08(6)\times10^{-9}$ is a measured constant. Table. 2.1 listed the energy and branching ratio of the γ-rays used to calibrate the absolute efficiency.

The efficiency curve shown in Fig. 2.3 is described by a fitting function as follows,

TABLE 2.1

ABSOLUTE EFFICIENCY OF GERMANIUM DETECTOR.

E_γ (keV)	I_γ (%)	ϵ(E) (%)
661.57	85.1(2)	$1.19(4)\times10^{-1}$
1522.3	2.8(2)	$3.55(49)\times10^{-2}$
1778.9	94.8(15)	$2.68(15)\times10^{-2}$
2838.9	5.5(4)	$1.88(20)\times10^{-2}$
3063.3	1.15(11)	$1.53(31)\times10^{-2}$
3123.7	0.70(7)	$1.97(44)\times10^{-2}$
4497.6	4.8(3)	$1.13(14)\times10^{-2}$
4608.4	4.5(4)	$8.71(123)\times10^{-3}$
4743	8.8(5)	$1.01(10)\times10^{-2}$
6019.9	6.0(5)	$6.85(92)\times10^{-3}$
6265.3	2.10(2)	$7.81(123)\times10^{-3}$
7924	4.3(4)	$5.11(86)\times10^{-3}$
7933.4	3.7(4)	$3.95(83)\times10^{-3}$
10762.9	76.6(15)	$2.50(17)\times10^{-3}$

Notes: Relative intensities of γ-rays and absolute efficiency from the ^{137}Cs source and ^{27}Al$(p,\gamma)^{28}$Si reaction at $E_p = 992$ keV. The relative intensities from the ^{27}Al$(p,\gamma)^{28}$Si reaction were taken from [6].

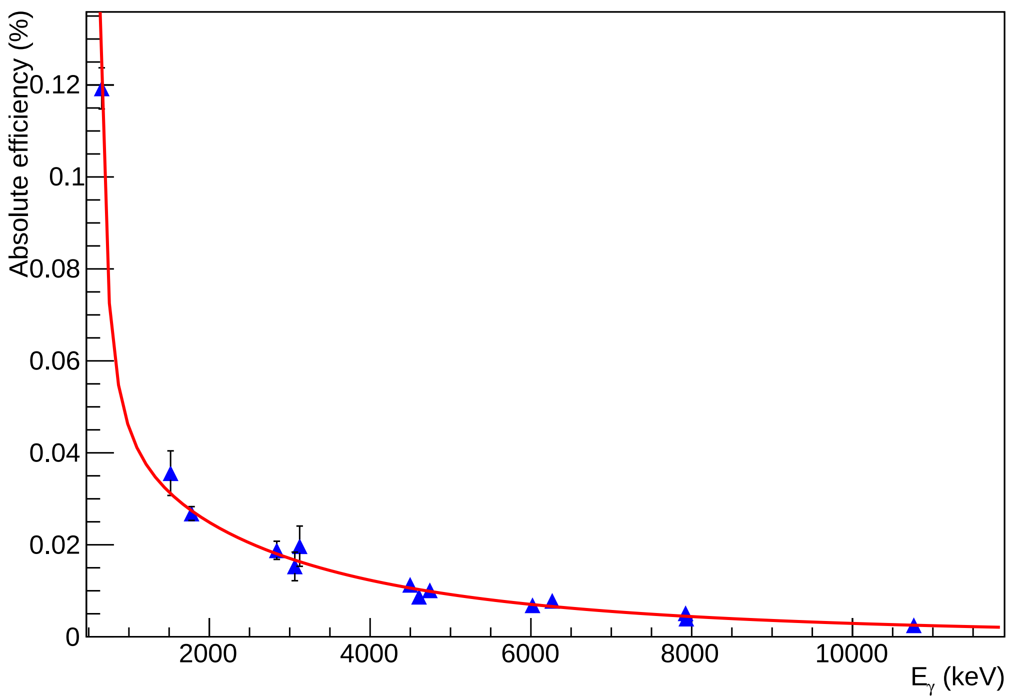


Figure 2.3. Full energy peak efficiency curve for HPGe at distance of d=215 mm and its fit function. The blue triangles represent the data points measured. The red line is the fit function best described the data.

$$\epsilon_{FEP}(d,E) = \frac{1 - exp(\frac{d+d_0}{a_0+b_0\cdot\sqrt{E}})}{(d+d_0)^2} \cdot exp(a + b\cdot \ln E + c\cdot (\ln E)^2) \tag{2.8}$$

where d is the distance between target and detector in unit of mm, E is the γ-ray energy in unit of keV, and a, b, c, a_0, b_0, c_0 are free parameters obtained from the fitting procedure. These parameters take the values as follows:

a = -34.93 ± 0.27	a_0 = 47.71 (fixed)
b = 7.36 ± 0.04	b_0 = -2.00 ± 0.01
c = -0.47 ± 0.01	d_0 = -214.78 ± 0.06.

2.3 Neutron Detectors

Neutrons can not be detected directly, since they have no charge. Instead of detecting neutrons, neutron detectors convert neutrons to detectable particles via neutron induced reactions. There are two main types of neutron detectors, proportional gas counters and scintillator detectors. In this thesis work, two types of neutron detectors are used to detect fast neutrons (>0.5 eV), a ^{3}He counter, and deuterated liquid scintillators.

2.3.1 ^{3}He Counter

The ^{3}He proportional counter used in this work was made up of 20 ^{3}He tubes forming two concentric rings. The ^{3}He tubes were embedded in a block of polyethylene moderator. A single ^{3}He counter consisted of an aluminum tube filled with high pressured ^{3}He gas. An anode wire operates at HV = 1100 V and was mounted at the center of the aluminum cylinder. As incident neutrons passed through the polyethylene moderator, and were converted to thermal neutrons, losing energy information via scattering events. The slow neutron induced $^3\text{He}(n,p)t$ ($Q = 764$ keV) reaction results in protons with energy $E_p = 573$ keV and tritons with energy $E_t = 191$ keV. Due to the high cross section for $^3\text{He}(n,p)t$ reaction for thermal neutrons ($\sigma = 5530$ barn), the ^{3}He detector can reach high detection efficiency, as shown in Fig. 2.4.

In an ideal case, a total energy of 764 keV is deposited in the ^{3}He counter, if the reaction products p and t are completely stopped in the active volume of the detector. In practice, while one nuclei is fully absorbed in the gas, the other one might hit the wall of the detector and only deposit a fraction of its energy, due to the finite size of the tube. The energy deposited in the gas can range from (E_t+0) to (E_t+E_p). This phenomenon is known as wall effect [65]. Fig. 2.5 (a) illustrates the wall effect in an idealized ^{3}He counter spectrum, which features two plateaus in the continuum. The measurement from a ^{3}He counter used in this work is shown in Fig. 2.5 (b), where

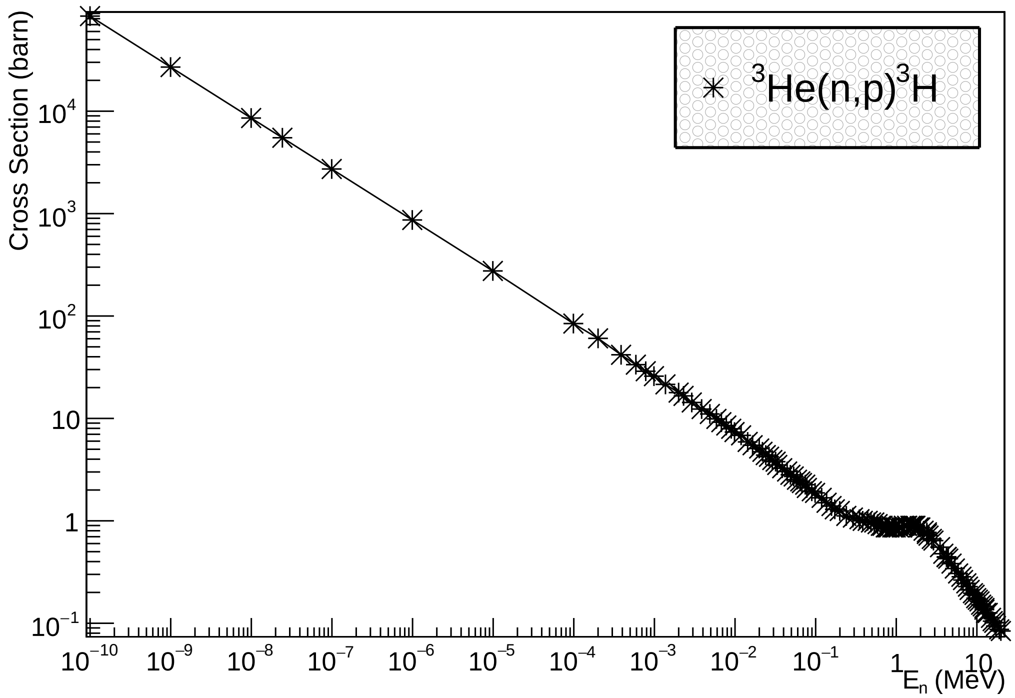


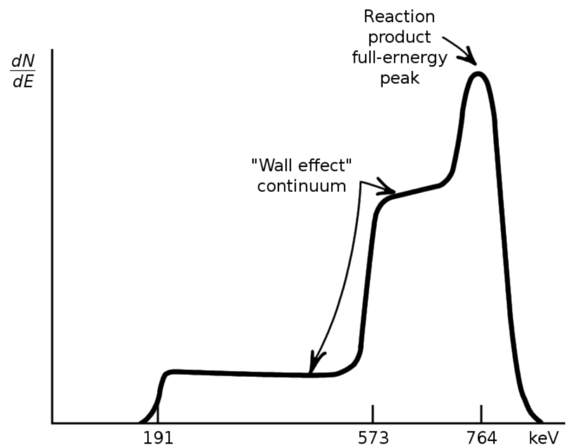
Figure 2.4. Cross section of ^{3}He(n,p)^{3}H from the ENDF compilation [3].

the region in red is a result of incident neutron events.

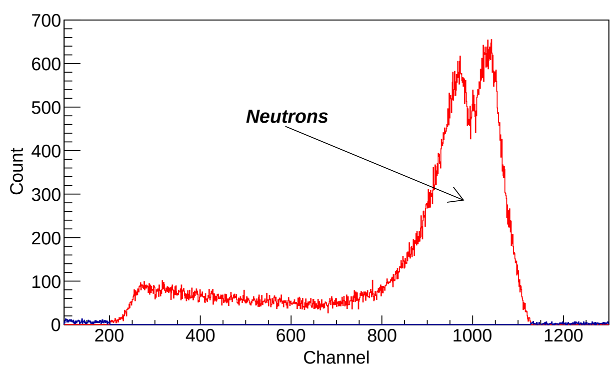
2.3.1.1 Detection Efficiency

The efficiency of the ^{3}He counter was determined with the activation reaction ^{51}V(p,n)^{51}Cr, which produces approximately mono-energetic neutrons. 9.910(10)% of the radioactive product ^{51}Cr decays to the first excited state of ^{51}V through electron capture with a half-life of $T_{\frac{1}{2}} = 27.7$ d [104], followed by the emission of a γ-ray with an energy of $E_\gamma = 320$ keV.

The neutron detection efficiency is determined as the ratio of the number of neutrons detected in the ^{3}He counter (N_d) to the number of neutrons produced during



(a)



(b)

Figure 2.5. Idealized (a) and measured (b) spectrum of a ^{3}He counter. Fig. (a) is taken from Knoll [65]. Fig. (a) features a full energy peak and two plateaus to its left corresponding to the partial energy deposition due to wall effect. Fig. (b) is a spectrum from the ^{3}He counter used in this work. It is not energy calibrated, and the few counts in lower channels is likely caused by gamma-ray-induced reactions.

the activation run (N_{tot}):

$$\epsilon = \frac{N_d}{N_{tot}}. \tag{2.9}$$

The total number of neutrons produced during the activation is equal to the number of ^{51}Cr nuclei, which can be determined by measuring the activity of its decay product ^{51}V in the counting station.

After an activation run from $t = 0$ to t_0, the number of ^{51}Cr nuclei (N_{Cr}) that decayed in time interval Δt are:

$$N_{Cr}(\Delta t) = \frac{N_{tot}}{t_0}(1 - e^{-\lambda t_0}) \int_t^{t+\Delta t} e^{-\lambda(t-t_0)} dt \tag{2.10}$$

where λ is the decay constant. The number of γ-rays emitted from ^{51}Cr (Nγ) from t_1 to t_2 are,

$$N_{Cr}(t_1) - N_{Cr}(t_2) = N_\gamma = \frac{N_c}{\epsilon_{HPGe} \cdot I} \tag{2.11}$$

where N$_c$ is the number of γ-rays recorded, ϵ_{HPGe} is the efficiency of HPGe at 320 keV, and I is the branching ratio. Hence, the total number of neutrons produced during the activation (N_{tot}) is given by,

$$N_{tot} = \frac{\lambda \cdot N_c \cdot t_0}{\epsilon_{HPGe} \cdot I \cdot (1 - e^{-\lambda t_0}) \cdot \left(e^{-\lambda(t_1-t_0)} - e^{-\lambda(t_2-t_0)}\right)}. \tag{2.12}$$

The γ-rays with E$_\gamma$ = 320 keV emitted from the ^{51}Cr nuclei were measured in a lead shielded counting station with well-calibrated efficiency of 1.157(21)% at source-detector distance of 2 cm.

The ^{3}He counter detector used in this work was carefully characterized in the earlier Ph.D. thesis work [18, 38], more details can be found there. Since this work uses the same detection system as [18], the neutron detection efficiency function from [18] was adopted. The efficiency curve, with an uncertainty of 5.4%, was parameterized by a fifth-order polynomial function, which is given by,

$$\epsilon(E) = 0.523 - 1.928 \cdot 10^{-4} \cdot E + 5.856 \cdot 10^{-8} \cdot E^2 - 5.574 \cdot 10^{-12} \cdot E^3 - 3.236 \cdot 10^{-15} \cdot E^4 \quad (2.13)$$

As a cross check, an efficiency measurement was repeated at a bombarding energy E_p = 510 keV and was found to be in good agreement with the previous result. The results of the measurement are shown in Table 2.2.

TABLE 2.2. NEUTRON DETECTION EFFICIENCY VERIFICATION

E_n[keV]	ϵ(this work)[%]	ϵ [18][%]
640	47.0±2.1	44.0±2.4

2.3.2 Deuterated Liquid Scintillator

2.3.2.1 Physics of Liquid Scintillator

An organic scintillator is a type of radiation detector that converts ionizing radiation to scintillation light. It is often coupled to an electronic light sensor, such as a photomultiplier tube (PMT), which collects the scintillation light and emits electrons via photoelectric effect.

The molecule of most organic scintillators has a special energy level structure as shown in Fig. 2.6. As charged particles pass through, the kinetic energy is transferred to the scintillator molecules and higher singlet levels are populated. These excited states will quickly de-excite to a lower lying level by radiationless transitions. As a result, excited molecules in S_{10} states are populated. Starting from the S_{10} state, photons can mainly be produced through three processes, prompt fluorescence, phosphorescence, and delayed fluorescence. Prompt fluorescence, which is the fastest among all three processes, is emitted through transitions between singlet states. Some S_{10} states can pass to triplet states T_1, and phosphorescence, with the longest time scale, is emitted from the T_1 to S_0 transition. However, some first excited triplet states can be thermally excited back to the first excited singlet state, and give rise to fluorescence light. This process is known as delayed fluorescence, and has the intermittent time scale. It is the prompt fluorescence and delayed fluorescence process that is important to neutron detection for organic scintillators.

There are two types of detectors used in this work, EJ301D and EJ315, both of them are organic liquid scintillators. This category of scintillator consists of three components, an organic scintillator(fluor), a solvent, and a wavelength shifter. EJ315 is a deuterated-benzene (C_6D_6)-based organic scintillator. EJ301D is a deuterated-xylene based (C_8D_10) organic scintillator. The two types of detectors utilize the same material, but slightly different composition.

Take EJ315 as an example, it exploits benzene as an aromatic (Ar) solvent, charged particles excite ground state Ar molecules to Ar*. Energized solvent molecules transfer energy to PPO (2,5-Diphenyloxazole), the primary fluor, which yields fluorescent light. POPOP, 1,4-bis(5-phenyloxazol-2-yl) benzene, serves as wavelength shifter, it absorbs light (360 nm) emitted by PPO and re-emits light with longer wavelength of 420 nm to better match the spectral sensitivity of a photomultipiler tube Knoll [65].

2.3.2.2 Elastic Scattering Cross Section of d+n

Deuterated liquid scintillator is based on elastic scattering of neutrons by deuterons. Part of the incident neutron energy is transferred to the deuteron (recoil nucleus) during the collision. According to the kinematics, the maximum energy transferred is

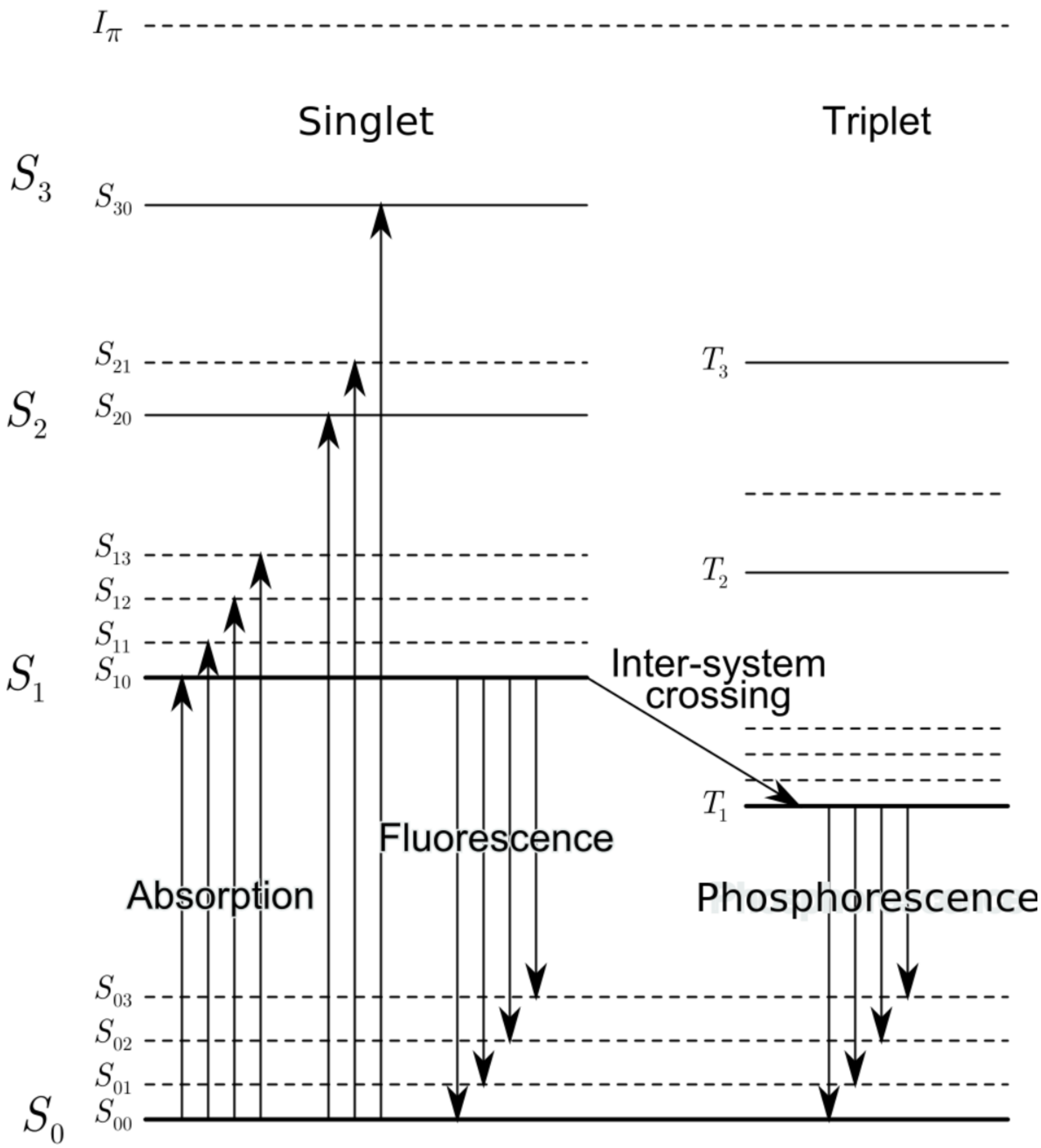


Figure 2.6. Electronic energy levels of an organic molecule. Figure taken from Knoll [65].

given by Eq. 2.14,

$$E_{R|max} = \frac{4A}{(1+A)^2} E_n, \tag{2.14}$$

where A is the ratio of the mass of the target nucleus and neutron mass, E_n is the kinetic energy of the incoming neutron in the laboratory frame, and E_R is the kinetic energy of recoil nucleus. In the case of deuterons, $E_{d|max} = 8/9 \times E_n$.

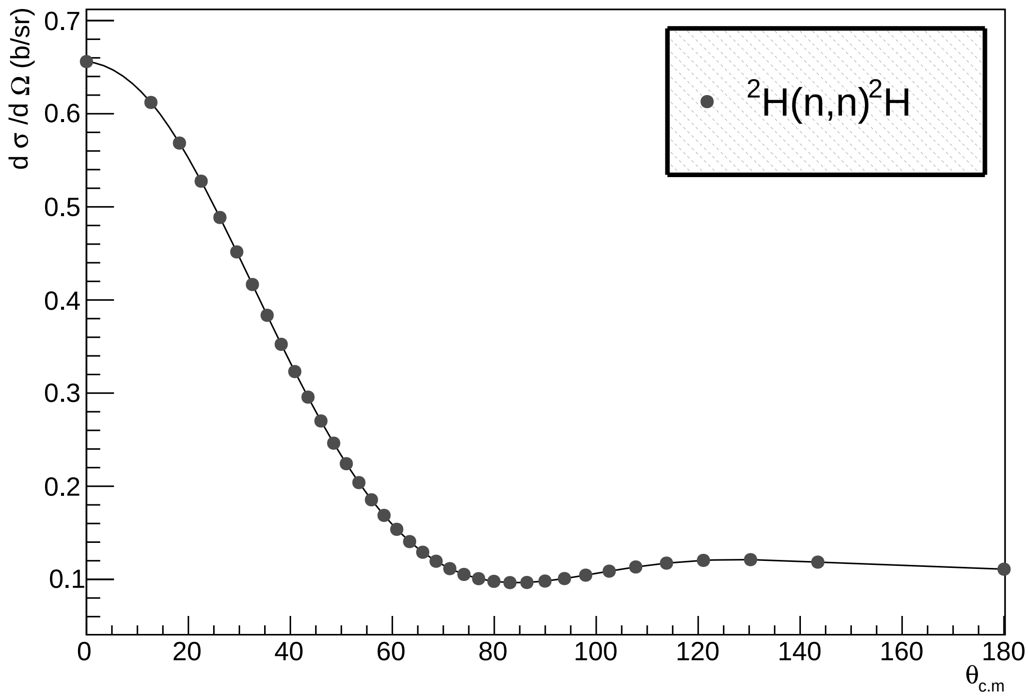


Figure 2.7. Differential cross section of the ^{2}H$(n,n)^2$H reaction from the ENDF compilation [3].

An example n+d differential cross section at $E_n = 3.0$ MeV is shown in Fig. 2.7. The asymmetric structure of the differential cross section gives rise to a response peak in the scintillator light spectrum, as shown in Fig. 2.8. In Fig. 2.8, two clear peaks corresponding to two neutron groups are visible, and the peak position is related to the incident neutron energy. With this underlying feature, deuterated liquid scintillator facilitates the ability to distinguish different neutron groups without

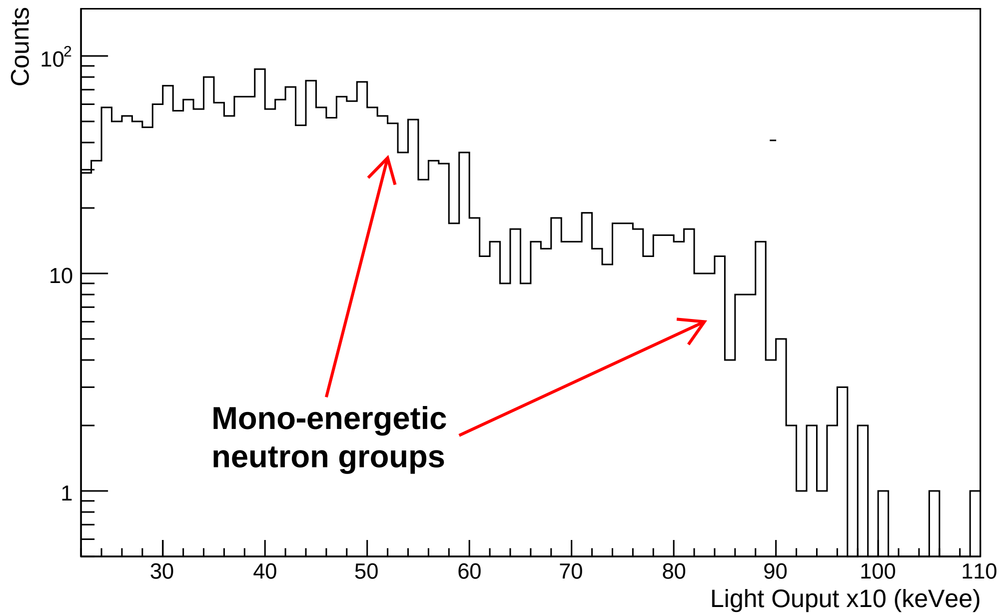


Figure 2.8. An example of light output spectrum from the deuterated liquid scintillator. 2 mono-energetic neutron groups are clearly seen.

the help of the time-of-flight technique.

2.3.2.3 Light Response

Deuterated liquid scintillator consists of low Z nuclei (deuterium and carbon). The cross sections of photoelectric effect and pair production are smaller than that of Compton scattering making it the dominant interaction mechanism for γ-rays. The energy transferred from a photon to an electron is governed by Eq. 2.15 and Eq. 2.16,

$$\frac{1}{E_f} - \frac{1}{E_i} = \frac{1}{m_e c^2}[1 - cos\theta_{cm}] \tag{2.15}$$

$$E_e = E_i - \frac{m_e c^2}{(1 - cos\theta_{cm}) + \frac{m_e c^2}{E_i}}, \tag{2.16}$$

where E_i is the incident γ-ray energy, E_f is the scattered γ-ray energy, $m_e c^2$ is the electron rest mass of 511 keV, and E_e is the energy of a recoil electron. In the light output spectrum, the only significant structure in the spectrum is the Compton edge. The light output spectrum is a pulse height spectrum in the unit of keVee or MeVee, which means the electron equivalent energy (1 MeV electron can generate 1 MeVee light yield). The Compton edge from a known gamma source measurement is useful for calibrating the energy scale of the light output spectrum.

Elastic and inelastic scattering are the primary interaction mechanisms for neutrons. The recoil charged particles lose energy (dE/dx) via ionizing nearby molecules as they travel through the material. The relation between light yield per unit path length, dL/dx, and dE/dx is given by the well-known Birks formula Knoll [65],

$$\frac{dL}{dx} = \frac{S\frac{dE}{dx}}{1 + kB\frac{dE}{dx} + C(\frac{dE}{dx})^2}, \tag{2.17}$$

where dE/dx is the energy loss per path length, S is the normal scintillation efficiency, kB is the quenching probability, and C is an additional fitting parameter needed to improve the empirical fit to experimental data. In the literature, many functional forms have been proposed to describe the light response function, including a rational of polynomials [67], power law [27] and simple quadratic. The function given by [37] is found to best capture the shape of experimental data for Z $\leq$ 6,

$$L(E_x) = aE_x + b(1 - e^{cE_x}), \tag{2.18}$$

where E_x is the initial energy of the electrons or recoil charged particles (protons, deuteron, ...), and a, b, and c are empirically fit parameters.

2.3.2.4 Pulse Shape Discrimination

A traditional way to discriminating photons and neutrons is the time-of-flight (TOF) technique. TOF can determine the particle types by their flight time. However, not many facilities can provide a pulsed beam with sufficient beam current for low cross section measurements. The organic liquid scintillators have exhibited an excellent ability to differentiate γ-rays and neutrons [82] without the help of TOF [89]. Specifically, different recoil particles generate different pulse shapes. This method is called pulse-shape discrimination (PSD). As mentioned in Sec. 2.3.2.1, the main contribution to the light yield is from prompt fluorescent (fast component) and there is a small fraction of contribution coming from the delayed fluorescent process (slow component). The light yielded by the latter process depends on the square of the density of the triplet states. As the energy deposited (dE/dx) by the recoil particles increases, the population of triplet states increases. As a result, heavier particles with more charge, like protons and deuterons, should have a more pronounced slow component than electrons. The PSD is done by comparing the ratio between the light from slow component and from the fast component, which leads to the discrimination of n/γ signals. A comparison of the pulse shape originating from different recoil particles is shown in Fig. 2.9.

The technique used for pulse shape discrimination is known as charge integration, which can be easily implemented in software system. Definitions for some parameters that are important in this method will be introduced. These terms are the long integral, short integral and PSD parameter, and a graphic explanation of them is shown in Fig. 2.10(a). The long integral is the charge pulse integration in the full signal length. The short integral is the charge integration in tail of the pulse, and it corresponds to the delayed fluorescence in scintillators. The PSD parameter is the ratio of the short integral and the long integral, which corresponds to the intensity

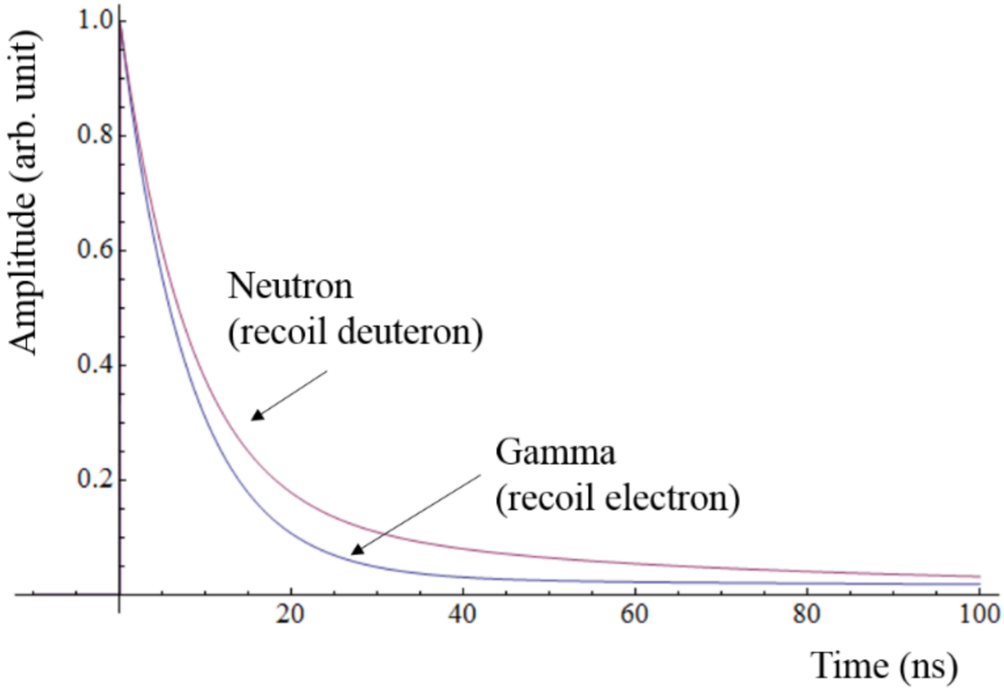
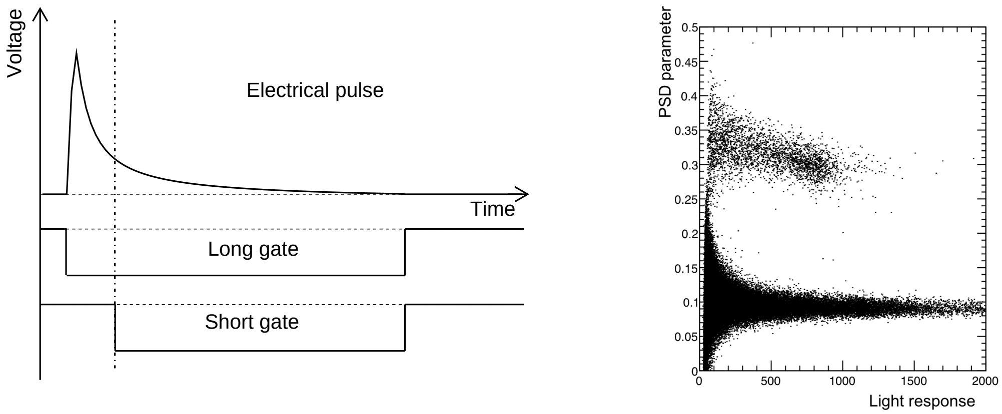


Figure 2.9. Scintillation pulses generated by recoil electrons and deuterons.

ratio of delayed fluorescence and prompt fluorescence.

$$PSD = \frac{\int_{tail:start}^{tail:end} Qdt}{\int_{fullsignal} Qdt} \tag{2.19}$$

An optimized choice of the position of a pulse tail will give rise to a decent pulse-shape discrimination, which is shown as two well seperated bands in Fig. 2.10(b). The events in the lower intense band represents electrons scattered by the incident γ-rays, and the events in the upper band are recoil deuterons scattered by incident neutrons.



(a) (b)

Figure 2.10. Time setting for short gate and long gate on an electrical pulse.

2.3.2.5 Unfolding Algorithm – MLEM

The pulse-height spectra (PHS) obtained from the deuterated Liquid Scintillator are in fact deuteron-recoil spectra, which result from the convolution of the incident neutron spectrum and the detector's response. This system can be described by a linear integral [85],

$$N(L) = \int R(L, E)\phi(E)dE, \tag{2.20}$$

where $N(L)$ is the measured pulse height spectra, $\phi(E)$ is the incident neutron spectra, and the kernel function $R(L, E)$ represents the detector response. Mathematically, this integral can be expressed as a discrete linear matrix multiplication approximation [81],

$$y_i = \sum_{j=1}^{J} r_{ij}x_j, i \;=\; 1, ..., I, \tag{2.21}$$

where y_i is the number of counts falling in the ith bin of the pulse-height spectra, x_j is the incident neutron flux with the same energy, and r_{ij} is an element of the response matrix R of the detector. Fig. 2.11 shows how the response matrix maps the incident neutron flux space to the light output (pulse-height spectra) space.

The process of extracting the neutron spectrum x is known as spectrum unfolding. A natural approach to solve the unfolding problem is direct matrix inversion, however, it usually introduces larger noise in the final unfolded neutron spectrum than that which is present in the data and results in non-physical negative count rate, since the response matrix R is highly ill-conditioned [87]. For this reason, various approximation approaches and codes have been developed to solve this problem. The method used in this work is known as maximum-likelihood expectation-maximization (MLEM). The MLEM method is based on the assumption that the binned number of counts in channel i of the pulse-height spectrum (PHS) follows a Poisson distribution, which takes the form of

$$P_i = \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!}, \tag{2.22}$$

where $\lambda_i = \sum_{j=1}^{J} r_{ij} x_j$ is the expected value of y_i. This natural assumption captures the photon statistical fluctuation in the PHS [81]. Hence the log-likelihood function can be established

$$P = \prod_{i=1}^{I} P_i \tag{2.23}$$

$$\ln P = \sum_{i=1}^{I} (y_i \ln \sum_{j=1}^{J} r_{ij} x_j - \ln y_i! - \sum_{j=1}^{J} r_{ij} x_j), \tag{2.24}$$

where P represents the probability of obtaining the measured pulse-height spectra. The next step is to take the partial derivative of $\ln P$ with respect to x_j, and find the maximum of the log-likelihood function when all the partial derivative terms vanish

$$\frac{\partial \ln P}{\partial x_j} = \sum_{i=1}^{I} [\frac{r_{ij} y_i}{\sum_{j=1}^{J} r_{ij} x_j} - r_{ij}], j = 1, ..., J. \tag{2.25}$$

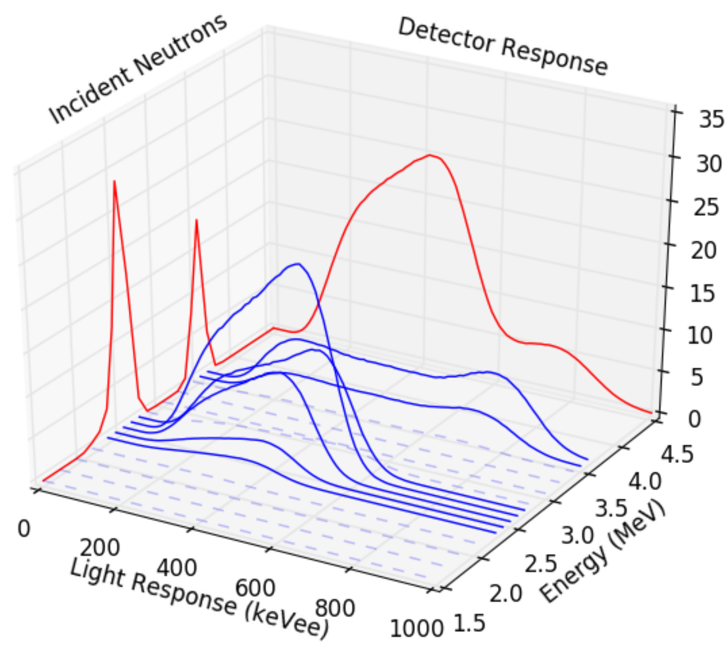


Figure 2.11. Graphical interpretation of Eq. 2.21

An approximate solution for Eq. 2.25 takes an iterative form,

$$x_j^{(k+1)} = \frac{x_j^{(k)}}{\sum_{i=1}^{I} r_{ij}} \sum_{i=1}^{I} r_{ij} \frac{y_i}{\sum_{l=1}^{J} r_{il} x_l^{(k)}}, j = 1, ..., J, \tag{2.26}$$

where $x_j^{(k)}$ stands for the estimation of the j^{th} neutron flux at the k^{th} iteration. The likelihood probability increases as the iteration goes.

The MLEM code was written in the C++ language. The inputs for the program are the response matrix, a light output spectrum to unfold, the energy threshold, and optionally additional a priori information to be given at the user's option.

2.3.2.6 Detection Efficiency

The low interaction probability between neutrons and surrounding matter allows neutrons to escape the target chamber, and all experimental setups are potential neutron scattering sources. Neutrons can be scattered several times before being detected by the deuterated liquid scintillator, which complicates the determination of the neutron detector efficiency. A Monte Carlo based neutron transport code, `MCNPX-PoliMi`, is capable of including scattering contributions from different elements. It is used to simulate the experimental setup and help compute the detection efficiency. An example of the simulated efficiency is shown in Fig. 2.12. The efficiency of the deuterated liquid scintillator is threshold dependent, and neutrons with energies lower than around 1 MeV were cut off because of the limitations of the PSD.

The detector response has been calibrated against the well known thick-target spectrum of the $^9\text{Be}(d, n)$ reaction to validate the simulation results. These calibration measurements were performed at the Edwards Accelerator Laboratory at Ohio University [40] using time-of-flight technique. By using a thick target, the neutrons produced during the reaction will gradually loss energy as they travel through the target material. Thus, one can detect neutrons with continuous energies arriving at the detector at different time. Comparing the detected thick-target spectrum to the measurement from Massey et al. [73] yields the detector efficiency.

The efficiency simulated using `MCNP-Polimi` [83] is shown together with the measurement in Fig. 2.12. The simulation is able produce a satisfactory reproduction of the measured efficiency.

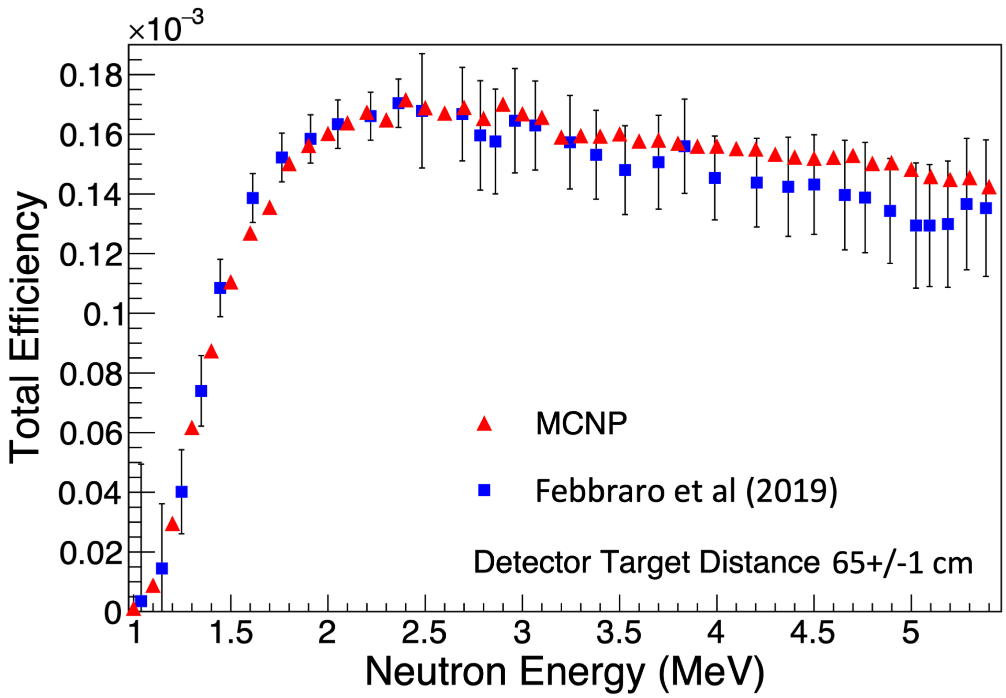


Figure 2.12. (Color online) Comparison of the efficiency curve of an 7.6 cm thick × 5.8 cm diameter EJ-315M deuterated liquid scintillator from a simulation using `MCNP-Polimi` (red triangles) and the efficiency curve measured using time-of-flight with the ^{9}Be(d, n) thick target method [73, 40] (blue squares).

CHAPTER 3

THE ^{11}B$(\alpha, n)^{14}$N REACTION

This chapter discusses the measurement of the ^{11}B$(\alpha, n)^{14}$N cross sections with the moderating ^{3}He proportional counter.

3.1 Experiment Setup

The ^{11}B$(\alpha, n)^{14}$N reaction (Q = 0.15789(1) MeV) was studied over α-particle beam energies E_α ranging from 520 to 2000 keV in steps of 20 keV or less and with beam intensities on target between 0.03 and 18-μA. The detection system consisted of a ^{3}He proportional counter, which is made up of 20 ^{3}He tubes encased in a polyethylene moderator. The detector is described in Best et al. [16, 17]. The target was placed at the center of the moderator to measure the angle integrated cross section.

The natural neutron background was 1.7 count/hr, which was determined by taking a 17 hrs room background run. It is negligibly compared to the yield of ^{11}B(α,n)^{14}N reaction in the energy range measured in this work. A PuBe source was used to identify the neutron region of interest on the ^{3}He detector spectrum. The energy dependence of the efficiency was modeled using the `MCNP` [98] code in a previous Ph.D thesis work by Best [18] and the absolute efficiency was obtained with the activation method using the ^{51}V$(p, n)^{51}$Cr reaction at E_p = 3.7 MeV. The resulting efficiency was found to be consistent with that found by Best et al. [16] to within 1% using a nearly identical setup. The small Q-value of the ^{11}B$(\alpha, n)^{14}$N reaction and the range of α-beam energies that were investigated results in neutrons that cover the range from $0.4 < E_n < 2.0$ MeV, taking into account the angular kinematic spread as well. The neutron detection efficiency of the ^{3}He counter varies from about 30 to 45% over this neutron energy range. The overall uncertainty in the neutron efficiency is estimated to be ≈5%. In addition, and rather critically, only the ground state in the ^{14}N final nucleus is energetically accessible. This ensures directly obtaining the cross section of ^{11}B$(\alpha, n_0)^{14}$N reaction without involving any complicated corrections for background reactions and detection efficiency.

3.2 Target Thickness

A ^{11}B enriched (>99%) target was created by electron sputtering enriched ^{11}B powder onto a clean tantalum backing 2 mm in thickness. The target thickness was measured by scanning over the well-known narrow resonance with a total width of Γ = 2.5(5) eV at E$_\alpha$ = 606.0(5) keV [107], as shown in Fig. 3.1. For a narrow resonance like this, the stopping power ϵ, the Broglie wavelength λ, and the partial widths Γ_i of the resonance can be considered constant over the resonance width. For a single narrow resonance, one can use the Breit-Wigner formula to approximate the cross section. Thus, the yield can be written as,

$$\begin{aligned} Y(E_0) &= \int_{(E_0-\Delta E)}^{E_0} \frac{1}{\epsilon(E)} \frac{\lambda^2}{4\pi} \omega \frac{\Gamma_a \Gamma_b}{(E_r - E)^2 + \Gamma^2/4} dE \\ &= \frac{\lambda_r^2}{2\pi} \frac{\omega\gamma}{\epsilon_r} \frac{\Gamma}{2} \int_{(E_0-\Delta E)}^{E_0} \frac{dE}{(E_r - E)^2 + \Gamma^2/4} \\ &= \frac{\lambda_r^2}{2\pi} \frac{\omega\gamma}{\epsilon_r} [arctan(\frac{E_0 - E_r}{\Gamma/2}) - arctan(\frac{E_0 - E_r - \Delta E}{\Gamma/2})] \end{aligned} \tag{3.1}$$

where ΔE is the total energy lost by the beam particles in the target. In other words, ΔE stands for the target thickness in energy units. After some algebra, one finds that the relationship between FWHM full width at half maximum Γ, and target thickness ΔE can be described by

$$FWHM = \sqrt{\Gamma^2 + \Delta E^2}. \tag{3.2}$$

Hence, the target thickness can be approximated with FWHM, when $\Gamma \ll \Delta E$. Using this approximation, the thickness of ^{11}B target was found to be 8.4(4) $\mu g/cm^2$.

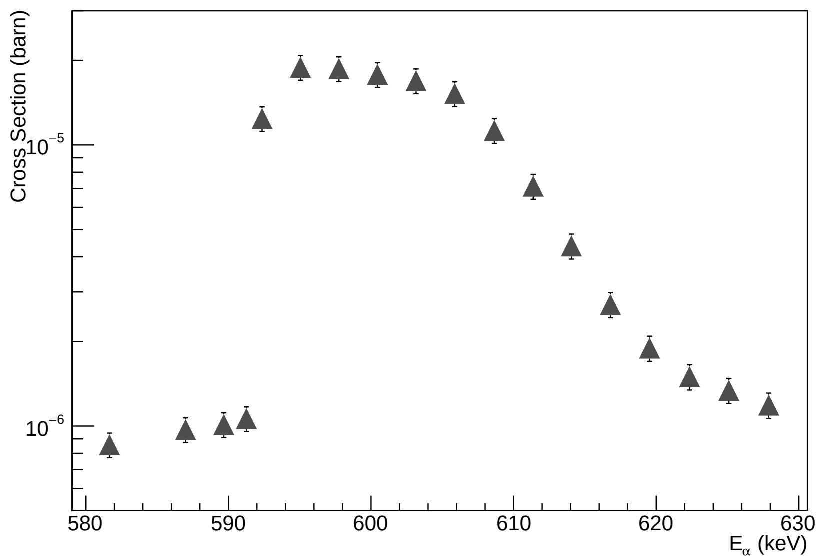


Figure 3.1. Thin target scan. Target was measured over the narrow resonance at $E_\alpha = 606.0(5)$ keV.

3.3 Reaction Yield

The sum of all the counts from the twenty ^{3}He tubes were normalized by the beam charge integration and target thickness assuming a thin target approximation,

$$\sigma \approx \frac{Y}{N_p N_t \epsilon} \tag{3.3}$$

where N_p are the number of beam particles made incident on target (found to be reproducible to 2%), N_t are the number of target atoms obtained by the target thicknesses, and ϵ is the ^{3}He counter efficiency. In the present experiment the range of neutron energies covered E_n = 0.36 to 2.0 MeV. This is a function of both the incident α-particle energy as well as the kinematic spread from the nearly 4π solid angle subtended by the ^{3}He counter. In this energy region, the efficiency dependency on neutron energy for the ^{3}He counter can be considered linear, as shown in Fig. 3.2. According to kinematics, the 0° outgoing neutrons with the maximum energy E_n=640 keV, have the minimum detection efficiency ϵ_{min} corresponding to the upper limit of the cross section, and the 180° neutrons, with the minimum energy E_n=371 keV, have the maximum efficiency ϵ_{max} corresponding to the lower limit of the cross section. The central value of the cross section was calculated as the mean of the upper and lower limits. This inherent uncertainty in the neutron energy and thus detection efficiency is reflected in the uncertainties of the data. The statistical uncertainty is between ±0.15% and ±6.13%. The contributions to the measurment systematic uncertainties are also considered: charge integration ±2% and relative efficiency ±5%.

The experimental data are shown in Fig. 3.3 compared to previous measurements by Wang et al. [107]. Over most of the energy range, the present measurements and those of Wang et al. [107] are in good agreement. However, there is an off-resonance region near $E_{\mathrm{c.m.}} \approx 0.5$ MeV where the present data are significantly smaller than those of Wang et al. [107]. This region is also just below the very strong narrow $7/2^-$ state at $E_{\mathrm{c.m.}} = 444.4(4)$ keV (not shown in the plot). This discrepancy could be the result of diffusion of some of the target material into the backing used by Wang et al. [107]. This type of over estimation of cross sections just above strong narrow resonances is a common issue (see, for example, Imbriani et al. [58]).

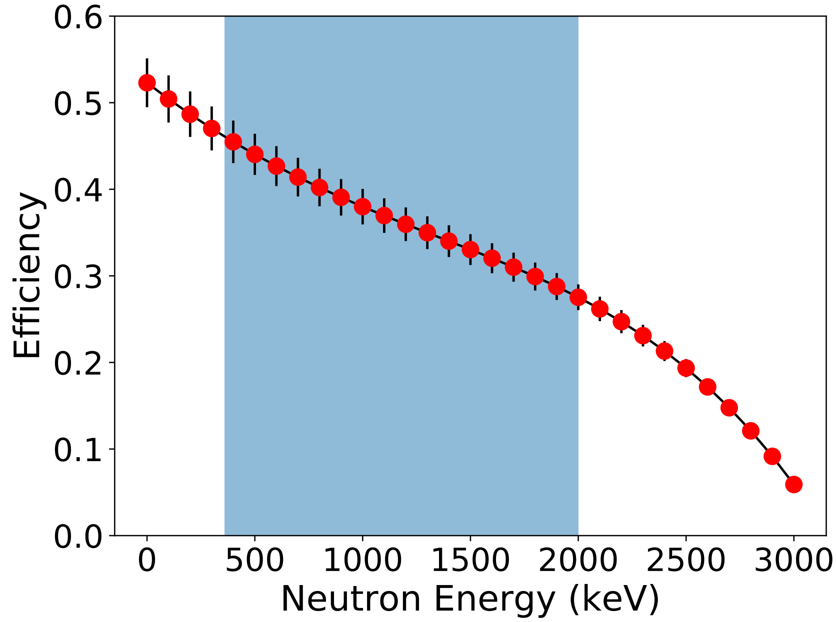


Figure 3.2. Detection efficiency of He^3 counter simulated using the MCNP. The blue shaded area represents the neutron energy range from 360 keV to 2000 keV covered in this work. This figure is adapted from the results of Best [18].

3.4 R-matrix Analysis

The low energy cross section of the ^{11}B$(\alpha, n)^{14}$N reaction has been analyzed using the phenomenological R-matrix approach [69, 33] using the code `AZURE2` [10, 100]. The alternate parameterization of Brune [21] has been used in order to work directly with physical parameters and to eliminate the need for boundary conditions. The physical constants used for the analysis of the ^{15}N system are given in Table 3.1.

At the energies under consideration, the ^{11}B$(\alpha, n)^{14}$N reaction has several open decay modes α-particle, neutron and proton emission. To achieve a physical solution using the phenomenological model, it is advantageous (if not necessary) to include other reaction data that can constrain the decay branchings to the other decay modes. For the ^{11}B$(\alpha, n)^{14}$N reaction and other reactions that populate the ^{15}N system a full

TABLE 3.1

PHYSICAL CONSTANTS AND R-MATRIX PARAMETERS FOR THE ^{15}N SYSTEM.

Parameter	Value
a_c (^{11}B+α)	5.1
a_c (^{14}C+p)	4.3
a_c (^{14}N+n)	4
M_p	1.00783
M_n	1.00866
M_α	4.00260
$M_{^{11}\mathrm{B}}$	11.0093
$M_{^{14}\mathrm{C}}$	14.0032
$M_{^{14}\mathrm{N}}$	14.0031

Notes: Channel radii are in units of fm and masses, taken from [7] , are in atomic masses.

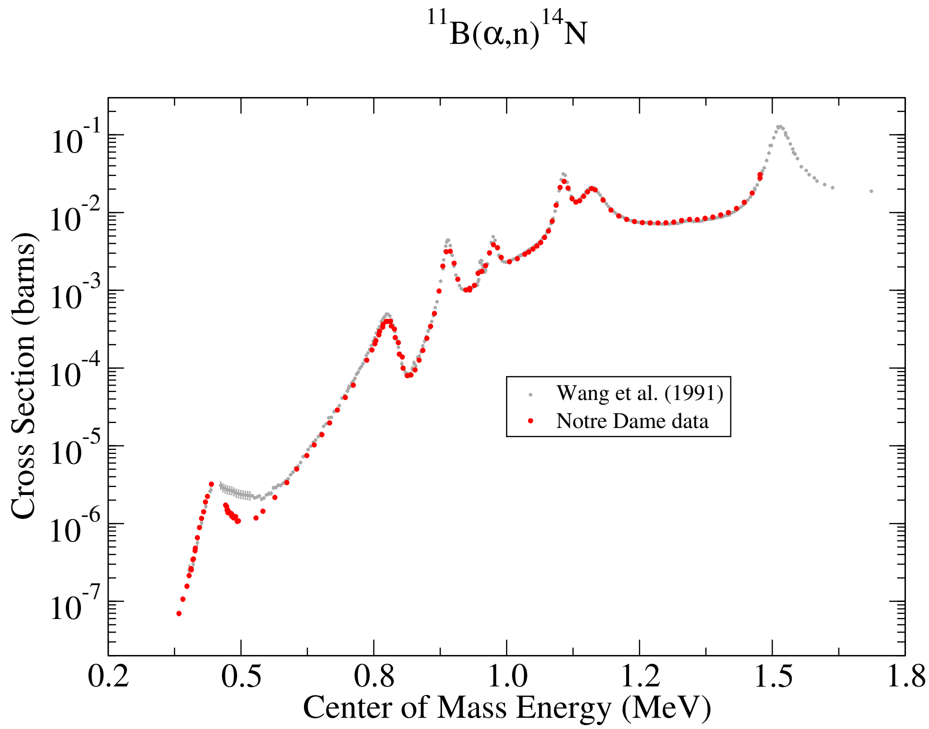


Figure 3.3. Comparison of the $^{11}B(\alpha, n)^{14}N$ data measured in the present work to that of Wang et al. [107]. The measurements are in good agreement with the exception of the region just above the lowest energy resonance near $E_{\text{c.m.}} = 0.5$ MeV.

multi-channel fit has been achieved.

3.4.1 Cross Section Correction for Target Effects

One powerful function of the `AZURE2` [10, 100] code is the target integration routine. By including the stopping power of beam particles in target material and the number of active nuclei in target, one is able to perform a direct fit of the target integrated R-matrix calculation to the yield curve.

The number of active target nuclei is determined as $4.61(21)\times10^{17}$ atoms/cm^2, which is converted from the target thickness 8.4(4) μg given in Sec. 3.2. The stopping power of ^{4}He in ^{11}B is given in the form of a fifth-order polynomial function,

$$\epsilon(E) = 2.443 \cdot 10^{-20} + 6.222 \cdot 10^{-20} \cdot E - 1.021 \cdot 10^{-19} \cdot E^2 + 6.846 \cdot 10^{-20} \cdot E^3 - 2.179 \cdot 10^{-20} \cdot E^4 + 2.683 \cdot 10^{-21} \cdot E^5. \tag{3.4}$$

Once the best fit is achieved, the cross section for a target of negligible thickness can be calculated as,

$$\sigma_t = \frac{\sigma_I}{Y_I} Y_t \tag{3.5}$$

where σ_t and Y_t are the cross section and yield for a target of negligible thickness, σ_I and Y_I are the cross section and yield for the best fit. The best fit here is not necessarily a physically meaningful fit, but is function that best describes the data and serves only as an interpolation function. All the cross sections data before and after the target effect correction are tabulated in Appendix. D.1.

3.4.2 ^{15}N System

The ^{11}B(α,n)^{14}N reaction populates states in the ^{15}N compound nucleus at an excitation energy range E_x = 11.2 to 12.6 MeV. There are three allowed particle decay modes: α-particle (S_α = 10.99118(1) MeV), neutron (S_n = 10.83330(1) MeV), and proton (S_p = 10.20742(1) MeV) [55, 106]. At these energies, only decays to the respective ground states in the final nuclei are possible.

The ^{15}N system has been subjected to R-matrix analysis previously [86, 80, 48, 62], which have provided starting parameters and guidance for the current fitting. Due to the complexity of both the data and the level structure, the global R-matrix analysis as presented must still be considered preliminary. While it satisfies the specific use here of obtaining a new estimation of the ^{11}B$(\alpha, n)^{14}$N reaction, a full evaluation of all the reaction data is still in progress [1]. In this work the ^{11}B$(\alpha, n)^{14}$N data of the present work is considered in addition to the previous measurement by Wang et al.

TABLE 3.2

ENERGY SHIFT IN R-MATRIX FIT FOR DIFFERENT DATASETS.

Reaction	Energy shifted (keV)
^{14}C(p,p)^{14}C [50]	2
^{14}C(p,p)^{14}C [53]	10
^{14}C$(p,n)^{14}$N [93]	-10
^{14}N$(n,p)^{14}$C [78]	9
^{14}N$(n,\alpha)^{10}$B [44]	-3
^{14}N(n,total)	10

[107]. Data for the reactions ^{14}C$(p,p)^{14}$C [50, 53], ^{14}C$(p,n)^{14}$N [93], ^{14}C$(p,\gamma)^{15}$N [14], ^{14}N(n,total) [54, 60], ^{14}N$(n,p)^{14}$C [44, 78, 59], and ^{14}N$(n,\alpha)^{10}$B [44, 78, 64, 59] are also included.

While some tension does exist between the different data sets, a universal fit was obtained that indicates general consistency. Most of the tension arises due to off-sets in the energy scaling of the data sets. These off-sets were generally obvious since the measurements are made with relatively good energy resolution and the energy shifts were significant compared with the widths of many of the resonances. The energy shifts applied to the different data sets are summarized in Table 3.2. The parameters for the best fit are given in Tables 3.3 and 3.4.

TABLE 3.3. R-MATRIX PARAMETERS FOR THE ANALYSIS OF THE ^{15}N SYSTEM.

This work					Niecke et al. (1977) [80]					Jurney et al. (1997) [62]				Hale et al. (1992) [48]			
J^π	E_x	l	Ch	Γ_i	J^π	E_x	l	Ch	Γ_i	J^π	E_x	Ch	Γ_i	J^π	E_x	Ch	Γ_i
3/2+	10.700	0	n	0.27						3/2+ or 3/2-	10.7019						
		2	p	0.01								p	0.2				
3/2+	10.800	0	n	0.00						3/2(+)	10.804						
												p	$(0.22\pm0.10)\times10^{-3}$				
7/2+	11.234	2	n	2.51										7/2+	11.238	n	2.4
																p	<0.1
1/2-	11.288	1	n	1.96	1/2-	11.292	1	n	(0.39)	1/2-	11.291			1/2-	11.291	n	1.8
							1	n	(0.39)								
		1	p	4.26			1	p	7.7			p	5.6			p	5.3
														1/2+	11.403	n	<5
																p	250
1/2+	11.425	1	a	0.00	1/2+	11.443				1/2+	11.4370			1/2+	11.425		
		0	n	21.10			0	n	31.5							n	26
		0	p	12.70			0	p	6.9			p	6.8 ± 0.5			p	11

TABLE 3.3 CONTINUED

This work					Niecke et al. (1977) [80]					Jurney et al. (1997) [62]				Hale et al. (1992) [48]			
J^π	E_x	l	Ch	Γ_i	J^π	E_x	l	Ch	Γ_i	J^π	E_x	Ch	Γ_i	J^π	E_x	Ch	Γ_i
1/2+	11.605	0	n	2.46	1/2+	11.628	0	n	2.1	1/2+	11.615						
							2	n	(0.1)								
		0	p	351.00			0	p	488			p	401+/-7				
							1	a	(0.001)								
3/2+	11.758	1	a	0.01	3/2+	11.783	1	a	0.22	3/2+	11.763			3/2+	11.766		
		2	n	7.49			2	n	(2.5)							n	33.7
		0	n	29.10			0	n	41.1								
							2	n	(1.4)								
		2	p	0.39			2	p	0.54			p	0.5			p	0.6
5/2-	11.872	2	a	0.04	5/2-	11.876				(3/2-)	11.875			5/2-	11.875		
		3	n	14.50			3	n	(0.5)							n	15.5
							1	n	23.5								
							3	n	(0.7)								
		3	p	0.04			3	p	0.069			p	0.03			p	<0.2
5/2+	11.935	1	a	0.01										5/2+	11.935		

TABLE 3.3 CONTINUED

This work					Niecke et al. (1977) [80]					Jurney et al. (1997) [62]				Hale et al. (1992) [48]			
J^π	E_x	l	Ch	Γ_i	J^π	E_x	l	Ch	Γ_i	J^π	E_x	Ch	Γ_i	J^π	E_x	Ch	Γ_i
		2	n	1.45												n	1.3
																p	<0.2
1/2-	11.957	2	a	0.10	1/2-	11.965				1/2-	11.965			1/2-	11.962		
		1	n	16.90			1	n	(15.7)							n	11.4
							1	n	(5.3)								
		1	p	0.32			1	p	0.58			p	0.3			p	0.4

Notes: The partial widths are in units of keV. Subthreshold state ANCs are in units of $\mathrm{fm}^{-1/2}$.

TABLE 3.4. R-MATRIX PARAMETERS FOR THE ANALYSIS OF THE ^{15}N SYSTEM.

This work					Niecke et al. (1977) [80]				
J^π	E_x	l	Ch	Γ_i	J^π	E_x	l	Ch	Γ_i
3/2+	12.051	1	a	0.92					
		2	n	49.00					
		0	n	186.00					
		2	p	1.57					
5/2+	12.089	1	a	0.35	5/2+	12.097	1	a	0.52
							3	a	(0.04)
		2	n	15.30			2	n	4.6
							2	n	14.2
		2	p	1.00			2	p	2.6
3/2-	12.142	0	a	2.92	3/2-	12.146	0	a	0.43
							2	a	0.85
		1	n	16.20			1	n	14.7
							1	n	9.1
							3	n	(3.2)
		1	p	25.70			1	p	15.9
5/2-	12.320	3	n	11.40	5/2-	12.328	3	n	(0.07)
		1	n	2.33			1	n	(0.14)
							3	n	21.5
		3	p	0.16			3	p	0.3
5/2+	12.496	1	a	6.10	5/2+	12.493	1	a	4.3
							3	a	4.3
		2	n	27.70			2	n	29.8

TABLE 3.4 CONTINUED

This work					Niecke et al. (1977) [80]				
J^π	E_x	l	Ch	Γ_i	J^π	E_x	l	Ch	Γ_i
		2	n	0.00			2	n	8.3
		2	p	0.47			2	p	0.39
5/2+	12.518	1	a	2.55	5/2+	12.522			
		2	p	87.60			2	p	58.4
1/2+	12.880	1	a	93.10					
		0	n	279.00					
		0	p	7.67					
5/2+	12.899	1	a	19.30	3/2-	12.921	0	a	15
							2	a	19
		2	n	6.77			1	n	7
							1	n	18
		2	p	7.24			1	p	9
3/2-	14.626	0	a	1150.00	3/2-	14.379	0	a	92
							2	a	894
		1	n	2030.00			1	n	28
							1	n	492
							3	n	292
		1	p	166.00			1	p	201

Notes: The partial widths are in units of keV. Subthreshold state ANCs are in units of $\mathrm{fm}^{-1/2}$.

For the $^{11}\mathrm{B}(\alpha, n)^{14}\mathrm{N}$ reaction, a number of resonances are present in the energy range observed as summarized by the corresponding levels in Table 3.3. While the

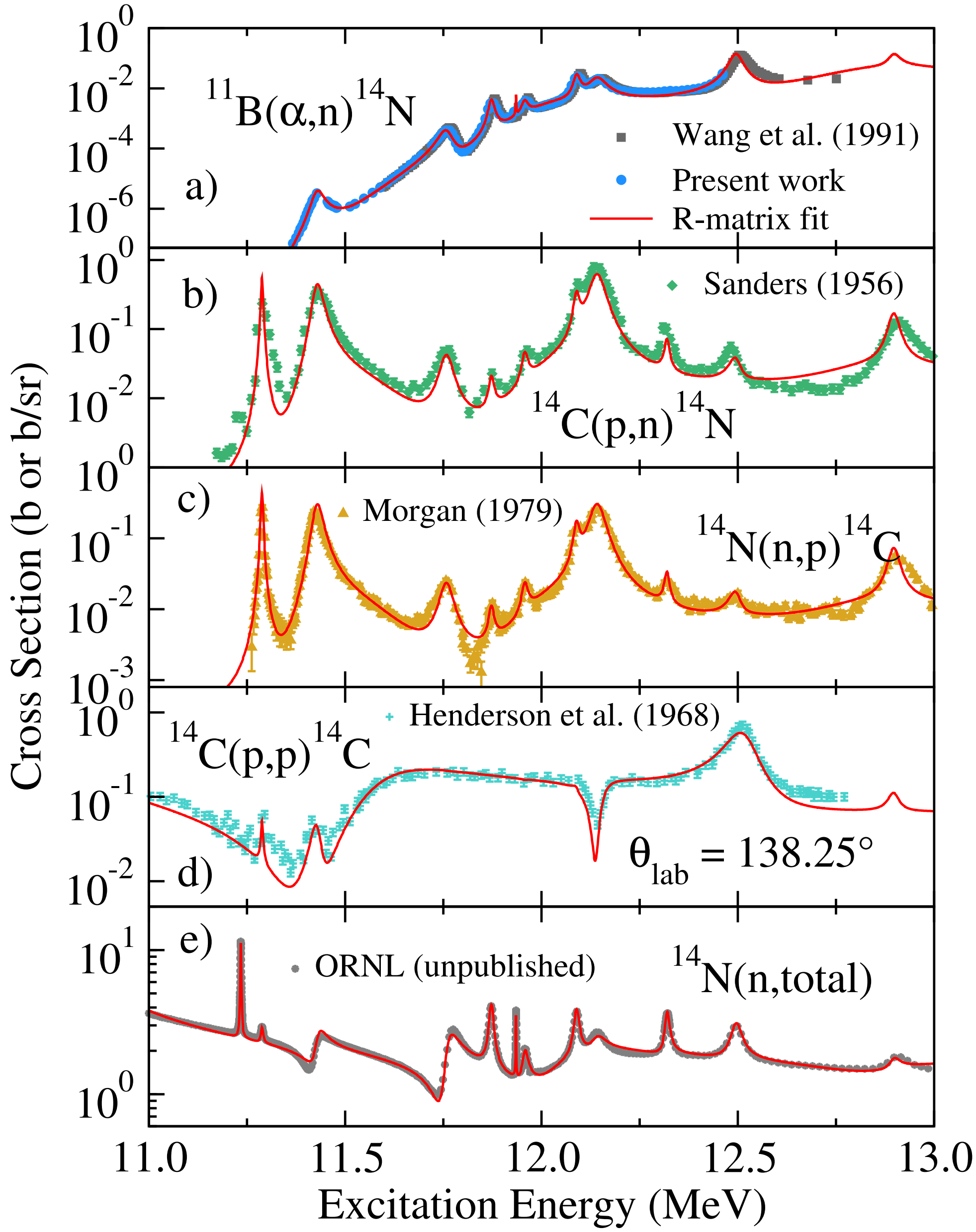


Figure 3.4. (Color Online) Example excitation functions from the global R-matrix fit of the ^{15}N system for the present ^{11}B$(\alpha,n)^{14}$N measurement (blue circles) and previous measurements of the ^{11}B$(\alpha,n)^{14}$N (Wang et al. [107], grey squares) ^{14}C$(p,n)^{14}$N (Sanders [93], green diamonds)), ^{14}N$(n,p)^{14}$C (Morgan [78], gold upward triangles), ^{14}C$(p,p)^{14}$C (Henderson et al. [53], turquoise plus), and ^{14}N(n,total) (ORNL (unpublished), grey stars) reactions.

same levels were observed in the previous measurement by Wang et al. [107], many of the level energies and resonance widths reported here are substantially different. This is the result of the inclusion of interference effects in the present analysis as the present analysis uses a full multichannel multilevel R-matrix formalism to fit the data, in contrast to the multilevel Briet-Wigner method used by Wang et al. [107] where interferences were neglected. As there are several broad resonances present, these interference effects are quite significant.

There have been at least three previous comprehensive R-matrix analyses [80, 48, 62] of the ^{15}N system over the same energy range as the present data. The earliest analysis by Niecke et al. [80] focuses on polarization measurements of the ^{14}C$(p,n)^{14}$N and ^{11}B$(\alpha,n)^{14}$N reactions. Unfortunately the R-matrix code used in the present work is not capable of calculating polarization observables, so this data could not be utilized. The works of Hale et al. [48] and Jurney et al. [62] focus on the ^{14}N+n reactions. All of these analyses were made before the publication of Wang et al. [107], therefore only include a limited amount of ^{11}B+α data. A comparison of the level parameters obtained from the R-matrix fit of this work are compared with those of Niecke et al. [80], Hale et al. [48], and Jurney et al. [62] in Table 3.3.

As a general remark, it was found that it was possible to reproduce the cross section data for the ^{14}C$(p,n)^{14}$N and its inverse reaction with previously reported levels in ^{15}N within the energy range of the data. On the other hand, it was impossible to reproduce the observed cross sections for the ^{11}B$(\alpha,n)^{14}$N and ^{14}N(n,total) reactions without the presence of background states with large partial widths. For the ^{11}B$(\alpha,n)^{14}$N reaction, this seems physically reasonable since higher energy data, see for example Refs. [113, 44, 78], observe very strong broad resonances at energies just above the region of the present measurements. For the ^{14}N(n,total) cross section, the saturation is less clear.

The ^{14}N(n,total) cross section was found to be particularly difficult to reproduce with only the resonances observed in the data region. In particular, the low energy region where the cross section is dominated by neutron scattering. The difficulty seems to stem from the contribution of a strong subthreshold state(s) as discussed by Hale et al. [48]. As in that work, we do not obtain an “entirely satisfactory” reproduction of the low energy cross section. In addition, the fit to the ^{14}N(n, total) data is extremely sensitive to the neutron channel radius and requires a value that is smaller than usual. Here a radius of 4 fm was used, the same as that used by Jurney et al. [62], but larger than the value of 2.6 fm used by Hale et al. [48]. In principle, any value for the channel radius can be used with appropriately compensating background levels. Usually a value close to $a_{\rm ch} \approx a_0(A_1^{1/3} + A_2^{1/3})$ works well (where $a_{\rm ch}$ is the channel radius, $a_0 \approx 1.4$ fm, and A_1 and A_2 are the unit masses of the particle partition), but in this case a smaller value is highly favored.

CHAPTER 4

^{10}B$(\alpha, n)^{13}$N REACTION

This chapter will discuss two measurements made of the ^{10}B$(\alpha, n)^{13}$N reaction ($Q = 1.05873(27)$ MeV) over different energy ranges, and they will be referred as low energy measurement covered beam energy range from 0.57 MeV to 2.5 MeV and high energy measurement covered beam energy range from 2.2 MeV to 4.9 MeV in the following sections. Both measurements were performed at the University of Notre Dame's Nuclear Science Laboratory using the Sta. Anna 5-MV accelerator.

4.1 Experiment Setup

4.1.1 Low Energy Measurement

The low energy measurement covered an energy range from $570 < E_\alpha < 2500$ keV. The accelerator produced a beam intensity on target between 0.1 and 60 μA. At high energies, on strong resonances, the beam intensity had to be limited to prevent dead time in the data acquisition system (DAQ). The detection system consisted of two deuterated liquid scintillator detectors, of type EJ315 [39] (size: 7.6 cm thick $\times$ 7.6 cm diameter) and EJ301D [15] (size: 7.6 cm thick $\times$ 5.8 cm diameter), and a high-purity germanium detector (HPGe) with a relative efficiency of 54%. As a reference detector, the EJ315 was fixed at an angle of 45° relative to the incoming α-beam. An excitation curve measurement was made with the EJ301D detector, which is a newly developed xylene-based scintillator with improved pulse-shape discrimination (PSD). An excitation function was measured at 0° in order to achieve the highest neutron energies at low beam energies. At higher energies, from 1.0 to 2.5 MeV, where the count rate was also high, the scintillators were placed in far geometry at a distance of 20.9 cm. As the beam energy went down below 1 MeV, the detector was moved into a close geometry of 5.5 cm to achieve sufficient yields. In addition, excitation curves were also measured for the ^{10}B$(\alpha,\alpha_1\gamma)^{10}$B ($E_x = 0.718380(11)$ MeV) and ^{10}B$(\alpha,p_{1,2,3}\gamma)^{13}$C reactions ($E_x = 3.089443(20)$, 3.684507(19), and 3.853807(19) MeV) using an HPGe detector at 130°. The distance between the target and HPGe detector face was 21.5 cm.

All measurements were taken with an enriched Boron target, which was made by evaporating ^{10}B enriched powder onto a clean 0.5 mm thick tantalum backing with an electron gun. While under beam bombardment, targets were cooled from the backside with a constant flow of deionized water. The thickness and uniformity of the target was measured at the National Institute for Standards and Technology (NIST) Center for Neutron Research (NCNR) using cold neutron depth profiling with the neutron standard ^{10}B$(n,\alpha)^7$Li reaction [35, 68]. See Sec. 4.4.2 for further discussion.

4.1.2 High Energy Measurment

The high energy measurement was studied for α-particle beam energies E_α ranging from $E_\alpha = 2.2$ to 4.9 MeV. Beam intensities between 0.1 and 20 eμA were impinged on target. The detection system consisted of five deuterated liquid scintillator detectors, one of type EJ-301D [15] (size: 7.6 cm thick $\times$ 7.6 cm diameter), one of type EJ-315 [39] (size: 7.6 cm thick $\times$ 5.8 cm diameter), and three of type EJ-315M (size: 7.6 cm thick $\times$ 5.8 cm diameter), and one high-purity germanium detector (HPGe) with an efficiency of 54% relative to a 3"$\times$3" NaI detector. The setup is shown in Fig. 4.1. As a reference detector, the EJ-315 was fixed 60(1)° at a distance (front face to target center) of 27(1) cm. The angular distribution measurements were made with the other four detectors, which were placed 63(1) cm from the target position, and rotated on a swing arm to 12 angular positions covering an angular range of $20 < \theta_{\text{lab}} < 150^\circ$ in the laboratory reference frame. The HPGe was used to monitor secondary γ-rays from the ^{10}B$(\alpha, p'\gamma)^{13}$C reaction and was placed at $\theta_{\text{lab}} = 130^\circ$ at a distance of 22.5(5) cm. The secondary γ-ray yields were used to aid in troubleshooting during the experiment. The setup is shown in Fig. 4.1.

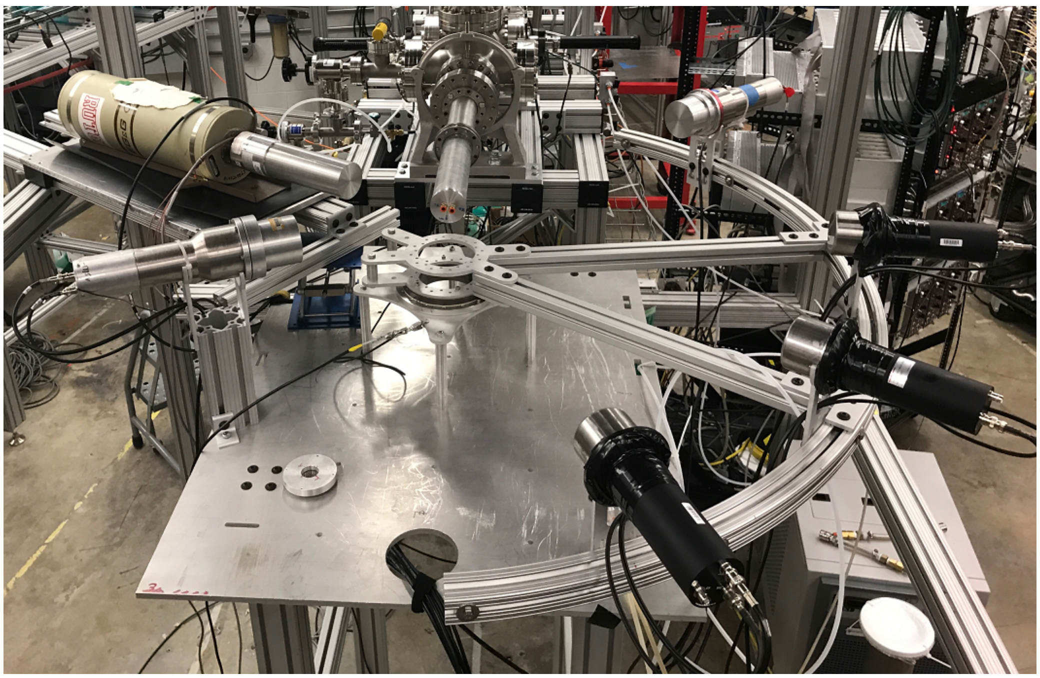


Figure 4.1. Experimental Setup for the differential cross section measurement of ^{10}B$(\alpha$,n$)^{13}$N reaction.

Targets were prepared by electron-gun sputtering a thin layer of enriched (96.2(5)%) ^{10}B powder onto a 0.5 mm thick tantalum backing. The target thickness was measured at NIST using the same technique described in Sec. 4.1.1. Twelve points, six with 3 mm diameter and three with 5 mm diameter, were measured across the target surface in order to gauge its homogeneity, where the individual measurements had an uncertainty of less than 1%. These measurements revealed that the target varied in thickness by $\pm 10\%$ with a mean value of 13(1) μg/cm^2. Because of the variation in the position of the beam spot on target, the effective target thickness could not be determined to better than this level of accuracy. See Sec. 4.5.2 for further discussion.

4.2 Detector Calibration

For both measurements, the energy calibration of the light response spectrum was carried out using a ^{137}Cs source, with the well-known single γ-ray at $E_\gamma = 661.7$ keV. The deuteron and electron recoil peaks from incident neutrons and γ-rays can be well separated using pulse shape discrimination down to about $E_n \approx 1$ MeV. A ^{252}Cf source was used to define the neutron gate, an example spectrum is shown in Fig. 4.2, where the vertical axis is the ratio of the tail pulse gate (180 ns) on the detector wave form to the total pulse gate (250 ns) (S/L). The neutron gate is bounded by the three red lines in Fig. 4.2, which represent the γ-ray upper bound, the neutron lower bound, and the neutron upper bound. All of them have the functional form of $\frac{a}{\sqrt{L}} + bL + c$, in which the $\frac{a}{\sqrt{L}}$ term represents the statistical fluctuation of photons. Each bound line represents a 5σ deviation from the centroids of the γ/neutron peak in the y-projection as shown of Fig. 4.2. This is better illustrated by projecting a 1D spectrum of S/L for a fixed total pulse gate value L as shown in Fig. 4.3.

4.3 Data Analysis

The work flow to extract useful quantities from the light response spectrum are listed below. Details of each step are described in the following sections.

1. Parameterize light response curve, resolution function, and PSD parameters for the detector

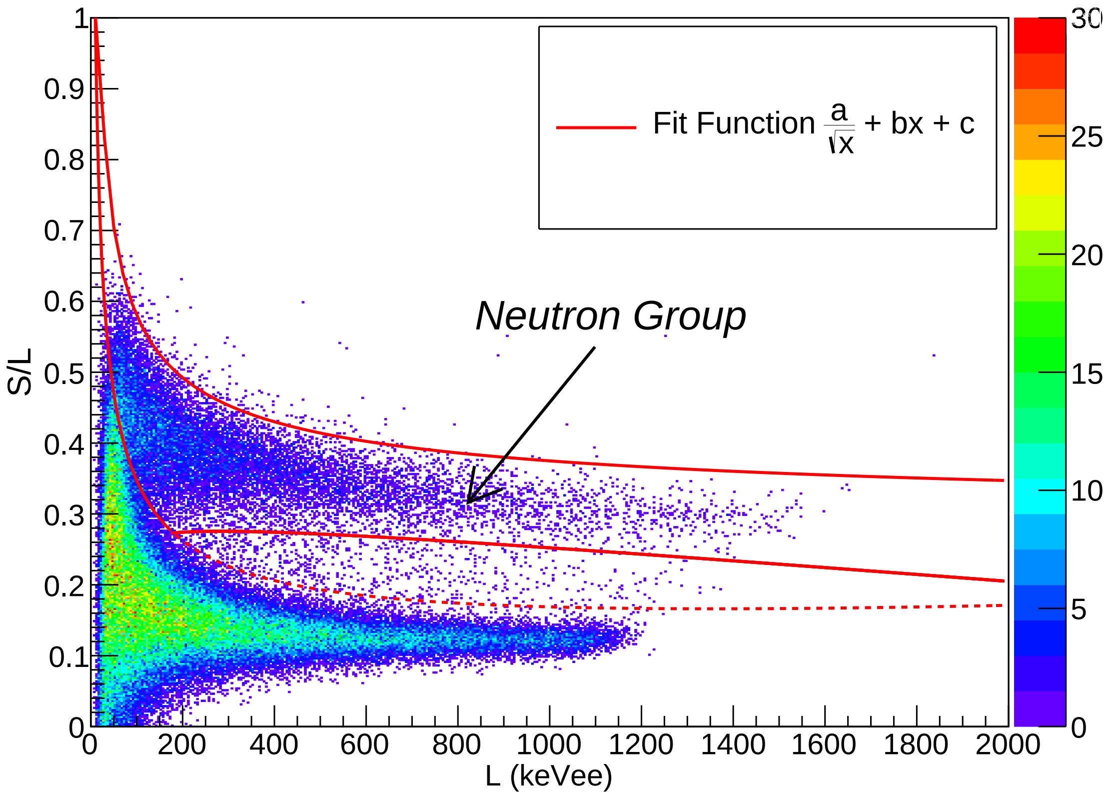


Figure 4.2. Pulse Shape Discrimination: S/L as a function of L. The neutron gate is defined by the region inside three red lines.

2. Re-create the experimental setup using `MCNPX-PoliMi` [98, 83] code to mimic neutron scatterings

3. Post-processing the output file generated by the previous step in an event-by-event fashion and generate response matrix

4. Unfold light response spectrum with MLEM algorithm to obtain neutron yield

5. Perform uncertainty analysis using Monte Carlo method

4.3.1 Parameterization of Light Response, Resolution and PSD

The response edge of the light response spectrum is the result of the convolution of the detector response and reaction cross section. The detector response is further

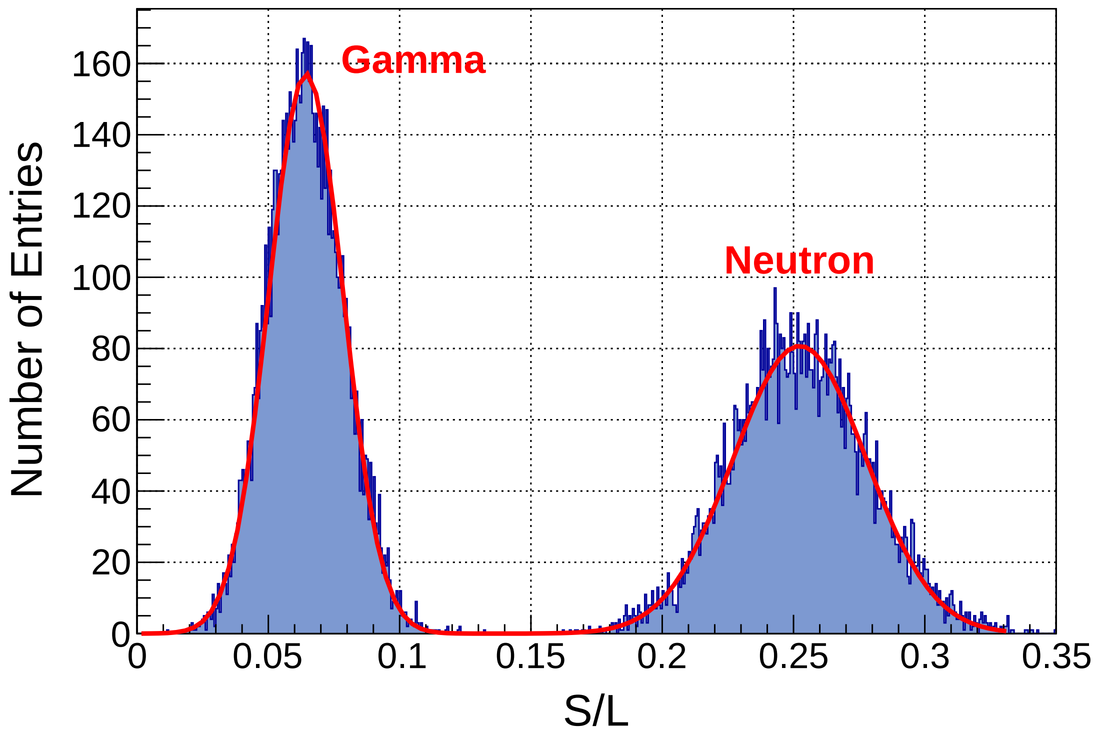


Figure 4.3. Projection of Fig. 4.2 at L = 350 keVee.

divided into two parts, the resolution of the detector and the energy converted to light from incident neutrons. A previous work by Febbraro [41] shows that a half Gaussian fit is able to describe the light response edge well. The ratio of the standard deviation σ and the location of the edge of the recoil peak was treated as the detector resolution. The location at 80% of the maximum Gaussian peak height was taken as the the edge of the light response peak.

The relationship between recoil energy of deuterium and the light output is given by Eq. 2.18. The measured light curve for the EJ-315M 3×2 scintillator is shown in Fig. 4.4. The parameterized fits for recoil deuterons for different detector types are shown in Table 4.1. The present data is in good agreement with previous measurements from Bildstein et al. [19].

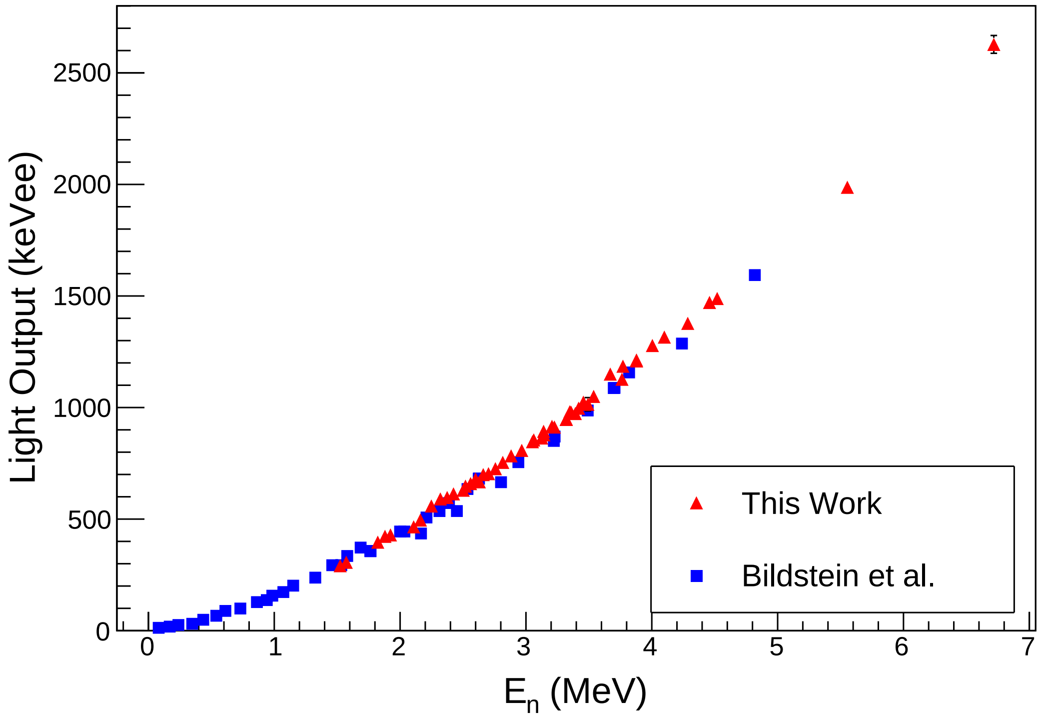


Figure 4.4. Comparison of light response curve for 3x2 EJ-315M liquid scintillator in this work and data taken from literature [19]. The uncertainty of the present data is less than the size of the point.

The resolution described by Eq. 4.1 is proved to be an empirical good fit to experimental data.[94]

$$\frac{\Delta L}{L} = \sqrt{\alpha^2 + \frac{\beta^2}{L} + \frac{\gamma^2}{L^2}} \tag{4.1}$$

where α, β, and γ can be obtained from a chi-square fitting routine. The results are listed in Table. 4.2.

The last piece of the puzzle to fully reproduce the physics processes inside the liquid scintillator is the PSD parameters. Intuitively, this information determines where a neutron event locates on Fig. 2.8. The PSD parameters are modeled by a

TABLE 4.1

DETECTOR LIGHT RESPONSE CONSTANTS

Scintillator	Size	a	b	c
EJ-301D	3x3	0.48(5)	0.80(33)	0.51(21)
EJ-315M-1	3x2	0.58(3)	1.38(22)	0.35(5)
EJ-315M-2	3x2	0.53(2)	1.05(14)	0.42(5)
EJ-315M-3	3x2	0.50(fixed)	1.22(1)	0.33(fixed)
EJ-315	3x3	0.59(2)	1.70(1)	0.30(1)

Notes: Detector light response parameters as fit to Eq. 2.18.

TABLE 4.2

DETECTOR RESOLUTION CONSTANTS

Scintillator	Size	γ	β	α
EJ-301D	3x3	0.19	3.29	2.53×10^{-4}
EJ-315M-1	3x2	0.02	3.43	7.89×10^{-2}
EJ-315M-2	3x2	0.36	4.18	7.07×10^{-4}
EJ-315M-3	3x2	0.07	2.96	1.62×10^{-1}
EJ-315	3x3	0.58	3.11	3.64×10^{-4}

Notes: Detector resolution constants as fit to Eq. 4.1.

TABLE 4.3

DETECTOR PSD CONSTANTS

Scintillator	Size	*psd*			*dpsd*	
		a	b	c	m	n
EJ-301D	3x3	0.27	-0.063	0.011	0.016	0.00025
EJ-315M-1	3x2	0.35	-0.060	0.0078	0.0094	0.0062
EJ-315M-2	3x2	0.35	-0.060	0.0082	0.010	0.0049
EJ-315M-3	3x2	0.27	-0.063	0.011	0.011	0.0024
EJ-315	3x3	0.22	-0.061	0.012	0.014	-0.0021

Notes: Detector psd constants as fit to Eq. 4.2.

function of the following form,

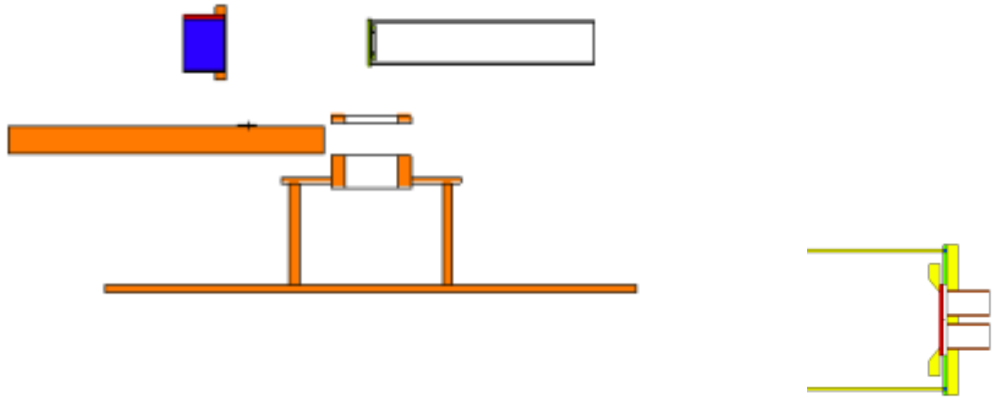
$$psd = a + b \times L + c \times L^2 \tag{4.2}$$

$$dpsd = \frac{m}{\sqrt{L}} + n \tag{4.3}$$

where *psd* stands for the center value of S/L in Fig. 4.2, and *dpsd* stands for the width of neutron band for given light response value L. The values of fitting parameters a, b, c, m, n are tabulated in Table. 4.3.

4.3.2 Response Matrix

The neutron detection response has been calibrated using a technique described previously (see Sec. 2.3.2.6). The detector response matrix is modeled using the `MCNP-Polomi` [98, 83] code and is tuned to match the experimental measurements



(a) (b)

Figure 4.5. Fig.(a) demonstrates the experimental setup geometry generated by MCNP code. Elements with different materials are represented in different colors. Fig.(b) shows the details of the target holder, water cooling bolts, and beamstop mounted at the end of the beampipe.

by way of phenomenological descriptions of the light response, resolution functions and PSD parameters (see Sec. 4.3.1). The simulation output file can then be used to interpolate the detector response to arbitrary energies.

A computer code (see Appendix. A) was developed to simulate the experimental setup. The simulated setup details are shown in Fig. 4.5. Since the neutron is neutrally charged, it will not be stopped easily. Thereby elastic and inelastic neutron scattering contributions from different elements should all be taken into account.

The most relevant quantity in the output file is the energy transferred from incoming neutrons (E_n) to the recoil deuterium nucleus (E_d). This information will go through a numerical process that is similar to the detector response. The process involves the following steps,

1. Converting recoil particle energy (E_d) to light response using the light response function determined in Sec. 4.3.1.

2. Performing Gaussian blur with the resolution function and modeling PSD parameters for given value of light response (see Sec. 4.3.1).
3. Applying the software determined pulse-shape discrimination to retain neutrons falling in the desired area.

All steps are coded in an MCNP post-processing file written in C++ language (see Appendix. B). The response matrix used in this work has the size of 1500×200, corresponding to 1500 bins with binwidth of 10 keVee in the light response spectrum and 200 bins with binwidth of 0.1 MeVee in the unfolded neutron spectrum. The solid angle efficiency and intrinsic efficiency of the detector are incorporated in the response matrix.

4.3.3 MLEM Unfolding

An example of neutron energy spectrum unfolding at $E_\alpha = 4.84$ MeV at $\theta_{lab} = 90°$ is shown in Fig. 4.6. The raw neutron energy spectrum is shown in the bottom panel of the figure by the red line. Neutron decays to the ground state and first excited state are clearly observed for the ^{10}B$(\alpha, n)^{13}$N reaction, as well as the ground state transition for the ^{11}B$(\alpha, n)^{14}$N reaction. The light output spectrum, shown by the red line in Fig. 4.2. The incident neutron spectrum, as shown in the top panel of Fig. 4.6, was extracted using the MLEM algorithm. The blue line in the bottom panel of Fig. 4.6 represents the estimation of the light output spectrum by forward-feeding the unfolded spectrum through Eq. 2.21. It serves as a double check of the unfolding process. Since the response matrix includes the detector's efficiency information, the counts in the unfolded neutron spectrum are efficiency corrected. A threshold of $E_n = 1.33$ MeV is imposed based on the PSD threshold limitations shown in Fig. 4.2.

4.3.4 Uncertainty Analysis

Although MLEM is a powerful method for spectrum unfolding, it is unable to provide a uncertainty estimation on the final unfolded spectrum. This is not particular

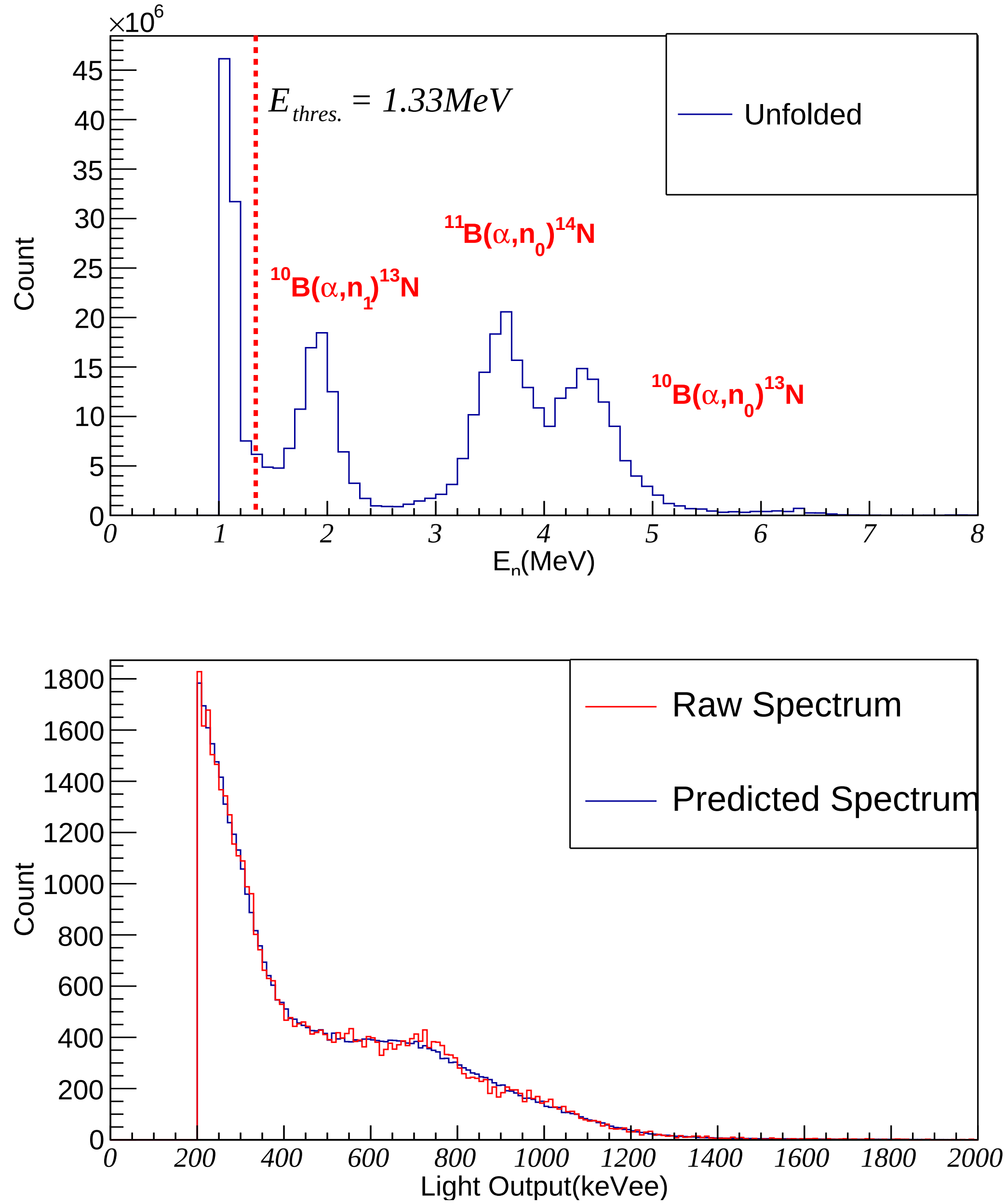


Figure 4.6. (Color online) Example of spectrum unfolding for a measurement at $E_\alpha = 4.84$ MeV and $\theta_{lab} = 90°$. The top panel shows the resulting unfolded neutron energy spectrum, which is unfolded using the well calibrated detector response model. The bottom panel shows the raw light output spectrum (red line) compared to the resulting neutron spectrum in the top panel folded with the detector response (blue line).

to the MLEM method, but is a common problem for many other unfolding algorithms [87]. Therefore, in order to estimate the uncertainty a Monte Carlo method is utilized. The underlying idea is to test how well the unfolding algorithm works given the statistics of each spectrum, by generating and unfolding numerous randomly generated pseudo light response spectra based on the measured raw spectra. The procedure of uncertainty estimation using this method are as follows:

1. Create a Poisson distribution and take the bin content in the light response spectrum as the mean value.
2. Reconstruct a new spectrum by generating random counts for each bin in the light response spectrum with a Poisson random number generator.
3. Repeat the above steps for a large number of iterations, and extract the neutron counts distribution as a histogram.
4. Fit a Gaussian distribution to the histogram in 3, then take the standard deviation as the estimated uncertainty.

The Monte Carlo uncertainties estimated using the above method increases as the counts (N) in the ^{10}B peak decreases. Fig. 4.7 shows a scatter plot of the Monte Carlo uncertainty and the statistical uncertainty ($1/\sqrt{N}$). In general, the relationship follows a linear trend (shown in red line). In the higher counts region, the points are tightly centered around the trendline. When the statistical error is larger than ∼6%, the points are more fluctuated. This can be partially explained by the fact that some of the measurements reaching low in energy where the neutron energies produced by the ^{10}B(α,n)^{13}N reaction are close to the energy threshold of the detector. It leads to an ill-defined response edge (see Fig. 4.6) in the light output spectrum, which is difficult for the algorithm to unfold.

4.4 Low Energy Measurement Results

Since the two measurements took place at different time, and focused on different energy ranges for different purposes, they will be discussed separately in the following

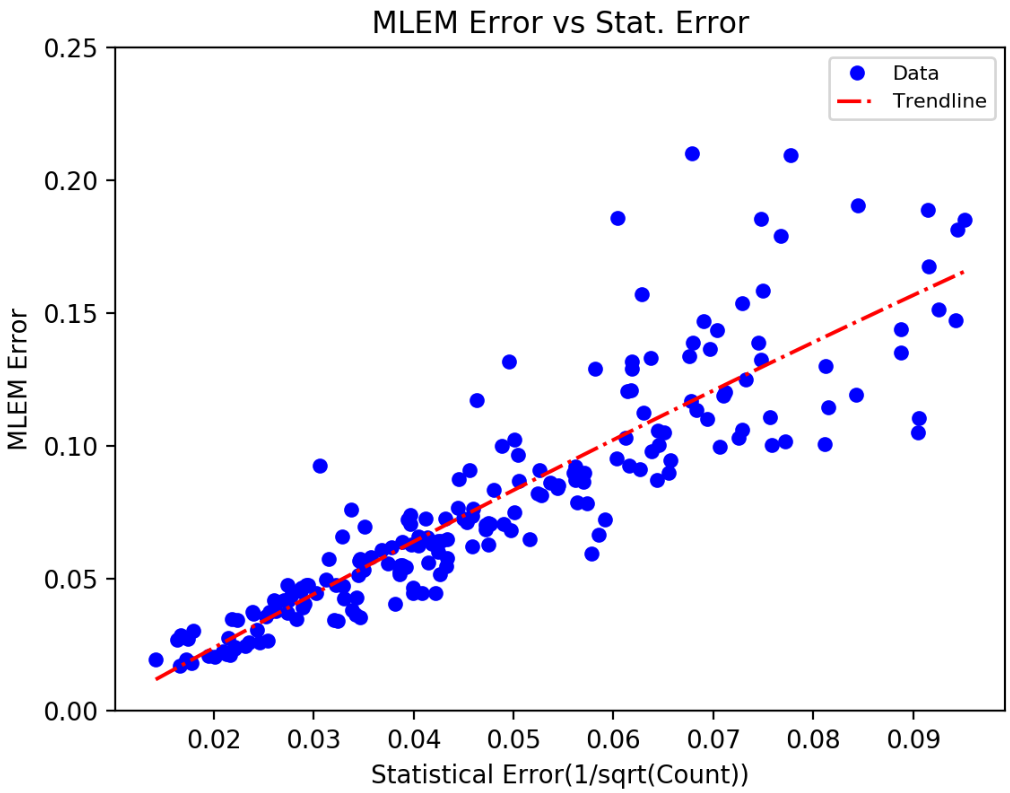


Figure 4.7. Monte Carlo uncertainty vs Statistical uncertainty. The red dashed line is a linear trendline of the data.

sections.

4.4.1 ^{10}B$(\alpha,\alpha_1\gamma)^{10}$B and ^{10}B$(\alpha,p_{123}\gamma)^{13}$C Channels

As described in Sec. 4.1, the deuterated liquid scintillator was paired with an HPGe detector to detect secondary γ-rays from reactions that can populate excited states in the final nucleus. Over the energy range of the present measurement these reactions are ^{10}B$(\alpha,\alpha_1\gamma)^{10}$B, ^{10}B$(\alpha,p_1\gamma)^{13}$C, ^{10}B$(\alpha,p_2\gamma)^{13}$C, and ^{10}B$(\alpha,p_3\gamma)^{13}$C corresponding to γ-ray decays to the ground states of the corresponding nuclei of E_γ = 0.718380(11), 3.089443(20), 3.684507(19), and 3.853807(19) MeV.

The efficiency of the HPGe detector was determined with standard sources ^{60}Co,

^{137}Cs as well as the ^{27}Al$(p,\gamma)^{28}$Si reaction at the well known narrow resonance at $E_p = 992$ keV [6]. The detector was at a distance where geometric effects and summing could be neglected compared to the precision of the measurements.

Prominent γ-ray lines were observed in the the HPGe spectrum at $E_\gamma = 3.089443(20)$, 3.684507(19), and 3.853807(19) MeV, while the line at $E_\gamma = 0.718380(11)$ MeV was only observed weakly at a few energies. Therefore, the cross sections for the ^{10}B$(\alpha, p_1\gamma)^{13}$C, ^{10}B$(\alpha, p_2\gamma)^{13}$C, and ^{10}B$(\alpha, p_3\gamma)^{13}$C were calculated while that of the ^{10}B$(\alpha, \alpha_1\gamma)^{10}$B reaction was neglected.

In order to calculate the ^{10}B$(\alpha, p_1\gamma)^{13}$C, ^{10}B$(\alpha, p_2\gamma)^{13}$C, and ^{10}B$(\alpha, p_3\gamma)^{13}$C cross sections, the γ-ray decay branching ratios [108] must be considered. In addition, since the angular distributions of the secondary cascade γ-rays are unknown, the differential cross sections measured at 130° are assumed to be proportional to the angle integrated cross section. This should be a reasonable approximation since secondary γ-ray angular distributions are symmetric about 90° and the second order Legendre polynomial is small at this angle of observation. Under these approximations, the branching ratios can be simply multiplied by the efficiency corrected yields and either added or subtracted to the yields of the different transitions to correct for feeding in or out. The yield curves of ^{10}B$(\alpha, p_{123}\gamma)^{13}$C reactions are shown in Fig. 4.19.

4.4.2 Target Degradation

The number of atoms in the target was determined using cold neutron depth profiling with the neutron standard ^{10}B$(n,\alpha)^7$Li reaction [35, 68]. Upon capturing a thermal neutron, an alpha particle with kinetic energy of 1472 keV and a recoil particle with kinetic energy of 840 keV are emitted. The product particles loose energy as they travel through the target material. The energy loss can be converted

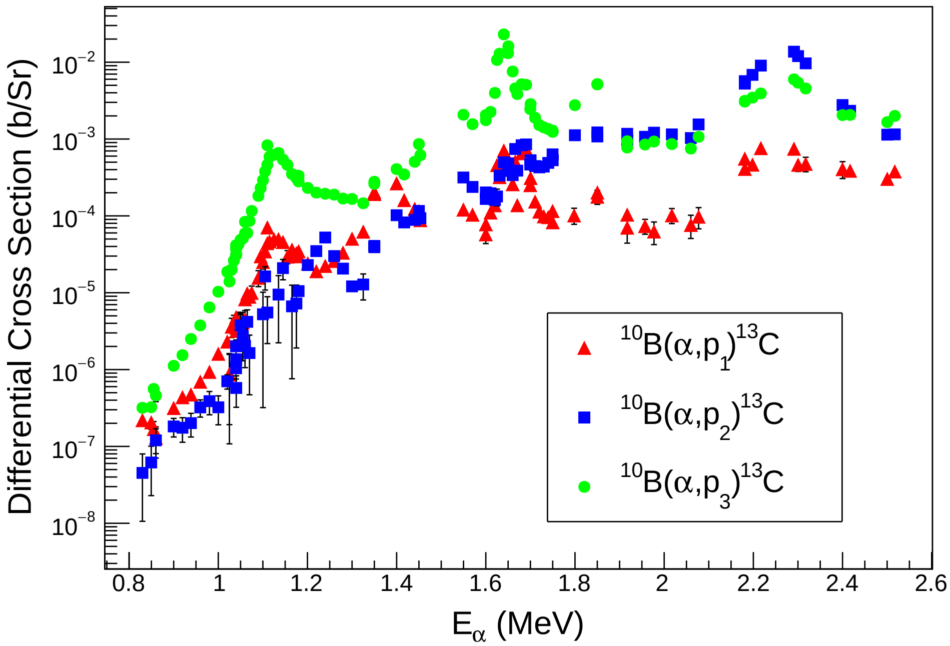


Figure 4.8. (Color Online) Differential cross sections of the present measurements. $^{10}\mathrm{B}(\alpha, p_1\gamma)^{13}\mathrm{C}$ (red triangles), $^{10}\mathrm{B}(\alpha, p_2\gamma)^{13}\mathrm{C}$ (blue squares) and $^{10}\mathrm{B}(\alpha, p_3\gamma)^{13}\mathrm{C}$ (green circles) at $\theta_{\mathrm{lab}} = 130°$.

to target depth by using Eq. 4.4

$$x = \int_{E(x)}^{E_0} dE/S(E), \tag{4.4}$$

where x is the path length traveled by the particle, $S(E)$ is the stopping power of the target material, E_0 and $E(x)$ are the initial and final kinetic energy of the particle respectively.

A photo of the target is shown in Fig. 4.9. There are small blisters on the target surface, which are a result of the α-beam bombardment on the target leading to the build up of helium bubbles in the Tantalum backing. This is one cause of target degradation. In the experiments at NIST, six points on the target surface are measured (see Fig. 4.9), three on the smooth area and three on the blistering area.

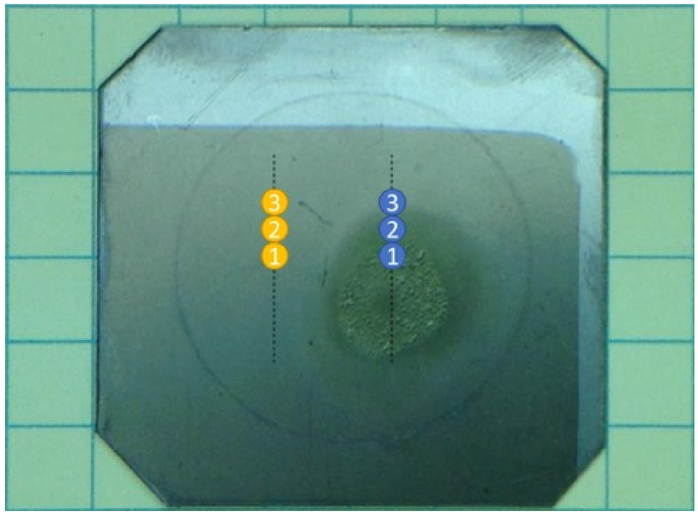
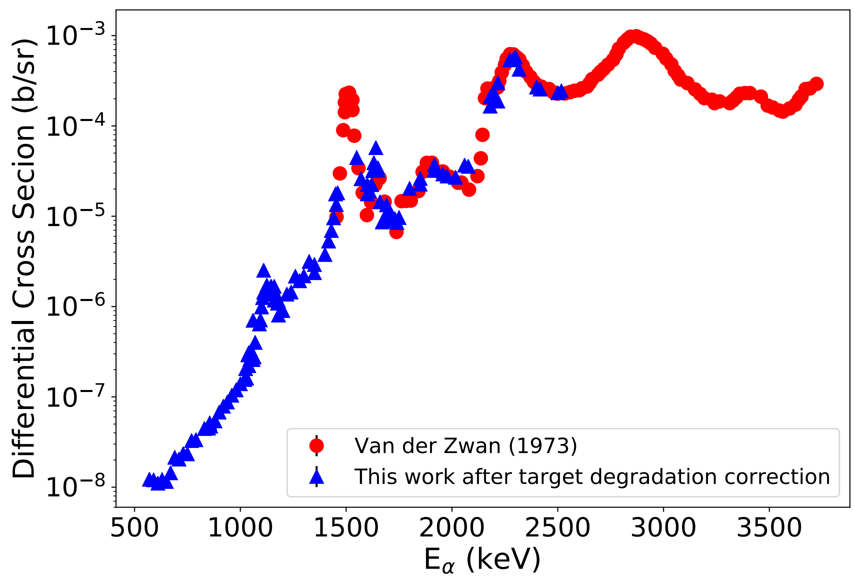


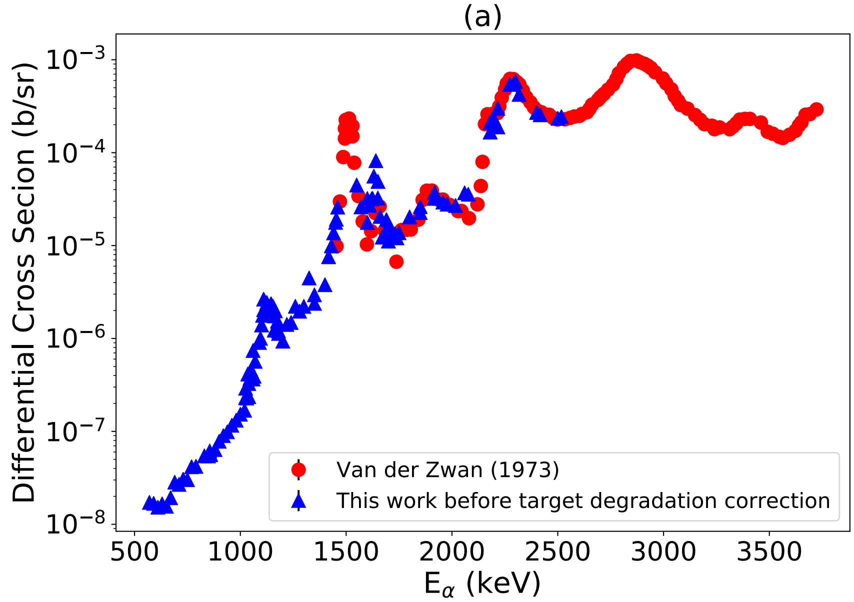
Figure 4.9. The blistering observed on the Boron target with Tantalum backing. The blistered area is visible as the darker colored region while the lighter colored and smooth region remains undamaged. The thickness of six points marked as 1, 2, 3 in both yellow and blue are measured at NIST.

These measurements revealed that the smooth area on the target has a homogeneous thickness of 0.703(9) μg/cm^2. The blistering area varied in thickness by $\pm$15% with a mean value of 0.492(3) μg/cm^2. Under the beam bombardment, the target is deteriorated by 30%. Thus, a model of the degradation as a function of the total amount of beam delivered to target is needed to correct the yield curve. The experiment consisted of 4 energy range sections, 1st run from 2.517 MeV to 1.55 MeV, 2nd run from 2.5 MeV to 0.57 MeV, 3rd run from 1.18 MeV to 1.03 MeV and 4th from 1.75 MeV to 1.46 MeV. Most of the damage occurred to the target for those runs where the Q was by far the largest and beam energy the lowest.

Fig. 4.10 shows the excitation curve before and after target degradation correction, with a comparison with the measurement from Van der Zwan and Geiger [101]. After applying the target degradation correction, the higher point on top of the resonance at about 1100 keV is now a lot closer for the low energy scan.



(a)



(b)

Figure 4.10. (Color Online) Comparison between the present measurements (blue triangle) before and after target degradation correction and those of Van der Zwan and Geiger [101] (red circle) at $\theta_{lab} = 0°$ at low energy.

4.4.3 Differential Cross Section

The energy loss of the beam through the target was calculated as 3.2 keV at E_α = 570 keV and 2.4 keV at E_α = 2000 keV. After the yields were corrected for target degradation effect, differential cross sections $\frac{d\sigma}{d\Omega}$ were calculated using Eq. 3.3. A comparison of the present $^{10}\mathrm{B}(\alpha,n)^{13}\mathrm{N}$ differential cross section data at $\theta_{\mathrm{lab}} = 0°$ with that of Van der Zwan and Geiger [101] is shown in Fig. 4.10 (b), where good agreement is observed over the overlapping energy range. In order to highlight the low energy nuclear structure, the experimental data are converted to differential S-factor (defined in Eq. 1.6), as shown in Fig. 4.11.

The increasing slope in the S-factor at low energies suggests the presence of a broad resonance at lower energy. The $^{10}\mathrm{B}(\alpha,p_1\gamma)^{13}\mathrm{C}$, $^{10}\mathrm{B}(\alpha,p_2\gamma)^{13}\mathrm{C}$, and $^{10}\mathrm{B}(\alpha,p_3\gamma)^{13}\mathrm{C}$ differential S-factors at $\theta_{\mathrm{lab}} = 130°$ are shown in Fig. 4.19.

4.5 High Energy Measurement Results

This section describes the second measurement of the $^{10}B(\alpha,n)^{13}N$ reaction. The results of this measurement include calibrating accelerator energy, extracting differential cross sections, performing Legendre Polynomial fit to angular distributions, obtaining the total cross sections, and calculating the thick target yields.

4.5.1 Accelerator Energy Calibration

To determine the constant k of the analyzing magnet(see Sec. 2.1), two well-known resonances at $E_\alpha = 1035$ keV and 1336 keV of the $^{13}\mathrm{C}(\alpha,\mathrm{n})^{16}\mathrm{O}$ are scanned before the measurement of the $^{10}B(\alpha,n)^{13}N$ reaction. The resulting excitation curve is shown in Fig. 4.12.

According to Eq. 2.2, the relationship between the magnetic field and k is linear. Besides these two energy calibration points from the resonances of the $^{13}\mathrm{C}(\alpha,\mathrm{n})^{16}\mathrm{O}$

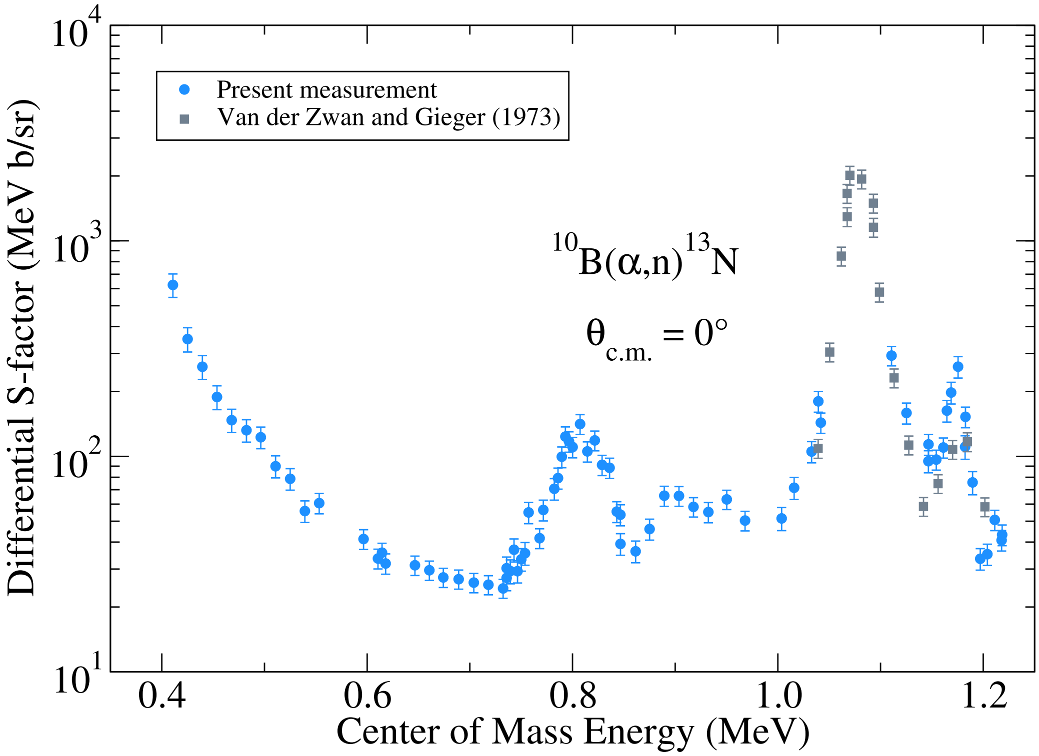


Figure 4.11. (Color Online) Comparison between the differential S-factors of the present measurements (blue circles) and those of Van der Zwan and Geiger [101] (gray squares) at $\theta_{\text{lab}} = 0°$.

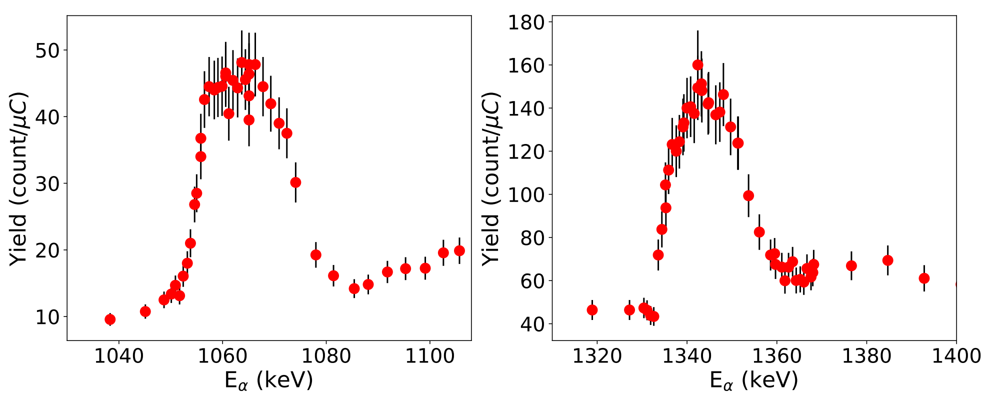


Figure 4.12. Yield of $E_\alpha = 1035$ keV (a) and $E_\alpha = 1336$ keV (b) resonances.

reaction, Eq. 2.2 requires the line to go through the origin. A least square fit to these points resulted in a calibration constant of the analyzing magnet with $k = 45.451(19)$, which results in beam energy variations of ≈1 keV in the energy range covered in this measurement.

4.5.2 Differential and Total Cross Section

As discussed in Sec. 4.1, the effective target thickness was found to vary by up to ±10%. Therefore, to obtain a more accurate absolute cross section, the present angular distributions were normalized to the differential cross section measurements of Van der Zwan and Geiger [101] at the common angle of 90°, where the expected variation is ≈5%. As the angular distributions were measured with a fixed monitor detector, they could be determined to a relative accurately of ≈10%. A thin target approximation was used for the calculation of the cross section as it varies slowly with energy compared to the energy losses through the target (10 to 17 keV). Therefore the beam energy has been taken as the energy of measurement (i.e. no target energy loss has been applied to the energies quoted here).

A comparison of the present angular distribution measurement at $E_\alpha = 4.63$ MeV is made to that of Van der Zwan and Geiger [101] in Fig. 4.13. This is one of the few energies where Van der Zwan and Geiger [101] made measurements at several additional angles, providing a good comparison with the present measurements. Over the remainder of the region of interest, Van der Zwan and Geiger [101] made extensive measurements as a function of energy but only at three angles. The complicated angular distributions that they observed at a few sample energies indicated that sampling at three angles was insufficient for an accurate determination of the angle integrated cross section in that work. The present measurements thus expand on the work of Van der Zwan and Geiger [101] by sampling twelve-point angular distributions at thirty-four energies between 2.2 and 4.9 MeV, allowing the angle integrated cross section to now be mapped over the full energy range of interest. Fig. 4.14 illustrates the complexity of the angular distributions and the need for these additional measurements. The differential cross sections for this work, after normalization to the 90° data of Van der Zwan and Geiger [101], are given in Appendix. D.3, D.4, and D.4.

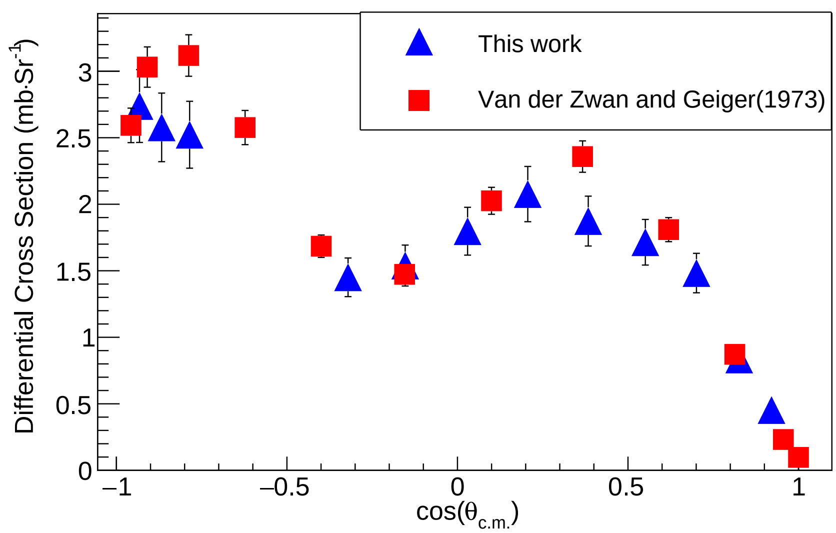


Figure 4.13. (Color online) Comparison of the angular distribution measurements at $E_\alpha = 4.63$ MeV from the present measurement (blue triangles) with those of Van der Zwan and Geiger [101] (red squares). Only statistical uncertainties are shown.

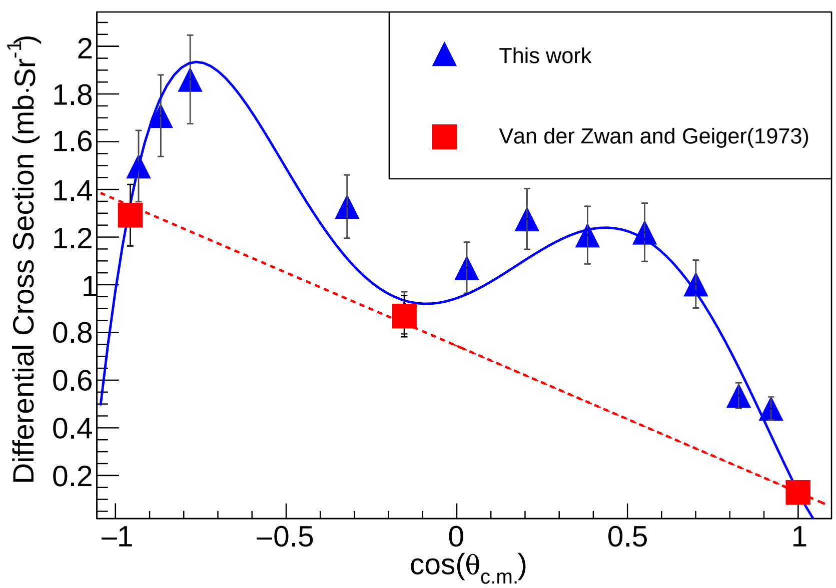


Figure 4.14. (Color online) Comparison of Legendre polynomial fits to angular distributions from the present work (blue triangles) to that of Van der Zwan and Geiger [101] (red squares) at $E_\alpha = 4.768$ MeV. The present measurements at twelve angles reveal a significantly more complex angular distribution than inferred from the three angle measurements of Van der Zwan and Geiger [101]. To illustrate, a first-order Legendre polynomial (dotted red line) is sufficient to fit the angular distribution of Van der Zwan and Geiger [101], while a fifth-order Legendre polynomial (blue solid line) is required to match the present data.

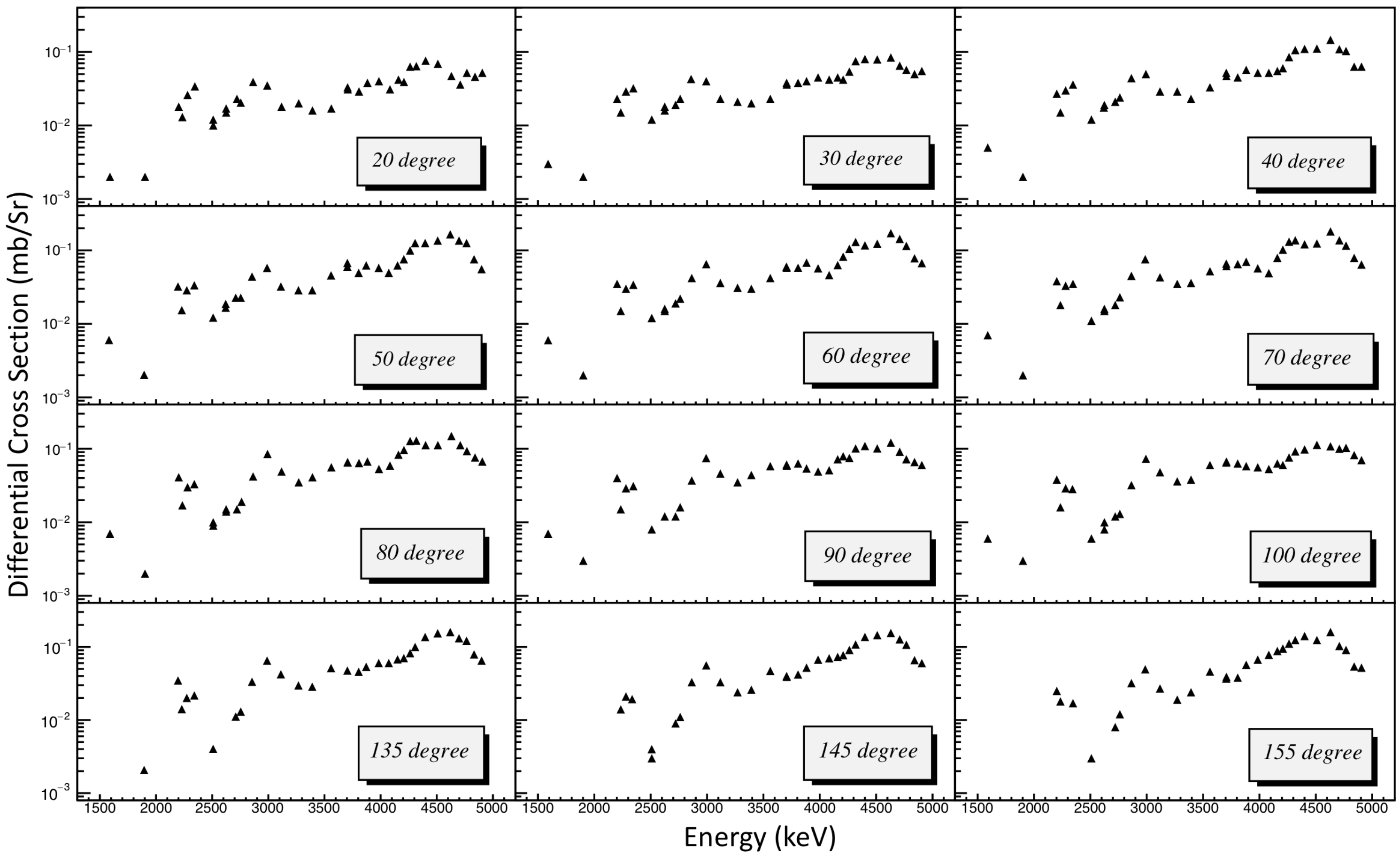


Figure 4.15. Differential cross section at 12 angles, covering 20°, 30°, 40°, 50°, 60°, 70°, 80°, 90°, 100°, 135°, 145°, and 155°.

To determine the angle-integrated cross section for the $^{10}\mathrm{B}(\alpha, n_0)^{13}\mathrm{N}$ reaction, the differential cross sections were fit with a Legendre polynomial expansion in the center-of-mass reference frame. The order of the polynomial fit was determined by χ^2 statistics tests. The differential cross section can be written as a partial wave expansion of the general form

$$\frac{d\sigma}{d\Omega}(\theta_{\text{c.m.}}) = \sum_{L=0}^{L_{\max}} a_L P_L \cos(\theta_{\text{c.m.}}), \tag{4.5}$$

where the parameters a_L were determined by performing a χ^2 fit to the experimental data at each energy and $L_{\max}$, the highest order Legendre polynomial of the fit, was determined using a p-value test. As in Van der Zwan and Geiger [101], the measured differential cross sections were observed to be both strongly varying with angle and highly asymmetric about $\theta_{\text{c.m.}} = 90°$ in the center-of-mass frame of reference. This indicates strong interference between broad overlapping resonances, which is reflected in the large odd-order angular distribution coefficients from the Legendre polynomial fits. Legendre polynomials of up to order $L = 5$ were necessary to describe some of the higher energy angular distributions, which is consistent with the previous findings of Van der Zwan and Geiger [101]. The angular distribution coefficients are given in Table 4.4, where the angle integrated cross section has been calculated as $\sigma = 4\pi a_0$.

For the angle integrated cross sections, the significant sources of uncertainty are from the Legendre polynomial fit (≈3%), and the normalization to the differential data of Van der Zwan and Geiger [101] (≈5% statistical, ≈15% systematic). In Van der Zwan and Geiger [101], the statistical uncertainties are not explicitly given but are quoted as being indicated by "the scatter of the measured points." From Fig. 2 of Van der Zwan and Geiger [101], over the relatively flat cross section around $E_\alpha \approx 3.75$ MeV, the statistical uncertainty is estimated to be ≈5%, which is also similar to the uncertainties that are given explicitly in Fig. 5 for the angular distributions of that work.

Measurements were made considering the previous data of Van der Zwan and Geiger [101], where excitation curves were mapped at three angles in very small energy steps. These data indicate a cross section that is dominated by resonances with widths of a few hundred keV. Therefore, the present measurements have been made with larger energy steps than that of Van der Zwan and Geiger [101], but still with a sufficient number to map the resonance structures. A cubic spline interpolation was then made for each of the angular distribution coefficients at each measured energy in order to obtain the cross section at an arbitrary energy throughout the region. The interpolation was used for the numerical integration of the cross section to compare with thick-target yields as will be described in Sec. 4.5.3.

A comparison to the data of Gibbons and Macklin [46] and Prior et al. [84] is shown in Fig. 4.16. The comparison is not one-to-one because the data of Gibbons and Macklin [46] and Prior et al. [84] are total cross section measurements, while those of the present measurement are of the ground state cross section only. Below $E_\alpha \approx 3.4$ MeV, the present measurements and those of Prior et al. [84] are in good agreement. At higher energies the measurements slowly diverge because below $E_\alpha \approx 3.4$ MeV the ground state cross section dominates the total cross section, while at higher energies the excited state cross sections increase rapidly to become dominant. This is supported by the general trend of the differential cross section measurements of Van der Zwan and Geiger [101] for the excited state transitions, which show a rapid increase in the n_2 and n_3 cross sections above $E_\alpha \approx 3.4$ MeV. The data of Gibbons and Macklin [46] are considerably larger in cross section than both the present data and that of Prior et al. [84] and also show a different energy dependence.

TABLE 4.4. ANGULAR DISTRIBUTION COEFFICIENTS

E_α(keV)	$\sigma_0 = 4\pi a_0$	a_1	a_2	a_3	a_4	a_5
1590	$7.3(4)\times10^{-1}$		$-4.6(3)\times10^{-2}$			
1902.0	$3.7(2)\times10^{-1}$	$-1.4(3)\times10^{-2}$				
2202	$5.4(3)\times10^{0}$	$-1.4(3)\times10^{-1}$	$-1.5(4)\times10^{-1}$			
2234.7	$2.5(1)\times10^{0}$	$-6.3(14)\times10^{-2}$				
2281	$4.3(2)\times10^{0}$		$-7.0(33)\times10^{-2}$			
2346	$4.7(2)\times10^{0}$		$-7.4(32)\times10^{-2}$			
2509	$1.3(1)\times10^{0}$	$3.8(7)\times10^{-2}$	$-2.0(9)\times10^{-2}$			
2510	$1.3(1)\times10^{0}$	$3.9(7)\times10^{-2}$	$-2.6(9)\times10^{-2}$	$-2.4(15)\times10^{-2}$		
2625	$1.5(2)\times10^{0}$	$1.5(4)\times10^{-1}$	$-1.1(4)\times10^{-1}$			
2625	$1.7(2)\times10^{0}$	$7.8(39)\times10^{-2}$	$-6.1(38)\times10^{-2}$			
2721	$2.4(1)\times10^{0}$	$4.1(11)\times10^{-2}$				$4.0(23)\times10^{-2}$
2762	$2.7(1)\times10^{0}$	$3.3(15)\times10^{-2}$		$-5.5(28)\times10^{-2}$		
2864	$6.0(3)\times10^{0}$	$-4.6(35)\times10^{-2}$		$-7.9(61)\times10^{-2}$		
2993	$1.0(0)\times10^{1}$	$-3.1(5)\times10^{-1}$	$-2.6(7)\times10^{-1}$			
3118	$6.4(3)\times10^{0}$	$-2.3(3)\times10^{-1}$	$-2.1(3)\times10^{-1}$	$1.3(4)\times10^{-1}$		
3271	$5.0(2)\times10^{0}$	$-1.3(3)\times10^{-1}$	$-1.5(3)\times10^{-1}$	$1.0(4)\times10^{-1}$	$-7.8(39)\times10^{-2}$	
3393	$5.2(2)\times10^{0}$	$-1.7(3)\times10^{-1}$	$-1.6(3)\times10^{-1}$	$8.5(34)\times10^{-2}$		
3560	$7.6(3)\times10^{0}$	$-3.2(3)\times10^{-1}$	$-1.8(3)\times10^{-1}$			
3706	$8.7(4)\times10^{0}$	$-1.8(3)\times10^{-1}$	$-3.1(4)\times10^{-1}$			
3707	$8.5(4)\times10^{0}$	$-1.9(3)\times10^{-1}$	$-2.8(4)\times10^{-1}$			
3807	$8.4(4)\times10^{0}$	$-1.9(3)\times10^{-1}$	$-2.2(4)\times10^{-1}$			
3883	$9.1(4)\times10^{0}$	$-2.5(5)\times10^{-1}$		$-1.9(6)\times10^{-1}$		

TABLE 4.4 CONTINUED

E_α(keV)	$\sigma_0 = 4\pi a_0$	a_1	a_2	a_3	a_4	a_5
3985	8.9(4)$\times10^{0}$	-3.2(5)$\times10^{-1}$	1.3(5)$\times10^{-1}$	-2.0(6)$\times10^{-1}$		
4083	8.8(4)$\times10^{0}$	-4.1(5)$\times10^{-1}$	1.4(6)$\times10^{-1}$	-1.7(6)$\times10^{-1}$		
4159	1.1(0)$\times10^{1}$	-4.0(6)$\times10^{-1}$		-2.8(11)$\times10^{-1}$	2.0(10)$\times10^{-1}$	
4208	1.1(0)$\times10^{1}$	-3.8(7)$\times10^{-1}$		-5.2(12)$\times10^{-1}$	3.4(10)$\times10^{-1}$	
4263	1.4(1)$\times10^{1}$	-3.6(8)$\times10^{-1}$		-6.6(15)$\times10^{-1}$	3.6(13)$\times10^{-1}$	
4318	1.7(1)$\times10^{1}$	-4.1(10)$\times10^{-1}$		-7.2(17)$\times10^{-1}$	3.8(16)$\times10^{-1}$	
4400	1.8(1)$\times10^{1}$	-6.9(11)$\times10^{-1}$	2.3(11)$\times10^{-1}$	-3.3(13)$\times10^{-1}$		
4510	1.9(1)$\times10^{1}$	-6.5(10)$\times10^{-1}$		-3.1(11)$\times10^{-1}$		
4630	2.3(1)$\times10^{1}$	-8.5(9)$\times10^{-1}$		-9.4(16)$\times10^{-1}$	-5.1(19)$\times10^{-1}$	5.6(13)$\times10^{-1}$
4709	1.7(1)$\times10^{1}$	-6.6(9)$\times10^{-1}$		-6.1(13)$\times10^{-1}$		2.5(19)$\times10^{-1}$
4768	1.6(1)$\times10^{1}$	-5.7(10)$\times10^{-1}$		-2.5(15)$\times10^{-1}$	-7.2(17)$\times10^{-1}$	4.2(13)$\times10^{-1}$
4840	1.1(1)$\times10^{1}$	-3.9(7)$\times10^{-1}$		-1.7(10)$\times10^{-1}$		
4905	1.0(0)$\times10^{1}$	-3.4(4)$\times10^{-1}$			-3.8(5)$\times10^{-1}$	
4909	9.7(4)$\times10^{0}$	-3.6(4)$\times10^{-1}$			-3.4(4)$\times10^{-1}$	

Notes: Angular distribution coefficients for fits to center-of-mass differential cross sections.

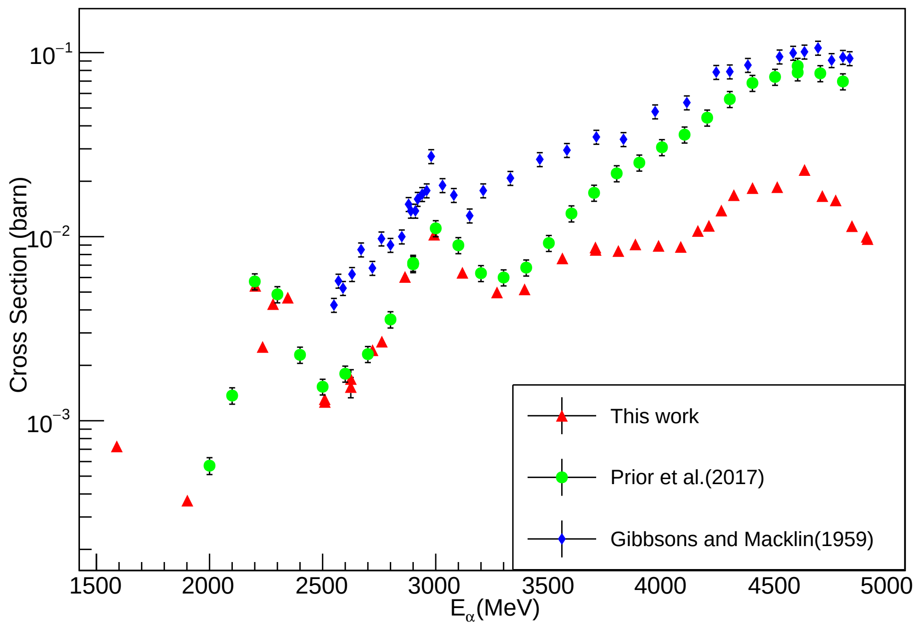


Figure 4.16. Comparison of the ^{10}B$(\alpha, n)^{13}$N angle integrated cross section measurements. Those of [46] and [84] are total cross section measurements (sum over all neutron de-excitations), while the present data are of just the ground state transition.

4.5.3 Thick Target Yield

The cross sections of this work can be compared to thick-target yield measurements using the methods described in Roughton et al. [91], for example. The thick target yield can be calculated from the cross section σ and the stopping power cross section $dE/dx(\rho x)$ by

$$Y(E) = \frac{N_A}{A_T} \int_0^E \frac{\sigma(E')}{dE/dx(\rho x)(E')} dE', \tag{4.6}$$

where E is the energy of the thick-target yield measurement and E' is the energy of the thin target cross section measurements. Stopping power cross sections and their uncertainties are taken from SRIM [112]. The uncertainties on the thick target yields are ≈8%, resulting from the uncertainties in the angle integrated cross section ≈6% and the stopping power ≈5%.

There are two thick target yield measurements that have been presented in the literature, those of Bair and Gomez del Campo [12] and Roughton et al. [92]. Those of Roughton et al. [92] were made using the activation method. As stated earlier, this method is only sensitive to the ground state transition of the ^{10}B$(\alpha, n)^{13}$N reaction, since the excited states are all proton unbound. Bair and Gomez del Campo [12], on the other hand, measured the thick target yield by the direct detection of neutrons, making their measurement sensitive to de-excitations to all final states. A comparison of the different thick-target yields is shown in Fig. 4.17. The large deviation of the two measurement methods is roughly consistent with the measurement of the total cross section versus only the ground state component, as discussed in Sec. 4.5.2 for the thin-target cross section measurements.

For comparison, the thick-target yields have also been calculated using the cross section data of Prior et al. [84]. These thick-target yields can be directly comparable with those of Bair and Gomez del Campo [12], and show a larger yield. This may be because the yields of Bair and Gomez del Campo [12] are not corrected for the changing neutron efficiency at higher energies. The thick-target yields calculated from the present data are in good agreement with those measured directly by Roughton et al. [92].

4.5.4 R-matrix Analysis

The low energy cross sections of the ^{10}B$(\alpha, n)^{13}$N reaction has been analyzed using the phenomenological R-matrix approach [69, 33] using the code `AZURE2` [10, 100].

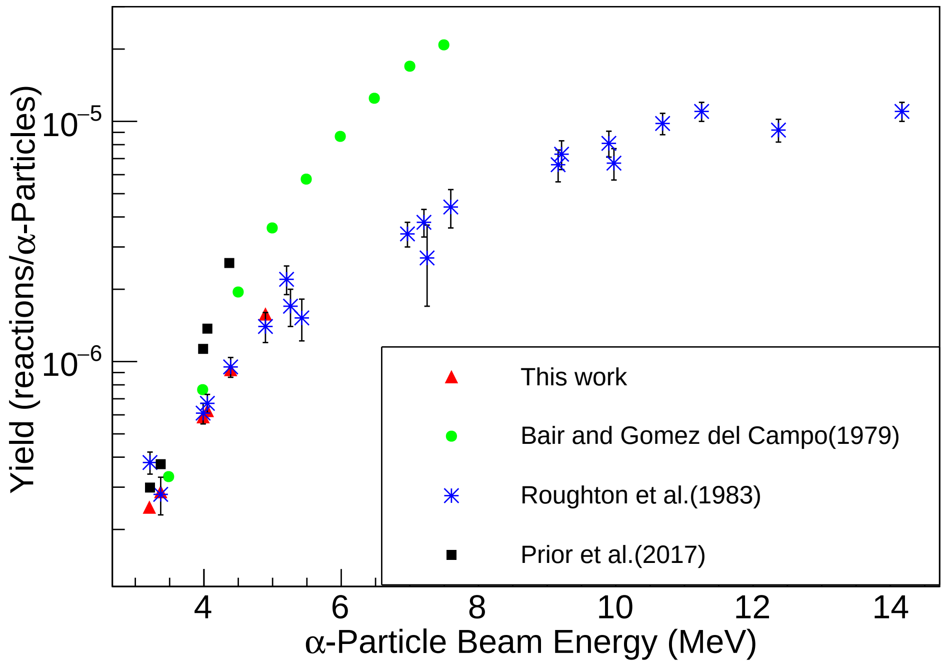


Figure 4.17. (Color online) Comparison of the thick target yields measured by Roughton *et al.* [92] (blue stars) and Bair and Gomez del Campo [12] (green circles) with those calculated from the cross section measurements of this work (red triangles). Note that the data of Roughton *et al.* [92] represent only the yield from ground state transition of the $^{10}\mathrm{B}(\alpha, n_0)^{13}\mathrm{N}$ reaction while that of Bair and Gomez del Campo [12] represents the yield from all transitions. Further, the yields of Bair and Gomez del Campo [12] are not corrected for the difference in the detection efficiency of neutrons emitted to different final states.

The alternate parameterization of Brune [21] has been used in order to work directly with physical parameters and to eliminate the need for boundary conditions. The physical constants used for the analysis of the ^{14}N system are given in Table 4.5.

The ^{10}B+α reactions populate the ^{14}N compound system (shown in Fig. 4.18) above the α-particle ($S_\alpha = 11.61211(1)$), neutron ($S_n = 10.55338(27)$ MeV), proton ($S_p = 7.55056(1)$ MeV), and deuteron ($S_d = 10.2723080(7)$ MeV) separation energies [55, 106]. In addition, the α-particle and proton decays can populate excited states in ^{10}B ($E_x = 0.718380(11)$ MeV) and ^{13}C ($E_x = 3.089443(20)$, $3.684507(19)$, and $3.853807(19)$ MeV) respectively [5, 4]. All tolled, this allows for eight possible particle decay modes.

In addition to the complication of several decay modes, there are very few measurements of the low energy cross sections of the ^{10}B+α reactions. A very complete study was made by Shire et al. [97] but the actual data were only shown in a very simplistic plot. Van der Zwan and Geiger [101] made a study of the differential cross section at three angles of $\theta_{\text{lab}} = 0$, 90, and 160°, but their measurements do not extend below the strong resonance at $E_\alpha = 1.51$ MeV (see Fig. 4.10 (b)). Similarly, the ^{10}B$(\alpha, p_0)^{13}$C data of Chen et al. [25] also only extend just below the corresponding resonance in the ground state proton decay mode. Mo and Weller [77] has made the lowest energy study of the ^{10}B$(\alpha,\alpha)^{10}$B reaction, but the lowest reported resonances are well above the region of interest for the present analysis. No previous ^{10}B$(\alpha,\alpha_1)^{10}$B measurements have been reported over this energy region, but the present measurements have found that this channel is negligibly small over the entire energy range under analysis.

Because of the very limited data for the other possible decay modes, those of Shire et al. [97] have been digitized from Fig. 2 of that work. Because no actual data points are shown, only a line, points were digitized in approximately equal spacing. For the fitting, uncertainties of 20% were arbitrarily assigned. The inclusion of this

TABLE 4.5

PHYSICAL CONSTANTS AND R-MATRIX PARAMETERS FOR THE ^{14}N SYSTEM

Parameter	Value
a_c (^{10}B+α)	5
a_c (^{13}C+p)	5
a_c (^{13}N+n)	5
a_c (^{12}C+d)	5
M_p	1.00783
M_n	1.00866
M_α	4.00260
M_d	2.01410
$M_{^{10}\mathrm{B}}$	10.0129
$M_{^{13}\mathrm{C}}$	13.0034
$M_{^{13}\mathrm{N}}$	13.0057
$M_{^{12}\mathrm{C}}$	12

Notes: Channel radii are in units of fm and masses, taken from Wang et al. [105], are in atomic masses.

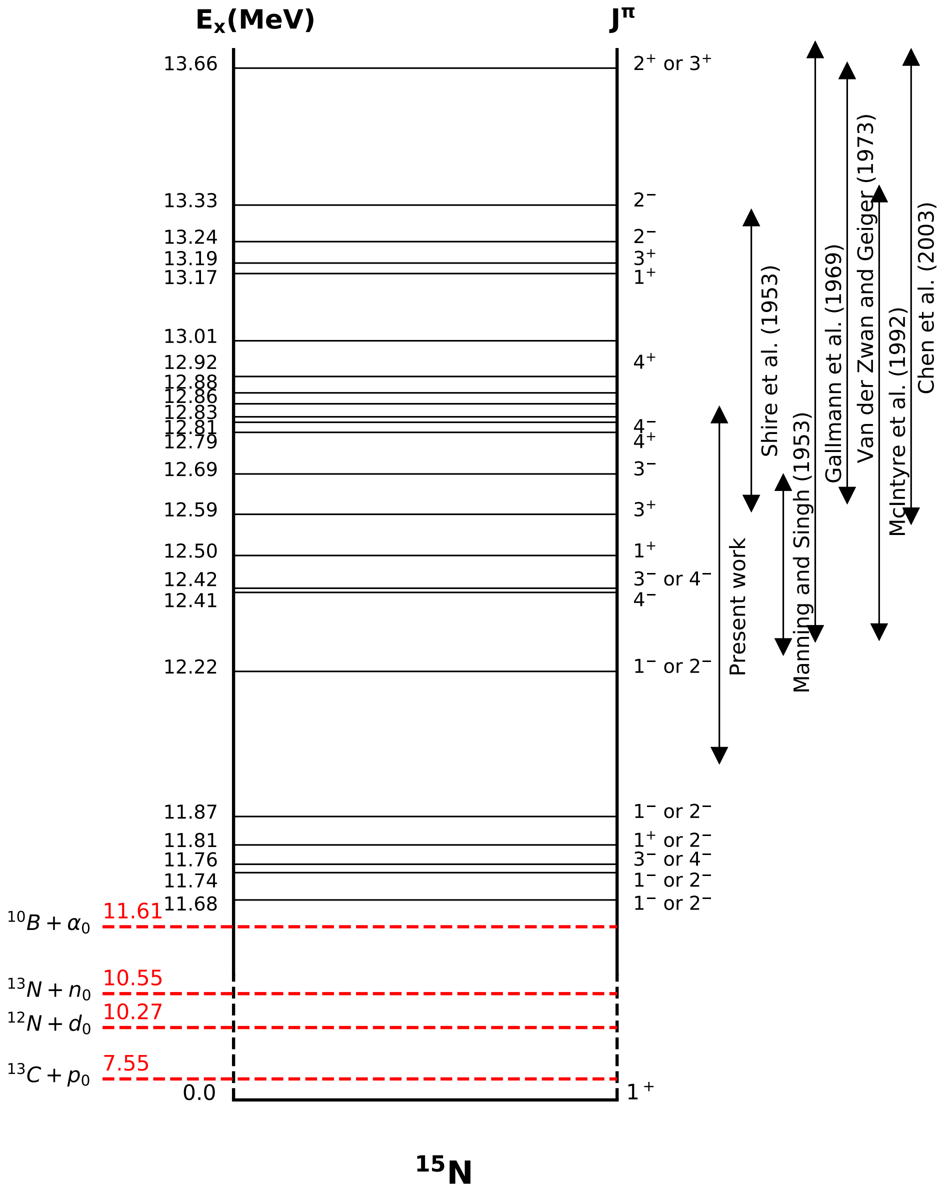


Figure 4.18. (Color Online) Level diagram of the ^{14}N system with level information pertinent to the present analysis.

data added significant constraint to the R-matrix analysis by providing data over two additional lower energy resonance in the ^{10}B$(\alpha, p_0)^{13}$C reaction and providing the only data for the ^{10}B$(\alpha, d)^{12}$C reaction as shown in Fig. 4.19.

A comparison of the present data for the ^{10}B+α population mode with those in the overlapping range from the literature [97, 25, 101] are shown in Fig. 4.19. The present measurement is shown in blue circles with those measured previous measurements from Shire et al. [97] (green stars), Van der Zwan and Geiger [101] in grey upward triangles, Chen et al. [25] in brown squares, and McIntyre et al. [74] in khaki downward triangles at low energy. The red solid curve represents the R-matrix fit described in the text. While there are few data sets, the combination of the present measurements and the literature data provide constraint for every decay mode but α_1 for each level used in the calculation. The data are plotted as differential astrophysical S-factor for clearer presentation. However, the data are usually limited to measurements at a single angle, so there is very limited constraint on the spin-parities of the states. Those values reported in the compilation have been utilized.

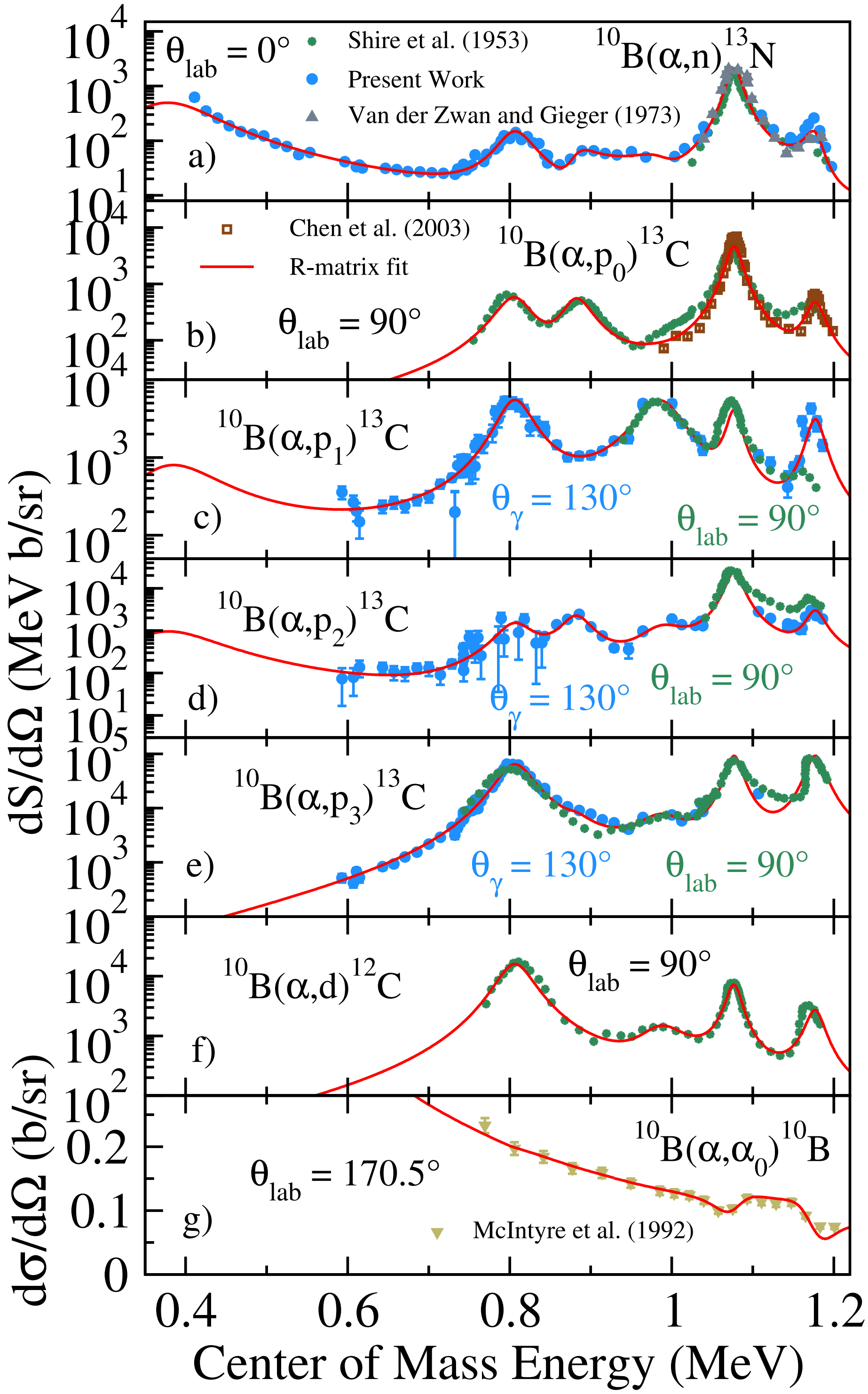


Figure 4.19. (Color Online) Comparison of the present ^{10}B+α measurements with previous measurements.

TABLE 4.6. R-MATRIX PARAMETERS FOR THE ^{14}N SYSTEM

Jpi	Ex	l	s	α	n	a1	p0	p1	p2	p3	d
1+	11.050	2	3	2.50×10^{2}fm$^{-1/2}$							
		1	0		2.20×10^{1}						
		1	1				4.40×10^{1}				
		0	1					2.20×10^{1}			
		1	1						3.53×10^{0} m$^{-1/2}$		
		0	1								2.90×10^{1}
2-	11.290	2	0		3.10×10^{1}						
		2	1				1.54×10^{2}				
		1	1					1.10×10^{1}			
		1	1								7.90×10^{1}
1+	11.390	1	0		4.00×10^{0}						
		1	0				9.00×10^{0}				
		0	1					5.00×10^{0}			
		2	1								1.50×10^{1}
3+	11.503	3	0		1.00×10^{0}						

TABLE 4.6 CONTINUED

Jpi	Ex	l	s	α	n	a1	p0	p1	p2	p3	d
		3	0				1.00×10^{0}				
		2	1					1.00×10^{0}			
		2	1								4.00×10^{0}
1+	11.800	0	1								5.00×10^{1}
2-	12.000	1	3	8.95×10^{-6}							
		2	0		4.02×10^{1}						
		1	1					1.28×10^{2}			
1-	12.200	3	3	4.41×10^{-5}							
		0	1				1.00×10^{2}				
		0	1						3.35×10^{1}		
		2	2						3.00×10^{1}		
		1	1								1.00×10^{0}
4-	12.417	1	3	2.89×10^{-1}							
		4	0		2.80×10^{-2}						
		4	0				4.55×10^{-1}				

TABLE 4.6 CONTINUED

Jpi	Ex	l	s	α	n	a1	p0	p1	p2	p3	d
		3	1					2.92×10^{0}			
		2	2						9.87×10^{-9}		
		1	3							3.41×10^{1}	
		3	1								9.26×10^{-7}
1+	12.493	2	3	2.52×10^{-1}							
		1	0		1.42×10^{-1}						
		0	1					2.73×10^{-2}			
		2	1					2.40×10^{0}			
		1	1						2.24×10^{1}		
		2	2							4.55×10^{1}	
3+	12.597	0	3	9.19×10^{0}							
		3	0		1.09×10^{-3}						
		2	1					1.10×10^{0}			
		1	2						1.62×10^{-1}		
		0	3							7.85×10^{-1}	

TABLE 4.6 CONTINUED

Jpi	Ex	l	s	α	n	a1	p0	p1	p2	p3	d
		2	1								3.41×10^{1}
3-	12.691	1	3	9.46×10^{0}							
		2	1		9.69×10^{-1}						
		2	1				1.76×10^{-1}				
		3	0					1.82×10^{-1}			
		2	1						1.30×10^{0}		
		1	2							6.70×10^{0}	
4+	12.796	2	3	1.01×10^{1}							
		3	1		2.45×10^{-1}						
		3	1				1.27×10^{-1}				
		4	0					5.83×10^{-1}			
		3	1						8.75×10^{-1}		
		2	2							2.80×10^{1}	
4-	12.820	1	3	4.33×10^{0}							
		4	0				1.11×10^{-2}				

TABLE 4.6 CONTINUED

Jpi	Ex	l	s	α	n	a1	p0	p1	p2	p3	d
		3	1					1.70×10^{-1}			
		4	1						2.68×10^{-1}		
		3	2							6.66×10^{0}	
1+	13.166	2	3	4.00×10^{-2}							
		1	0		1.40×10^{1}						
2-	13.243	1	3	1.00×10^{0}							
		2	0		9.10×10^{1}						
2+	13.656	2	3	8.00×10^{0}							
		1	1		9.00×10^{1}						
1+	13.740	2	3	1.00×10^{0}							
		1	0		1.80×10^{2}						

Notes: The resonance width are in unit of keV.

CHAPTER 5

CONCLUSION

In this work we measured the reactions ^{10,11}B(α,n)^{13,14}N covering different energy range with different techniques for different purposes.

The ^{11}B(α,n)^{14}N has been measured at α-particle beam energies E_α ranging from 520 to 2000 keV with a ^{3}He proportional counter. The present measurements largely confirm the previous data of [107] but improve on measurements over a low cross section region. A comprehensive R-matrix analysis of the reaction data that populate the ^{15}N system over a similar excitation energy range as the present data has been performed. Study of the low energy cross section will continue at the new CASPAR underground facility [88].

The ^{10}B(α,n)^{13}N reaction was measured twice with deuterated liquid scintillators that utilized unfolding algorithm to differentiate different neutron groups. The low energy measurements of the ^{10}B(α,n)^{13}N reaction extend to a previously unmeasured energy range $E_{\mathrm{c.m.}} < 1.0$ MeV (see Fig. 4.10). By performing neutron spectroscopy with better than 0.5 MeV energy resolution, the ground state portion of the cross section, which is the only component that produces ^{13}N, was separated from the excited states. A preliminary R-matrix analysis was performed as described in Sec. 4.5.4, but it should be emphasized that the limited amount of data available does not provide enough constraint for a unique solution. In addition to the previously observed strong resonance at $E_{\mathrm{c.m.}} \approx 1.08$ MeV, new features are observed. Two weak structure has been observed at $E_{\mathrm{c.m.}} \approx 0.89$ MeV and at $E_{\mathrm{c.m.}} \approx 0.806$ MeV. The S-factor in the $^{10}\mathrm{B}(\alpha,n)^{13}\mathrm{N}$ reaction rises significantly at low energies. In the present analysis, the rise in cross section is reproduced by the lowest energy level given in Table 4.6. Given the limited data, especially the limited angular distributions, the J^π of this state is not well constrained by the data.

The high energy measurements of the ^{10}B$(\alpha, n_0)^{13}$N reaction have been made over the energy range $E_\alpha = 2.2$ MeV to 4.9 MeV, which is the typical temperature conditions of a NIF shot to calculate the yield of ^{13}N. By using the same techniques as the low energy measurements, the angle integrated cross section was obtained for the first time. A large discrepancy in the measured cross section exists between the present results and Gibbons and Macklin [46], which were previously adopted in the HYDRA code for simulating ^{13}N yields from this reaction in a NIF implosion. Preliminary calculations demonstrated that the present measurements result in an approximately 85% reduction in the predicted ^{13}N yield from the ^{10}B$(\alpha, n_0)^{13}$N. Further, the improved uncertainty in the cross section of the present measurements increases the accuracy of this estimate. This demonstrates a clear need for an improvement in the accuracy of the cross section data used in NIF simulations.

APPENDIX A

MCNP INPUT FILE

```
c cell cards
c EJ−301D detector
200 1 −1.023 −200 −204
     *trcl=(0 0 −35.87 0 90 90 90 0 90 90 90 0) imp:n=1
201 9 −2.700 −201 #200 #203 #204
     *trcl=(0 0 −35.87 0 90 90 90 0 90 90 90 0) imp:n=1
202 9 −2.700 −202 #201 #200 #203 #204 imp:n=1
     *trcl=(0 0 −35.87 0 90 90 90 0 90 90 90 0) imp:n=1
203 4 −2.230 −203
     *trcl=(0 0 −35.87 0 90 90 90 0 90 90 90 0) imp:n=1
204 10 −0.001165 −200 204
     *trcl=(0 0 −35.87 0 90 90 90 0 90 90 90 0) imp:n=1
2 0 101          imp:n=0
3 0 −101 #200 #201 #202 #203 #204 103 #6 #7 #8 #9 #10 #11 #12 #13
    #14 #15 #17 123 124 125 126 127 128 129 130 131 #27
    133 134 135 136 137 138 140 imp:n=1
6 0 −109 110 −107 imp:n=1
7 11 −8.07 −106 #6 −109 110    imp:n=1
8 9 −2.700 −105 −109 108 #7 #6 #9 #10 imp:n=1
9 0 −109 110 −112 imp:n=1
10 11 −8.07 −111 #9 −109 110   imp:n=1
11 0 −116 −117 109 imp:n=1
12 9 −2.700 −115 #11 −117 109 imp:n=1
13 2 −1.800 −114 #12 −117 109 #11 imp:n=1
14 9 −2.700 −113 #13 −117 109 #12 #11 imp:n=1
```

```
15 7 -16.654 -118 #17 #11 #12 imp:n=1
c 15 7 -16.654 -118 imp:n=1
c 16 0 -120 -121 122 imp:n=1
17 9 -2.7000 -119 120 -121 122 imp:n=1
4 9 -2.7000 -103 104      imp:n=1
5 0 -104 #17  #15  imp:n=1
18 9 -2.7000 -123 imp:n=1
19 9 -0.797  -124 imp:n=1
20 9 -0.797 -125 imp:n=1
21 9 -0.797 -126 imp:n=1
22 9 -0.797 -127 imp:n=1
23 9 -1.009 -128 imp:n=1
24 9 -1.009 -129 imp:n=1
25 9 -1.009 -130 imp:n=1
26 9 -1.009 -131 imp:n=1
27 9 -1.058 -132 imp:n=1
29 9 -2.700 -133 imp:n=1
30 9 -2.700 -134 imp:n=1
31 9 -2.700 -135 imp:n=1
32 9 -2.700 -136 imp:n=1
33 9 -2.700 -137 138 imp:n=1
 34 9 -2.700 -138 139 imp:n=1
 35 0 -139 imp:n=1
 36 9 -2.700 -140 141 imp:n=1
 37 0 -141 imp:n=1

c Surface card
101 box -200 -200 -200 400 0 0 0 400 0 0 0 400
102 rcc 0 0 -28.25 0 0 -7.62 3.81
103 rcc 0 0 0 0 0 42.545 3
104 rcc 0 0 0 0 0 42.545 2.857
105 cz 3.175
```

```
106 c/z −0.714 0 0.556
107 c/z −0.714 0 0.476
108 pz −0.635
109 pz −0.16
110 pz −1.375
111 c/z 0.714 0 0.556
112 c/z 0.714 0 0.476
113 cz 3.175
114 cz 3.
115 cz 2.85
116 cz 1.5
117 pz 0.
118 box −1.5 −1.5 0. 3 0 0 0 3 0 0 0 0.091
119 cz 2.38125
120 kz −.85 1.33
121 pz 0.51
122 pz 0.0921
123 box −38.1 −33.178 −50.8 76.2 0 0 0 −0.9525 0 0 0 101.6
124 box −30.4 −34.1305 −31.4 7.62 0 0 0 −141.9225 0 0 0 7.62
125 box 30.4 −34.1305 −31.4 −7.62 0 0 0 −141.9225 0 0 0 7.62
126 box −30.4 −34.1305 45 7.62 0 0 0 −141.9225 0 0 0 −7.62
127 box 30.4 −34.1305 45 −7.62 0 0 0 −141.9225 0 0 0 −7.62
128 box −22.6 −92.55 −23.66 −3.81 0 0 0 −7.62 0 0 0 60.96
129 box −22.6 −92.55 37.4 45.2 0 0 0 −7.62 0 0 0 3.81
130 box −22.6 −92.55 −28 45.2 0 0 0 −7.62 0 0 0 3.81
131 box 22.6 −92.55 −23.66 3.81 0 0 0 −7.62 0 0 0 60.96
132 box −1.9 −15.24 −9 3.81 0 0 0 3.81 0 0 0 −60.325
133 rcc −14.605 −33.178 0  0 13.97 0 1
134 rcc 14.605 −33.178 0  0 13.97 0 1
135 rcc 0 −33.178 −14.605 0 13.97 0 1
136 rcc 0 −33.178 14.605 0 13.97 0 1
137 box −17.145 −19.208 0 17.145 0 17.145 0 0.635 0 17.145 0 −17.145
```

```
138 rcc 0 −19.843 0 0 4.445 0 7.62
139 rcc 0 −19.843 0 0 4.445 0 5.08
140 rcc 0 −10.953 0 0 0.9525 0 7.62
141 rcc 0 −10.953 0 0 0.9525 0 5.08
200 rcc 0 0 0 0 0 7.620 3.810
201 rcc 0 0 −0.152 0 0 8.300 3.950
202 rcc 0 0 6.148 0 0 2.000 5.010
203 rcc 0 0 7.620 0 0 0.318 3.810
204 py 3.286
mode n
c Materials
M1        1002.          0.4919
          1001.          0.0039
          6000.          0.5042
M2        1001.          0.009417
          6000.          0.280555
          9019.          0.710028
M4        5000        −0.040064
          8016        −0.539562
          11023       −0.028191
          13027       −0.011644
          14000       −0.377220
          19000       −0.003321
M6        32000.          1.000
M7        73000.          −1.00000
M8        6000           −0.000400
          14000          −0.005000
          15031          −0.000230
          16000          −0.000150
          24000          −0.190000
          25055          −0.010000
          26000          −0.701730
```

```
        28000              -0.092500
M9    12000              -0.010000
        13027              -0.972000
        14000              -0.006000
        22000              -0.000880
        24000              -0.001950
        25055              -0.000880
        26000              -0.004090
        29000              -0.002750
        30000              -0.001460
M10   7014               1.0000000
M11     26000          -0.000868
        29000          -0.665381
        30000          -0.325697
        50000          -0.002672
        82000          -0.005377
M12      79197          -1.000
PHYS:N J 20.
PHYS:P 0 1 1
CUT:N 20
CUT:P 20 J 0
sdef POS 0 0 0.092 ERG=d1 PAR=1
sp1 1
NPS  1E8
IPOL 0 0 0 0 2J 1 200
FILES 21 DUMN1_0
```

APPENDIX B

MCNP POST-PROCESSING CODES

```
void mcnp_scan(){
    int cell_num = 1;
    char const path[200] = "path_to_file";

    const char* prefix = "dumn1";
    const char* ext1 = ".spe";
    const char* ext2 = ".root";
    const char* inDir = "path_to_file";
    char* dir = gSystem->ExpandPathName(inDir);
    void* dirp = gSystem->OpenDirectory(dir);
    const char* entry;
    char* filename[500];
    string rootpath, outputpath;
    const char* dirfilename[1000];
    int num;
    Int_t n = 0;
    TString str;

    while((entry = (char*)gSystem->GetDirEntry(dirp))) {
        str = entry;
        if(str.Contains(prefix)) {
            rootpath = "";
            outputpath = "";
            dirfilename[n] = gSystem->ConcatFileName(dir, entry);
            filename[n] = entry;
```

```
    TString temp_str = entry;
    TString run_num(temp_str(6,8));

    ifstream infile(dirfilename[n]);
    if (infile.is_open()) {
        cout<<"File opened."<<endl;

        TString rootfilename = "R" + run_num + ".root";
        cout<<rootfilename<<endl;
        TString outfilename = "" + run_num + ".spe";
        ofstream fout(outfilename);

        string token;
        int bin_num = 1500, hist_range = 15000;
        int Target, cell, i = 0, temp, neutFlag,
            dFlag, pFlag, ncount = 0;
        float Edep, time, Lo, L, FWHM, dpsd, psd,
              paraL, paraS, Ggate, NgateL, NgateU,
              bottome_line;

        float A[3] = {1.88708,0.0305906,7.22758e-06},
              C[3] = {-1.26074,0.28593,-3.89199e-05};

        TFile *ff = new TFile(rootfilename, "RECREATE");
        TTree *tt = new TTree("T","Digitizer waveforms");

        tt->Branch("L",&paraL,"L/f");
        tt->Branch("N",&neutFlag,"N/i");
        tt->Branch("PSD",&psd,"S/f");
        tt->Branch("D_Recoil", &dFlag, "D/i");
        tt->Branch("P_Recoil", &pFlag, "P/i");
        tt->Branch("target", &Target, "target/i");
```

```
string line;
while (getline(infile, line)){
    i++;
    istringstream iss(line);
    int j = 0;
    while (j < 4) {
        iss >> temp;
        j++;
    }
    iss >> Target >> cell >> Edep >> time;

    /** Light response for EJ-301D for d **/
    if (cell == cell_num && Target == 1002) {
        L = 0.475651*Edep - 0.801497*(1 -
            TMath::Exp(-0.512168*Edep));
        dFlag=1;pFlag=0;
    }

    TRandom r(0);
    /** Resolution for EJ-301D **/
    if (cell == cell_num && Target == 1002)
    {
        duration = ( std::clock() - start );
        FWHM = (L*1E3)*TMath::Sqrt(pow(0.0002527,2) +
                (pow(3.2885,2)/(L*1E3)) +
                (pow(-0.1937,2)/TMath::Power((L*1E3),2)));
        L = L*1E3 + (TMath::ErfInverse(2*r.Rndm()-1)
        *TMath::Sqrt(2*TMath::Power(FWHM/2.35,2)));
    }

    /** PSD parameters for EJ301D */
```

```
                if (cell == cell_num && Target == 1002)
                {
                    L=L/1000.0;
                    dpsd = 0.0196457/TMath::Sqrt(L) - 0.00345068;
                    psd = 0.275474 + (-0.0653279*L) + 0.0113781*L*L;
                    psd = psd + (TMath::ErfInverse(2*r.Rndm() - 1)
                    *TMath::Sqrt(2*TMath::Power(dpsd,2)));
                    paraL = L*1000;

                    Ggate = A[0]/TMath::Sqrt(paraL)+A[1]+A[2]*paraL;
                    NgateL = C[0]/TMath::Sqrt(paraL)+C[1]+C[2]*paraL;
                    if(psd > Ggate && psd > NgateL&& psd < 1)
                    { neutFlag = 1; ncount++;}
                    else{neutFlag = 0;}
                    tt->Fill();
                }
            tt->Fill();
            }
            ff->cd(); tt->Write();
            infile.close();
            TH1F *h1 = new TH1F("h1", "L", bin_num,0,hist_range);
            T->Draw("L>>h1", "PSD>0&&L>0&&PSD<1&&N==1");
            for (int i = 0; i < bin_num; i++) {
                fout << h1->GetBinContent(i) << endl;
            }
            ff->Close();
        }
        n++;
    }
  }
}
```

APPENDIX C

UNCERTAINTY ESTIMATION CODE FOR MLEM

```
{
    #include<iostream>
    #include<fstream>
    #include<sstream>
    #include<string>
    #include<cstdlib>
    #include <ctime>
    #include "TRandom.h"
    #include "TH1.h"
    using namespace std;

    float sigma, sample, new_sample;
    int i = 0;
    string openfile, line, outfile;

    std::clock_t start;
    double duration;
    start = std::clock();

    cout << "File to read: ";
    cin >> openfile;
    //generate more input files
    while(i < 1000){
        ifstream fp(openfile.c_str());
        getline(fp, line);
```

```
    outfile = "";
    outfile += "tranform";
        if(i < 10){ outfile += "00"; }
        if(i >= 10 && i < 100) { outfile += "0"; }

    std::stringstream ss;
    ss << i;
    std::string str;
    ss >> str;
    outfile += str;
    outfile += ".spe";
    cout<<outfile<<endl;
    ofstream fout(outfile.c_str());
    fout<<line<<endl;

    //tranform data
    while (fp >> sample){
        TRandom *r = new TRandom();
        duration = ( std::clock() - start );
        r->SetSeed(duration);

        sigma = TMath::Sqrt(sample);
        new_sample = sample + (TMath::ErfInverse(2*r->Rndm()-1))
        *(TMath::Sqrt(2*TMath::Power(sigma,2)));
        while(new_sample < 0){
            new_sample = sample + (TMath::ErfInverse(2*r->Rndm()-1))
            *(TMath::Sqrt(2*TMath::Power(sigma,2)));
        }
        fout<<new_sample<<endl;
    }
    i++;}
}
```

APPENDIX D

TABLES

TABLE D.1. CROSS SECTIONS OF THE ^{11}B$(\alpha, n)^{14}$N REACTION

512.7	6.97×10^{-8}	4.15×10^{-9}	9.61×10^{-8}	5.72×10^{-9}
522.8	1.07×10^{-7}	6.35×10^{-9}	1.49×10^{-7}	8.89×10^{-9}
533.2	1.57×10^{-7}	9.08×10^{-9}	2.41×10^{-7}	1.39×10^{-8}
538.7	2.15×10^{-7}	1.27×10^{-8}	3.14×10^{-7}	1.85×10^{-8}
543.5	2.63×10^{-7}	1.48×10^{-8}	3.99×10^{-7}	2.24×10^{-8}
544.5	2.54×10^{-7}	1.37×10^{-8}	4.20×10^{-7}	2.26×10^{-8}
548.5	3.44×10^{-7}	1.91×10^{-8}	5.17×10^{-7}	2.87×10^{-8}
549.7	3.50×10^{-7}	1.87×10^{-8}	5.51×10^{-7}	2.94×10^{-8}
554	4.47×10^{-7}	2.46×10^{-8}	6.95×10^{-7}	3.83×10^{-8}
554.7	4.83×10^{-7}	2.58×10^{-8}	7.22×10^{-7}	3.86×10^{-8}
560	6.58×10^{-7}	3.47×10^{-8}	9.70×10^{-7}	5.12×10^{-8}
565.3	8.85×10^{-7}	4.75×10^{-8}	1.31×10^{-6}	7.00×10^{-8}
570.6	1.16×10^{-6}	6.15×10^{-8}	1.75×10^{-6}	9.23×10^{-8}
575.9	1.42×10^{-6}	7.58×10^{-8}	2.27×10^{-6}	1.22×10^{-7}
580.6	1.89×10^{-6}	1.01×10^{-7}	2.75×10^{-6}	1.48×10^{-7}
585.8	2.24×10^{-6}	1.19×10^{-7}	3.17×10^{-6}	1.69×10^{-7}
596	3.21×10^{-6}	1.71×10^{-7}	3.24×10^{-6}	1.73×10^{-7}
632.4	1.73×10^{-6}	9.15×10^{-8}	1.44×10^{-6}	7.65×10^{-8}
635.1	1.65×10^{-6}	8.78×10^{-8}	1.38×10^{-6}	7.34×10^{-8}

TABLE D.1 CONTINUED

E_α (keV)	σ (barn)	$\Delta\sigma$	σ_{corr} (barn)	$\Delta\sigma_{corr}$
636.2	1.49×10^{-6}	7.95×10^{-8}	1.36×10^{-6}	7.26×10^{-8}
638.2	1.49×10^{-6}	7.88×10^{-8}	1.32×10^{-6}	6.99×10^{-8}
638.4	1.38×10^{-6}	7.40×10^{-8}	1.31×10^{-6}	7.04×10^{-8}
640.7	1.40×10^{-6}	7.46×10^{-8}	1.27×10^{-6}	6.81×10^{-8}
643.5	1.37×10^{-6}	7.30×10^{-8}	1.23×10^{-6}	6.59×10^{-8}
646.3	1.35×10^{-6}	7.22×10^{-8}	1.20×10^{-6}	6.40×10^{-8}
647.9	1.25×10^{-6}	6.66×10^{-8}	1.18×10^{-6}	6.30×10^{-8}
648.6	1.32×10^{-6}	7.06×10^{-8}	1.17×10^{-6}	6.26×10^{-8}
652	1.19×10^{-6}	6.36×10^{-8}	1.14×10^{-6}	6.07×10^{-8}
655.1	1.19×10^{-6}	6.29×10^{-8}	1.12×10^{-6}	5.92×10^{-8}
658.1	1.23×10^{-6}	6.56×10^{-8}	1.10×10^{-6}	5.86×10^{-8}
662.3	1.07×10^{-6}	5.69×10^{-8}	1.08×10^{-6}	5.77×10^{-8}
665.4	1.09×10^{-6}	5.80×10^{-8}	1.07×10^{-6}	5.71×10^{-8}
710.3	1.18×10^{-6}	6.27×10^{-8}	1.33×10^{-6}	7.07×10^{-8}
728	1.44×10^{-6}	7.65×10^{-8}	1.62×10^{-6}	8.59×10^{-8}
758.8	2.17×10^{-6}	1.15×10^{-7}	2.45×10^{-6}	1.30×10^{-7}
789.4	3.37×10^{-6}	1.78×10^{-7}	3.90×10^{-6}	2.06×10^{-7}
814.8	5.03×10^{-6}	2.65×10^{-7}	5.81×10^{-6}	3.06×10^{-7}
840.7	7.49×10^{-6}	3.96×10^{-7}	8.80×10^{-6}	4.65×10^{-7}
859.8	1.03×10^{-5}	5.45×10^{-7}	1.20×10^{-5}	6.37×10^{-7}
879.4	1.39×10^{-5}	7.33×10^{-7}	1.66×10^{-5}	8.77×10^{-7}
899.2	1.97×10^{-5}	1.04×10^{-6}	2.32×10^{-5}	1.23×10^{-6}
919.2	2.89×10^{-5}	1.52×10^{-6}	3.29×10^{-5}	1.74×10^{-6}
939.5	4.19×10^{-5}	2.21×10^{-6}	4.77×10^{-5}	2.52×10^{-6}

TABLE D.1 CONTINUED

E_α (keV)	σ (barn)	$\Delta\sigma$	σ_{corr} (barn)	$\Delta\sigma_{corr}$
960	6.02×10^{-5}	3.18×10^{-6}	7.12×10^{-5}	3.77×10^{-6}
994.6	1.26×10^{-4}	6.66×10^{-6}	1.54×10^{-4}	8.13×10^{-6}
1007.8	1.71×10^{-4}	9.00×10^{-6}	2.15×10^{-4}	1.13×10^{-5}
1015.5	2.06×10^{-4}	1.08×10^{-5}	2.62×10^{-4}	1.38×10^{-5}
1017.9	2.25×10^{-4}	1.19×10^{-5}	2.79×10^{-4}	1.47×10^{-5}
1025.7	2.67×10^{-4}	1.40×10^{-5}	3.37×10^{-4}	1.77×10^{-5}
1027.6	2.96×10^{-4}	1.55×10^{-5}	3.51×10^{-4}	1.84×10^{-5}
1035.5	3.35×10^{-4}	1.76×10^{-5}	4.02×10^{-4}	2.12×10^{-5}
1036.5	3.66×10^{-4}	1.92×10^{-5}	4.07×10^{-4}	2.13×10^{-5}
1044.3	3.99×10^{-4}	2.09×10^{-5}	4.15×10^{-4}	2.18×10^{-5}
1047.7	3.98×10^{-4}	2.09×10^{-5}	4.00×10^{-4}	2.10×10^{-5}
1055.6	4.00×10^{-4}	2.10×10^{-5}	3.28×10^{-4}	1.72×10^{-5}
1057.6	3.47×10^{-4}	1.82×10^{-5}	3.05×10^{-4}	1.60×10^{-5}
1065.5	3.17×10^{-4}	1.67×10^{-5}	2.15×10^{-4}	1.13×10^{-5}
1067.7	2.47×10^{-4}	1.30×10^{-5}	1.93×10^{-4}	1.02×10^{-5}
1075.6	2.12×10^{-4}	1.12×10^{-5}	1.31×10^{-4}	6.92×10^{-6}
1078.1	1.51×10^{-4}	7.88×10^{-6}	1.18×10^{-4}	6.14×10^{-6}
1085.8	1.39×10^{-4}	7.27×10^{-6}	8.95×10^{-5}	4.69×10^{-6}
1087.9	9.98×10^{-5}	5.24×10^{-6}	8.51×10^{-5}	4.46×10^{-6}
1098.4	7.95×10^{-5}	4.22×10^{-6}	7.91×10^{-5}	4.19×10^{-6}
1108.1	8.18×10^{-5}	4.31×10^{-6}	9.10×10^{-5}	4.79×10^{-6}
1118.6	9.45×10^{-5}	5.00×10^{-6}	1.17×10^{-4}	6.18×10^{-6}
1129	1.26×10^{-4}	6.67×10^{-6}	1.58×10^{-4}	8.37×10^{-6}
1138.5	1.69×10^{-4}	8.93×10^{-6}	2.20×10^{-4}	1.17×10^{-5}

TABLE D.1 CONTINUED

E_α (keV)	σ (barn)	$\Delta\sigma$	σ_{corr} (barn)	$\Delta\sigma_{corr}$
1148.9	2.41×10^{-4}	1.28×10^{-5}	3.19×10^{-4}	1.69×10^{-5}
1158.5	3.44×10^{-4}	1.82×10^{-5}	4.60×10^{-4}	2.44×10^{-5}
1168.3	5.04×10^{-4}	2.67×10^{-5}	7.21×10^{-4}	3.82×10^{-5}
1180.1	9.75×10^{-4}	5.16×10^{-5}	1.51×10^{-3}	8.02×10^{-5}
1189.8	2.05×10^{-3}	1.07×10^{-4}	3.26×10^{-3}	1.71×10^{-4}
1198.5	3.14×10^{-3}	1.64×10^{-4}	4.04×10^{-3}	2.12×10^{-4}
1208.9	3.18×10^{-3}	1.67×10^{-4}	2.30×10^{-3}	1.21×10^{-4}
1218.8	2.23×10^{-3}	1.16×10^{-4}	1.48×10^{-3}	7.71×10^{-5}
1228.2	1.39×10^{-3}	7.30×10^{-5}	1.19×10^{-3}	6.24×10^{-5}
1248.8	1.01×10^{-3}	5.30×10^{-5}	1.06×10^{-3}	5.52×10^{-5}
1258.7	1.06×10^{-3}	5.59×10^{-5}	1.10×10^{-3}	5.81×10^{-5}
1258.7	1.01×10^{-3}	5.30×10^{-5}	1.10×10^{-3}	5.79×10^{-5}
1270.9	1.16×10^{-3}	6.06×10^{-5}	1.26×10^{-3}	6.60×10^{-5}
1280.2	1.66×10^{-3}	8.70×10^{-5}	1.39×10^{-3}	7.32×10^{-5}
1289.7	1.75×10^{-3}	9.15×10^{-5}	1.90×10^{-3}	9.96×10^{-5}
1299.6	2.08×10^{-3}	1.09×10^{-4}	2.69×10^{-3}	1.41×10^{-4}
1309.6	3.03×10^{-3}	1.59×10^{-4}	3.98×10^{-3}	2.09×10^{-4}
1319.7	3.86×10^{-3}	2.03×10^{-4}	3.87×10^{-3}	2.03×10^{-4}
1330	3.53×10^{-3}	1.85×10^{-4}	2.91×10^{-3}	1.53×10^{-4}
1339.6	2.66×10^{-3}	1.40×10^{-4}	2.49×10^{-3}	1.31×10^{-4}
1360.8	2.33×10^{-3}	1.22×10^{-4}	2.42×10^{-3}	1.27×10^{-4}
1381	2.55×10^{-3}	1.34×10^{-4}	2.71×10^{-3}	1.42×10^{-4}
1400.1	2.91×10^{-3}	1.52×10^{-4}	3.10×10^{-3}	1.62×10^{-4}
1410.3	3.11×10^{-3}	1.63×10^{-4}	3.36×10^{-3}	1.76×10^{-4}

TABLE D.1 CONTINUED

E_α (keV)	σ (barn)	$\Delta\sigma$	σ_{corr} (barn)	$\Delta\sigma_{corr}$
1421.1	3.39×10^{-3}	1.78×10^{-4}	3.69×10^{-3}	1.93×10^{-4}
1431.1	3.71×10^{-3}	1.94×10^{-4}	4.08×10^{-3}	2.14×10^{-4}
1440.9	4.10×10^{-3}	2.15×10^{-4}	4.61×10^{-3}	2.42×10^{-4}
1450.6	4.79×10^{-3}	2.51×10^{-4}	5.45×10^{-3}	2.86×10^{-4}
1460.9	5.81×10^{-3}	3.05×10^{-4}	7.07×10^{-3}	3.70×10^{-4}
1470.6	7.73×10^{-3}	4.06×10^{-4}	1.03×10^{-2}	5.41×10^{-4}
1480.7	1.25×10^{-2}	6.52×10^{-4}	1.79×10^{-2}	9.35×10^{-4}
1490.7	2.11×10^{-2}	1.10×10^{-3}	2.72×10^{-2}	1.42×10^{-3}
1500.9	2.52×10^{-2}	1.32×10^{-3}	2.31×10^{-2}	1.21×10^{-3}
1511	2.06×10^{-2}	1.08×10^{-3}	1.68×10^{-2}	8.78×10^{-4}
1521.3	1.52×10^{-2}	7.95×10^{-4}	1.42×10^{-2}	7.45×10^{-4}
1531.1	1.37×10^{-2}	7.14×10^{-4}	1.41×10^{-2}	7.37×10^{-4}
1541.2	1.43×10^{-2}	7.45×10^{-4}	1.56×10^{-2}	8.14×10^{-4}
1551.2	1.62×10^{-2}	8.48×10^{-4}	1.82×10^{-2}	9.50×10^{-4}
1561.2	1.85×10^{-2}	9.68×10^{-4}	2.06×10^{-2}	1.08×10^{-3}
1571.3	2.04×10^{-2}	1.07×10^{-3}	2.08×10^{-2}	1.08×10^{-3}
1580.9	1.96×10^{-2}	1.03×10^{-3}	1.87×10^{-2}	9.81×10^{-4}
1601.1	1.45×10^{-2}	7.58×10^{-4}	1.33×10^{-2}	6.97×10^{-4}
1621.4	1.08×10^{-2}	5.65×10^{-4}	1.03×10^{-2}	5.39×10^{-4}
1641.4	9.00×10^{-3}	4.73×10^{-4}	8.86×10^{-3}	4.66×10^{-4}
1661.3	8.18×10^{-3}	4.28×10^{-4}	8.14×10^{-3}	4.26×10^{-4}
1681.6	7.65×10^{-3}	4.03×10^{-4}	7.77×10^{-3}	4.09×10^{-4}
1701.8	7.43×10^{-3}	3.90×10^{-4}	7.60×10^{-3}	3.99×10^{-4}
1722.1	7.37×10^{-3}	3.87×10^{-4}	7.56×10^{-3}	3.97×10^{-4}

TABLE D.1 CONTINUED

E_α (keV)	σ (barn)	$\Delta\sigma$	σ_{corr} (barn)	$\Delta\sigma_{corr}$
1742.1	7.34×10^{-3}	3.86×10^{-4}	7.59×10^{-3}	3.99×10^{-4}
1762.2	7.40×10^{-3}	3.89×10^{-4}	7.68×10^{-3}	4.04×10^{-4}
1782.5	7.58×10^{-3}	3.96×10^{-4}	7.82×10^{-3}	4.09×10^{-4}
1802.6	7.95×10^{-3}	4.16×10^{-4}	8.00×10^{-3}	4.19×10^{-4}
1822.9	8.18×10^{-3}	4.29×10^{-4}	8.22×10^{-3}	4.31×10^{-4}
1842.7	8.10×10^{-3}	4.27×10^{-4}	8.47×10^{-3}	4.46×10^{-4}
1862.8	8.40×10^{-3}	4.44×10^{-4}	8.79×10^{-3}	4.65×10^{-4}
1882.7	8.78×10^{-3}	4.61×10^{-4}	9.19×10^{-3}	4.82×10^{-4}
1903	9.38×10^{-3}	4.92×10^{-4}	9.76×10^{-3}	5.12×10^{-4}
1922.6	9.98×10^{-3}	5.27×10^{-4}	1.06×10^{-2}	5.60×10^{-4}
1943.5	1.13×10^{-2}	5.95×10^{-4}	1.21×10^{-2}	6.33×10^{-4}
1963.5	1.36×10^{-2}	7.15×10^{-4}	1.47×10^{-2}	7.74×10^{-4}
1983.7	1.78×10^{-2}	9.38×10^{-4}	2.05×10^{-2}	1.08×10^{-3}
2003.9	2.78×10^{-2}	1.47×10^{-3}	3.55×10^{-2}	1.87×10^{-3}
2003.9	3.08×10^{-2}	1.63×10^{-3}	3.55×10^{-2}	1.87×10^{-3}

Notes: Cross sections before and after correction for target effects of the ^{11}B$(\alpha,n)^{14}$N reaction, in the laboratory frame of reference. Alpha particle energies in the unit of keV, and cross sections in the unit of barn.

TABLE D.2. DIFFERENTIAL CROSS SECTIONS FOR THE ^{10}B$(\alpha,n)^{13}$N REACTION AT 0°

570.1	1.22×10^{-8}	3.92×10^{-10}	1.72×10^{-8}	5.52×10^{-10}
590.1	1.20×10^{-8}	3.89×10^{-10}	1.67×10^{-8}	5.42×10^{-10}

E_α (keV)	σ (barn)	$\Delta\sigma$	σ_{corr} (barn)	$\Delta\sigma_{corr}$
610.2	1.10×10^{-8}	3.68×10^{-10}	1.52×10^{-8}	5.08×10^{-10}
630.2	1.21×10^{-8}	3.93×10^{-10}	1.65×10^{-8}	5.36×10^{-10}
650.2	1.15×10^{-8}	3.80×10^{-10}	1.56×10^{-8}	5.14×10^{-10}
670.3	1.44×10^{-8}	4.42×10^{-10}	1.93×10^{-8}	5.91×10^{-10}
689.9	2.13×10^{-8}	5.90×10^{-10}	2.83×10^{-8}	7.82×10^{-10}
710	2.04×10^{-8}	5.91×10^{-10}	2.68×10^{-8}	7.77×10^{-10}
729.9	2.34×10^{-8}	6.56×10^{-10}	3.05×10^{-8}	8.54×10^{-10}
750	2.33×10^{-8}	6.51×10^{-10}	3.01×10^{-8}	8.42×10^{-10}
769.5	3.27×10^{-8}	7.93×10^{-10}	4.15×10^{-8}	1.01×10^{-9}
789.9	3.34×10^{-8}	8.00×10^{-10}	4.19×10^{-8}	1.00×10^{-9}
829.9	4.44×10^{-8}	1.02×10^{-9}	5.44×10^{-8}	1.25×10^{-9}
849.6	4.49×10^{-8}	1.03×10^{-9}	5.43×10^{-8}	1.25×10^{-9}
855.1	5.13×10^{-8}	1.19×10^{-9}	6.13×10^{-8}	1.43×10^{-9}
860.1	4.73×10^{-8}	1.18×10^{-9}	5.60×10^{-8}	1.40×10^{-9}
880.2	5.40×10^{-8}	1.27×10^{-9}	6.35×10^{-8}	1.49×10^{-9}
900	6.75×10^{-8}	1.48×10^{-9}	7.81×10^{-8}	1.71×10^{-9}
919.7	7.86×10^{-8}	1.71×10^{-9}	8.98×10^{-8}	1.95×10^{-9}
938.7	8.75×10^{-8}	1.89×10^{-9}	9.88×10^{-8}	2.13×10^{-9}
959.6	1.04×10^{-7}	2.21×10^{-9}	1.16×10^{-7}	2.47×10^{-9}
980.2	1.19×10^{-7}	2.60×10^{-9}	1.31×10^{-7}	2.87×10^{-9}
1000.1	1.39×10^{-7}	2.92×10^{-9}	1.53×10^{-7}	3.21×10^{-9}
1020.2	1.55×10^{-7}	3.21×10^{-9}	1.68×10^{-7}	3.48×10^{-9}
1025	2.05×10^{-7}	7.15×10^{-9}	2.90×10^{-7}	1.02×10^{-8}
1025	1.62×10^{-7}	6.08×10^{-9}	2.31×10^{-7}	8.63×10^{-9}

E_α (keV)	σ (barn)	$\Delta\sigma$	σ_{corr} (barn)	$\Delta\sigma_{corr}$
1030	1.60×10^{-7}	6.23×10^{-9}	2.28×10^{-7}	8.86×10^{-9}
1035	2.91×10^{-7}	9.46×10^{-9}	4.13×10^{-7}	1.34×10^{-8}
1039.9	3.07×10^{-7}	1.05×10^{-8}	3.26×10^{-7}	1.12×10^{-8}
1039.9	2.21×10^{-7}	6.98×10^{-9}	2.36×10^{-7}	7.45×10^{-9}
1045	2.60×10^{-7}	8.30×10^{-9}	3.69×10^{-7}	1.18×10^{-8}
1050	3.23×10^{-7}	1.01×10^{-8}	4.58×10^{-7}	1.44×10^{-8}
1055	2.64×10^{-7}	8.68×10^{-9}	3.74×10^{-7}	1.23×10^{-8}
1060	7.02×10^{-7}	2.65×10^{-8}	7.40×10^{-7}	2.79×10^{-8}
1060	2.58×10^{-7}	8.09×10^{-9}	3.65×10^{-7}	1.15×10^{-8}
1065	2.76×10^{-7}	8.94×10^{-9}	3.90×10^{-7}	1.27×10^{-8}
1070	3.99×10^{-7}	9.53×10^{-9}	5.65×10^{-7}	1.35×10^{-8}
1090	6.39×10^{-7}	1.99×10^{-8}	9.03×10^{-7}	2.81×10^{-8}
1095	7.09×10^{-7}	2.10×10^{-8}	1.00×10^{-6}	2.97×10^{-8}
1100.2	9.86×10^{-7}	2.77×10^{-8}	1.39×10^{-6}	3.91×10^{-8}
1105.1	1.25×10^{-6}	3.29×10^{-8}	1.77×10^{-6}	4.64×10^{-8}
1110	1.44×10^{-6}	3.72×10^{-8}	2.03×10^{-6}	5.25×10^{-8}
1110.1	2.51×10^{-6}	6.94×10^{-8}	2.64×10^{-6}	7.30×10^{-8}
1114.9	1.44×10^{-6}	3.72×10^{-8}	2.03×10^{-6}	5.25×10^{-8}
1125.1	1.72×10^{-6}	4.21×10^{-8}	2.43×10^{-6}	5.93×10^{-8}
1135.1	1.37×10^{-6}	3.56×10^{-8}	1.93×10^{-6}	5.02×10^{-8}
1145.1	1.68×10^{-6}	4.06×10^{-8}	2.36×10^{-6}	5.73×10^{-8}
1155.1	1.56×10^{-6}	3.97×10^{-8}	2.21×10^{-6}	5.59×10^{-8}
1160.1	1.67×10^{-6}	5.47×10^{-8}	1.74×10^{-6}	5.72×10^{-8}
1160.2	1.17×10^{-6}	3.61×10^{-8}	1.22×10^{-6}	3.77×10^{-8}

TABLE D.2 CONTINUED

E_α (keV)	σ (barn)	$\Delta\sigma$	σ_{corr} (barn)	$\Delta\sigma_{corr}$
1165.3	1.40×10^{-6}	3.64×10^{-8}	1.98×10^{-6}	5.13×10^{-8}
1175.2	1.09×10^{-6}	2.97×10^{-8}	1.53×10^{-6}	4.18×10^{-8}
1180.1	1.16×10^{-6}	3.16×10^{-8}	1.20×10^{-6}	3.28×10^{-8}
1180.2	8.01×10^{-7}	2.38×10^{-8}	1.13×10^{-6}	3.36×10^{-8}
1200.8	9.00×10^{-7}	2.68×10^{-8}	9.32×10^{-7}	2.77×10^{-8}
1219.8	1.37×10^{-6}	3.71×10^{-8}	1.41×10^{-6}	3.83×10^{-8}
1239.9	1.44×10^{-6}	3.83×10^{-8}	1.48×10^{-6}	3.94×10^{-8}
1259.9	2.16×10^{-6}	5.22×10^{-8}	2.21×10^{-6}	5.34×10^{-8}
1279.7	1.92×10^{-6}	4.45×10^{-8}	1.96×10^{-6}	4.53×10^{-8}
1300	2.18×10^{-6}	5.69×10^{-8}	2.21×10^{-6}	5.76×10^{-8}
1325.1	3.14×10^{-6}	6.97×10^{-8}	4.47×10^{-6}	9.91×10^{-8}
1350	2.36×10^{-6}	6.65×10^{-8}	2.38×10^{-6}	6.70×10^{-8}
1350	2.91×10^{-6}	6.73×10^{-8}	2.94×10^{-6}	6.81×10^{-8}
1399.9	3.76×10^{-6}	1.30×10^{-7}	3.78×10^{-6}	1.31×10^{-7}
1416.9	5.29×10^{-6}	1.71×10^{-7}	7.55×10^{-6}	2.44×10^{-7}
1430	6.92×10^{-6}	2.19×10^{-7}	9.88×10^{-6}	3.13×10^{-7}
1440.6	9.57×10^{-6}	2.77×10^{-7}	1.37×10^{-5}	3.95×10^{-7}
1449.9	1.76×10^{-5}	4.73×10^{-7}	1.77×10^{-5}	4.76×10^{-7}
1453.3	1.34×10^{-5}	3.46×10^{-7}	1.91×10^{-5}	4.94×10^{-7}
1460	1.81×10^{-5}	4.86×10^{-7}	2.59×10^{-5}	6.95×10^{-7}
1549.7	4.45×10^{-5}	1.01×10^{-6}	4.47×10^{-5}	1.01×10^{-6}
1570.2	2.59×10^{-5}	7.27×10^{-7}	2.60×10^{-5}	7.30×10^{-7}
1599.9	1.76×10^{-5}	5.60×10^{-7}	1.77×10^{-5}	5.62×10^{-7}
1600.1	2.24×10^{-5}	6.08×10^{-7}	3.19×10^{-5}	8.68×10^{-7}

TABLE D.2 CONTINUED

E_α (keV)	σ (barn)	$\Delta\sigma$	σ_{corr} (barn)	$\Delta\sigma_{corr}$
1610.6	1.90×10^{-5}	4.62×10^{-7}	2.70×10^{-5}	6.60×10^{-7}
1620.4	2.23×10^{-5}	5.61×10^{-7}	3.17×10^{-5}	8.00×10^{-7}
1625.1	3.19×10^{-5}	9.80×10^{-7}	3.20×10^{-5}	9.83×10^{-7}
1630.9	3.90×10^{-5}	1.22×10^{-6}	5.57×10^{-5}	1.73×10^{-6}
1640.4	5.73×10^{-5}	1.70×10^{-6}	8.17×10^{-5}	2.42×10^{-6}
1649.9	3.23×10^{-5}	1.08×10^{-6}	3.24×10^{-5}	1.08×10^{-6}
1650.5	3.42×10^{-5}	9.35×10^{-7}	4.88×10^{-5}	1.33×10^{-6}
1660.2	1.44×10^{-5}	5.47×10^{-7}	2.05×10^{-5}	7.79×10^{-7}
1670.6	8.61×10^{-6}	2.56×10^{-7}	1.23×10^{-5}	3.65×10^{-7}
1680.3	8.90×10^{-6}	2.62×10^{-7}	1.27×10^{-5}	3.73×10^{-7}
1690.4	1.33×10^{-5}	3.55×10^{-7}	1.90×10^{-5}	5.06×10^{-7}
1699.7	1.11×10^{-5}	2.97×10^{-7}	1.11×10^{-5}	2.97×10^{-7}
1700.4	1.18×10^{-5}	3.24×10^{-7}	1.69×10^{-5}	4.62×10^{-7}
1710.5	9.68×10^{-6}	2.77×10^{-7}	1.38×10^{-5}	3.94×10^{-7}
1720	9.53×10^{-6}	2.64×10^{-7}	1.36×10^{-5}	3.75×10^{-7}
1730	9.38×10^{-6}	2.74×10^{-7}	1.33×10^{-5}	3.90×10^{-7}
1740	8.49×10^{-6}	2.56×10^{-7}	1.21×10^{-5}	3.63×10^{-7}
1750	9.63×10^{-6}	2.73×10^{-7}	1.37×10^{-5}	3.89×10^{-7}
1799.8	2.02×10^{-5}	7.23×10^{-7}	2.03×10^{-5}	7.25×10^{-7}
1850.1	2.61×10^{-5}	9.45×10^{-7}	2.61×10^{-5}	9.47×10^{-7}
1850.2	2.25×10^{-5}	7.71×10^{-7}	2.26×10^{-5}	7.72×10^{-7}
1917.1	3.24×10^{-5}	1.17×10^{-6}	3.24×10^{-5}	1.17×10^{-6}
1917.1	3.71×10^{-5}	1.23×10^{-6}	3.71×10^{-5}	1.23×10^{-6}
1917.1	3.28×10^{-5}	8.40×10^{-7}	3.28×10^{-5}	8.41×10^{-7}

TABLE D.2 CONTINUED

E_α (keV)	σ (barn)	$\Delta\sigma$	σ_{corr} (barn)	$\Delta\sigma_{corr}$
1917.2	3.20×10^{-5}	8.92×10^{-7}	3.21×10^{-5}	8.93×10^{-7}
1957.1	2.93×10^{-5}	8.22×10^{-7}	2.93×10^{-5}	8.23×10^{-7}
1977.2	2.78×10^{-5}	8.95×10^{-7}	2.78×10^{-5}	8.96×10^{-7}
2017.2	2.69×10^{-5}	8.87×10^{-7}	2.69×10^{-5}	8.88×10^{-7}
2059.8	3.66×10^{-5}	1.23×10^{-6}	3.69×10^{-5}	1.24×10^{-6}
2077.2	3.55×10^{-5}	1.24×10^{-6}	3.56×10^{-5}	1.24×10^{-6}
2180.8	1.65×10^{-4}	4.63×10^{-6}	1.66×10^{-4}	4.65×10^{-6}
2180.8	2.10×10^{-4}	4.75×10^{-6}	2.11×10^{-4}	4.77×10^{-6}
2198.1	2.30×10^{-4}	5.48×10^{-6}	2.30×10^{-4}	5.48×10^{-6}
2216.8	2.99×10^{-4}	6.97×10^{-6}	2.99×10^{-4}	6.97×10^{-6}
2217	1.88×10^{-4}	4.94×10^{-6}	1.89×10^{-4}	4.97×10^{-6}
2272.7	5.38×10^{-4}	1.18×10^{-5}	5.41×10^{-4}	1.18×10^{-5}
2291.1	5.52×10^{-4}	1.27×10^{-5}	5.52×10^{-4}	1.27×10^{-5}
2300.3	5.78×10^{-4}	1.27×10^{-5}	5.81×10^{-4}	1.28×10^{-5}
2317.4	4.22×10^{-4}	9.86×10^{-6}	4.22×10^{-4}	9.86×10^{-6}
2400	2.68×10^{-4}	5.69×10^{-6}	2.69×10^{-4}	5.72×10^{-6}
2417	2.55×10^{-4}	5.94×10^{-6}	2.55×10^{-4}	5.94×10^{-6}
2500.3	2.32×10^{-4}	4.89×10^{-6}	2.33×10^{-4}	4.92×10^{-6}
2517.2	2.44×10^{-4}	5.94×10^{-6}	2.44×10^{-4}	5.94×10^{-6}

Notes: Differential cross sections at 0° for the ^{10}B$(\alpha, n)^{13}$N reaction, in the laboratory frame of reference. Alpha particle energies in unit of keV, and cross sections in unit of barn.

TABLE D.3. DIFFERENTIAL CROSS SECTIONS FOR THE ^{10}B$(\alpha,n)^{13}$N REACTION FROM 20° TO 50°

E_α (keV)	20	30	40	50
1590	$2.0(1)\times10^{-2}$	$4.5(1)\times10^{-2}$	$6.2(2)\times10^{-2}$	$7.3(2)\times10^{-2}$
1902	$2.1(1)\times10^{-2}$	$2.2(1)\times10^{-2}$	$2.4(1)\times10^{-2}$	$2.5(1)\times10^{-2}$
2202	$2.3(1)\times10^{-1}$	$3.0(1)\times10^{-1}$	$3.5(1)\times10^{-1}$	$4.1(1)\times10^{-1}$
2235	$1.6(0)\times10^{-1}$	$1.9(0)\times10^{-1}$	$2.0(1)\times10^{-1}$	$1.9(0)\times10^{-1}$
2281	$3.4(1)\times10^{-1}$	$3.8(1)\times10^{-1}$	$3.9(1)\times10^{-1}$	$3.7(1)\times10^{-1}$
2346	$4.4(1)\times10^{-1}$	$4.1(1)\times10^{-1}$	$4.6(1)\times10^{-1}$	$4.3(1)\times10^{-1}$
2509	$1.5(0)\times10^{-1}$	$1.5(0)\times10^{-1}$	$1.5(0)\times10^{-1}$	$1.6(0)\times10^{-1}$
2510	$1.2(0)\times10^{-1}$	$1.6(0)\times10^{-1}$	$1.6(0)\times10^{-1}$	$1.6(0)\times10^{-1}$
2625	$2.2(1)\times10^{-1}$	$2.3(1)\times10^{-1}$	$2.4(1)\times10^{-1}$	$2.3(1)\times10^{-1}$
2625	$2.0(1)\times10^{-1}$	$2.1(1)\times10^{-1}$	$2.2(1)\times10^{-1}$	$2.1(1)\times10^{-1}$
2721	$3.0(1)\times10^{-1}$	$2.5(1)\times10^{-1}$	$2.7(1)\times10^{-1}$	$2.8(1)\times10^{-1}$
2762	$2.5(1)\times10^{-1}$	$3.0(1)\times10^{-1}$	$3.1(1)\times10^{-1}$	$2.9(1)\times10^{-1}$
2864	$5.0(1)\times10^{-1}$	$5.5(1)\times10^{-1}$	$5.7(1)\times10^{-1}$	$5.5(1)\times10^{-1}$
2993	$4.5(1)\times10^{-1}$	$5.2(1)\times10^{-1}$	$6.4(2)\times10^{-1}$	$7.2(2)\times10^{-1}$
3118	$2.3(1)\times10^{-1}$	$3.0(1)\times10^{-1}$	$3.7(1)\times10^{-1}$	$4.0(1)\times10^{-1}$
3271	$2.6(1)\times10^{-1}$	$2.7(1)\times10^{-1}$	$3.7(1)\times10^{-1}$	$3.7(1)\times10^{-1}$
3393	$2.1(1)\times10^{-1}$	$2.6(1)\times10^{-1}$	$3.0(1)\times10^{-1}$	$3.6(1)\times10^{-1}$
3561	$2.2(1)\times10^{-1}$	$3.0(1)\times10^{-1}$	$4.3(1)\times10^{-1}$	$5.8(1)\times10^{-1}$
3706	$4.2(1)\times10^{-1}$	$4.9(1)\times10^{-1}$	$6.7(2)\times10^{-1}$	$8.4(2)\times10^{-1}$
3707	$4.0(1)\times10^{-1}$	$4.7(1)\times10^{-1}$	$6.0(2)\times10^{-1}$	$7.5(2)\times10^{-1}$
3807	$3.7(1)\times10^{-1}$	$4.9(1)\times10^{-1}$	$5.8(1)\times10^{-1}$	$6.2(2)\times10^{-1}$
3883	$4.9(1)\times10^{-1}$	$5.2(1)\times10^{-1}$	$7.4(2)\times10^{-1}$	$7.8(2)\times10^{-1}$
3985	$5.2(1)\times10^{-1}$	$5.8(1)\times10^{-1}$	$6.8(2)\times10^{-1}$	$7.4(2)\times10^{-1}$
4083	$4.0(1)\times10^{-1}$	$5.4(1)\times10^{-1}$	$6.7(2)\times10^{-1}$	$6.1(2)\times10^{-1}$

TABLE D.3 CONTINUED

E_α (keV)	20	30	40	50
4159	$5.4(1)\times10^{-1}$	$5.9(2)\times10^{-1}$	$7.1(2)\times10^{-1}$	$7.7(2)\times10^{-1}$
4208	$5.1(1)\times10^{-1}$	$5.4(1)\times10^{-1}$	$7.8(2)\times10^{-1}$	$9.7(3)\times10^{-1}$
4263	$8.2(2)\times10^{-1}$	$7.0(2)\times10^{-1}$	$1.1(0)\times10^{0}$	$1.2(0)\times10^{0}$
4318	$8.3(2)\times10^{-1}$	$9.7(2)\times10^{-1}$	$1.4(0)\times10^{0}$	$1.6(0)\times10^{0}$
4401	$9.8(3)\times10^{-1}$	$1.0(0)\times10^{0}$	$1.4(0)\times10^{0}$	$1.6(0)\times10^{0}$
4510	$8.9(2)\times10^{-1}$	$1.0(0)\times10^{0}$	$1.4(0)\times10^{0}$	$1.7(0)\times10^{0}$
4630	$6.1(2)\times10^{-1}$	$1.1(0)\times10^{0}$	$1.9(0)\times10^{0}$	$2.1(1)\times10^{0}$
4709	$4.7(1)\times10^{-1}$	$8.4(2)\times10^{-1}$	$1.4(0)\times10^{0}$	$1.7(0)\times10^{0}$
4768	$6.7(2)\times10^{-1}$	$7.3(2)\times10^{-1}$	$1.3(0)\times10^{0}$	$1.6(0)\times10^{0}$
4841	$6.0(2)\times10^{-1}$	$6.4(2)\times10^{-1}$	$8.1(2)\times10^{-1}$	$9.6(2)\times10^{-1}$
4905	$6.7(2)\times10^{-1}$	$7.1(2)\times10^{-1}$	$8.1(2)\times10^{-1}$	$7.1(2)\times10^{-1}$
4909		$6.6(2)\times10^{-1}$	$9.0(2)\times10^{-1}$	$7.5(2)\times10^{-1}$

Notes: Differential cross sections, in the laboratory frame of reference, for the ^{10}B$(\alpha,n)^{13}$N reaction. Uncertainties in the α-particle beam energies are 3.0 keV. The tabulated uncertainties reflect the unfolding uncertainty, which is estimated to be ≈10%. The present data have a systematic uncertainty for measurements at different energies of ≈20%, therefore they have been normalized to the data of Van der Zwan and Geiger [101] as described in the text.

TABLE D.4. DIFFERENTIAL CROSS SECTIONS FOR THE ^{10}B$(\alpha,n)^{13}$N REACTION FROM 60° TO 90°

E_α (keV)	60	70	80	90
1590	$8.1(2)\times10^{-2}$	$8.6(2)\times10^{-2}$	$8.5(2)\times10^{-2}$	$8.5(2)\times10^{-2}$
1902	$2.4(1)\times10^{-2}$	$3.1(1)\times10^{-2}$	$3.0(1)\times10^{-2}$	$3.4(1)\times10^{-2}$
2202	$4.6(1)\times10^{-1}$	$4.9(1)\times10^{-1}$	$5.3(1)\times10^{-1}$	$5.2(1)\times10^{-1}$

TABLE D.3 CONTINUED

E_α (keV)	60	70	80	90
2235	$2.0(1)\times10^{-1}$	$2.3(1)\times10^{-1}$	$2.2(1)\times10^{-1}$	$1.9(0)\times10^{-1}$
2281	$3.9(1)\times10^{-1}$	$4.3(1)\times10^{-1}$	$3.9(1)\times10^{-1}$	$3.7(1)\times10^{-1}$
2346	$4.4(1)\times10^{-1}$	$4.6(1)\times10^{-1}$	$4.2(1)\times10^{-1}$	$4.0(1)\times10^{-1}$
2509	$1.5(0)\times10^{-1}$	$1.4(0)\times10^{-1}$	$1.3(0)\times10^{-1}$	$1.1(0)\times10^{-1}$
2510	$1.5(0)\times10^{-1}$	$1.4(0)\times10^{-1}$	$1.2(0)\times10^{-1}$	$1.1(0)\times10^{-1}$
2625	$2.1(1)\times10^{-1}$	$2.0(1)\times10^{-1}$	$1.9(0)\times10^{-1}$	$1.6(0)\times10^{-1}$
2625	$2.0(1)\times10^{-1}$	$1.9(0)\times10^{-1}$	$1.8(0)\times10^{-1}$	$1.5(0)\times10^{-1}$
2721	$2.4(1)\times10^{-1}$	$2.4(1)\times10^{-1}$	$1.9(0)\times10^{-1}$	$1.6(0)\times10^{-1}$
2762	$2.8(1)\times10^{-1}$	$2.9(1)\times10^{-1}$	$2.4(1)\times10^{-1}$	$2.0(1)\times10^{-1}$
2864	$5.5(1)\times10^{-1}$	$5.8(2)\times10^{-1}$	$5.4(1)\times10^{-1}$	$4.8(1)\times10^{-1}$
2993	$8.4(2)\times10^{-1}$	$9.6(2)\times10^{-1}$	$1.1(0)\times10^{0}$	$9.5(2)\times10^{-1}$
3118	$4.7(1)\times10^{-1}$	$5.5(1)\times10^{-1}$	$6.3(2)\times10^{-1}$	$5.9(2)\times10^{-1}$
3271	$4.0(1)\times10^{-1}$	$4.6(1)\times10^{-1}$	$4.5(1)\times10^{-1}$	$4.5(1)\times10^{-1}$
3393	$3.9(1)\times10^{-1}$	$4.6(1)\times10^{-1}$	$5.2(1)\times10^{-1}$	$5.7(1)\times10^{-1}$
3561	$5.4(1)\times10^{-1}$	$6.7(2)\times10^{-1}$	$7.2(2)\times10^{-1}$	$7.5(2)\times10^{-1}$
3706	$7.5(2)\times10^{-1}$	$8.5(2)\times10^{-1}$	$8.5(2)\times10^{-1}$	$7.9(2)\times10^{-1}$
3707	$7.4(2)\times10^{-1}$	$7.9(2)\times10^{-1}$	$8.3(2)\times10^{-1}$	$7.6(2)\times10^{-1}$
3807	$7.5(2)\times10^{-1}$	$8.4(2)\times10^{-1}$	$8.3(2)\times10^{-1}$	$8.2(2)\times10^{-1}$
3883	$8.8(2)\times10^{-1}$	$9.0(2)\times10^{-1}$	$8.7(2)\times10^{-1}$	$6.9(2)\times10^{-1}$
3985	$7.3(2)\times10^{-1}$	$7.4(2)\times10^{-1}$	$6.9(2)\times10^{-1}$	$6.3(2)\times10^{-1}$
4083	$6.0(2)\times10^{-1}$	$6.3(2)\times10^{-1}$	$7.6(2)\times10^{-1}$	$6.5(2)\times10^{-1}$
4159	$8.1(2)\times10^{-1}$	$1.0(0)\times10^{0}$	$1.1(0)\times10^{0}$	$9.3(2)\times10^{-1}$
4208	$1.1(0)\times10^{0}$	$1.3(0)\times10^{0}$	$1.2(0)\times10^{0}$	$1.0(0)\times10^{0}$
4263	$1.4(0)\times10^{0}$	$1.7(0)\times10^{0}$	$1.6(0)\times10^{0}$	$9.6(2)\times10^{-1}$

TABLE D.3 CONTINUED

E_α (keV)	60	70	80	90
4318	$1.7(0)\times10^{0}$	$1.8(0)\times10^{0}$	$1.7(0)\times10^{0}$	$1.3(0)\times10^{0}$
4401	$1.5(0)\times10^{0}$	$1.6(0)\times10^{0}$	$1.5(0)\times10^{0}$	$1.4(0)\times10^{0}$
4510	$1.6(0)\times10^{0}$	$1.6(0)\times10^{0}$	$1.4(0)\times10^{0}$	$1.3(0)\times10^{0}$
4630	$2.2(1)\times10^{0}$	$2.3(1)\times10^{0}$	$1.9(0)\times10^{0}$	$1.6(0)\times10^{0}$
4709	$1.8(0)\times10^{0}$	$1.8(0)\times10^{0}$	$1.4(0)\times10^{0}$	$1.2(0)\times10^{0}$
4768	$1.5(0)\times10^{0}$	$1.5(0)\times10^{0}$	$1.2(0)\times10^{0}$	$9.3(2)\times10^{-1}$
4841	$1.0(0)\times10^{0}$	$1.0(0)\times10^{0}$	$9.9(3)\times10^{-1}$	$8.5(2)\times10^{-1}$
4905	$8.7(2)\times10^{-1}$	$8.3(2)\times10^{-1}$	$8.7(2)\times10^{-1}$	$7.7(2)\times10^{-1}$
4909	$7.9(2)\times10^{-1}$	$7.1(2)\times10^{-1}$	$8.3(2)\times10^{-1}$	$7.2(2)\times10^{-1}$

TABLE D.5. DIFFERENTIAL CROSS SECTIONS FOR THE ^{10}B$(\alpha, n)^{13}$N REACTION FROM 100° TO 155°

E_α (keV)	100	135	145	155
1590	$7.2(2)\times10^{-2}$			
1902	$3.3(1)\times10^{-2}$	$3.0(1)\times10^{-2}$		
2202	$4.9(1)\times10^{-1}$	$4.4(1)\times10^{-1}$		$3.2(1)\times10^{-1}$
2235	$2.1(1)\times10^{-1}$	$1.8(0)\times10^{-1}$	$1.8(0)\times10^{-1}$	$2.3(1)\times10^{-1}$
2281	$3.7(1)\times10^{-1}$	$2.6(1)\times10^{-1}$	$2.7(1)\times10^{-1}$	
2346	$3.7(1)\times10^{-1}$	$2.7(1)\times10^{-1}$	$2.4(1)\times10^{-1}$	$2.2(1)\times10^{-1}$
2509	$8.0(2)\times10^{-2}$		$4.9(1)\times10^{-2}$	$3.9(1)\times10^{-2}$
2510	$8.2(2)\times10^{-2}$	$5.1(1)\times10^{-2}$	$4.4(1)\times10^{-2}$	
2625	$1.1(0)\times10^{-1}$			

E_α (keV)	100	135	145	155
2625	$1.2(0){\times}10^{-1}$			
2721	$1.5(0){\times}10^{-1}$	$1.4(0){\times}10^{-1}$	$1.2(0){\times}10^{-1}$	$1.1(0){\times}10^{-1}$
2762	$1.7(0){\times}10^{-1}$	$1.6(0){\times}10^{-1}$	$1.5(0){\times}10^{-1}$	$1.5(0){\times}10^{-1}$
2864	$4.1(1){\times}10^{-1}$	$4.2(1){\times}10^{-1}$	$4.2(1){\times}10^{-1}$	$4.2(1){\times}10^{-1}$
2993	$9.5(2){\times}10^{-1}$	$8.3(2){\times}10^{-1}$	$7.3(2){\times}10^{-1}$	$6.2(2){\times}10^{-1}$
3118	$6.2(2){\times}10^{-1}$	$5.3(1){\times}10^{-1}$	$4.3(1){\times}10^{-1}$	$3.5(1){\times}10^{-1}$
3271	$4.6(1){\times}10^{-1}$	$3.7(1){\times}10^{-1}$	$3.1(1){\times}10^{-1}$	$2.5(1){\times}10^{-1}$
3393	$5.0(1){\times}10^{-1}$	$3.6(1){\times}10^{-1}$	$3.4(1){\times}10^{-1}$	$3.1(1){\times}10^{-1}$
3561	$7.7(2){\times}10^{-1}$	$6.6(2){\times}10^{-1}$	$6.0(2){\times}10^{-1}$	$6.0(2){\times}10^{-1}$
3706	$8.4(2){\times}10^{-1}$	$6.1(2){\times}10^{-1}$	$5.1(1){\times}10^{-1}$	$4.8(1){\times}10^{-1}$
3707	$8.6(2){\times}10^{-1}$	$5.9(2){\times}10^{-1}$	$5.1(1){\times}10^{-1}$	$5.1(1){\times}10^{-1}$
3807	$8.2(2){\times}10^{-1}$	$5.7(1){\times}10^{-1}$	$5.4(1){\times}10^{-1}$	$4.9(1){\times}10^{-1}$
3883	$7.4(2){\times}10^{-1}$	$6.9(2){\times}10^{-1}$	$6.7(2){\times}10^{-1}$	$7.4(2){\times}10^{-1}$
3985	$7.2(2){\times}10^{-1}$	$7.5(2){\times}10^{-1}$	$8.6(2){\times}10^{-1}$	$8.7(2){\times}10^{-1}$
4083	$6.8(2){\times}10^{-1}$	$7.7(2){\times}10^{-1}$	$9.0(2){\times}10^{-1}$	$1.0(0){\times}10^{0}$
4159	$8.1(2){\times}10^{-1}$	$8.5(2){\times}10^{-1}$	$9.5(2){\times}10^{-1}$	$1.1(0){\times}10^{0}$
4208	$7.8(2){\times}10^{-1}$	$8.7(2){\times}10^{-1}$	$9.9(3){\times}10^{-1}$	$1.2(0){\times}10^{0}$
4263	$9.8(3){\times}10^{-1}$	$1.0(0){\times}10^{0}$	$1.2(0){\times}10^{0}$	$1.4(0){\times}10^{0}$
4318	$1.2(0){\times}10^{0}$	$1.3(0){\times}10^{0}$	$1.4(0){\times}10^{0}$	$1.6(0){\times}10^{0}$
4401	$1.3(0){\times}10^{0}$	$1.7(0){\times}10^{0}$	$1.8(0){\times}10^{0}$	$1.8(0){\times}10^{0}$
4510	$1.5(0){\times}10^{0}$	$1.9(1){\times}10^{0}$	$1.9(0){\times}10^{0}$	$1.6(0){\times}10^{0}$
4630	$1.4(0){\times}10^{0}$	$2.0(1){\times}10^{0}$	$2.0(1){\times}10^{0}$	$2.1(1){\times}10^{0}$
4709	$1.3(0){\times}10^{0}$	$1.7(0){\times}10^{0}$	$1.6(0){\times}10^{0}$	$1.3(0){\times}10^{0}$
4768	$1.3(0){\times}10^{0}$	$1.6(0){\times}10^{0}$	$1.4(0){\times}10^{0}$	$1.2(0){\times}10^{0}$

TABLE D.3 CONTINUED

E_α (keV)	100	135	145	155
4841	$1.1(0)\times10^{0}$	$1.0(0)\times10^{0}$	$8.5(2)\times10^{-1}$	$6.9(2)\times10^{-1}$
4905	$9.0(2)\times10^{-1}$	$8.3(2)\times10^{-1}$	$7.7(2)\times10^{-1}$	$6.7(2)\times10^{-1}$
4909	$9.1(2)\times10^{-1}$	$8.9(2)\times10^{-1}$	$8.1(2)\times10^{-1}$	$7.0(2)\times10^{-1}$

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